IHP SG13G2 · OPEN SOURCE RFIC EM WORKFLOW

gds2palace FEM workflow for IHP

User's Guide

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1 What's New

This chapter gives a brief overview of major features added since the previous edition of this guide. For the complete, dated change log, see `CHANGES.md` in the repository.

- **settings['conductor_model']** chooses the conductor model in one setting: `'surface'` (default, surface impedance), `'solve_inside'` (conductors meshed as volumes) or `'solve_inside_skin'` (volumes with thin layers along their faces for the skin effect). With the surface model, the peak Q of coils on TopMetal1/TopMetal2 with 2-8 µm traces is 18-30 % too high; for Q and loss, use `'solve_inside'` while designing and `'solve_inside_skin'` for final numbers. See "Choosing the conductor model" in chapter "Conductor loss modelling".
- `conductor_model` replaces **settings['filled_metals']**. Scripts that set it still work: `True` is the same as `'solve_inside'`, `False` the same as `'surface'`.
- **settings['z_thickness_factor'] was removed.** Scripts that still set it get a warning, and the full metal thickness is used on all conductor faces. Use `conductor_model` for accurate conductor loss.
- **Chapter "Dielectric loss modelling" redone** with `'solve_inside_skin'`. The earlier agreement with the measured line up to 40 GHz came from the surface model's extra loss in the Metal1 ground plane. With the right conductor loss, a SiO2 loss tangent of about 0.015 explains most of the measured line loss from 20 to 110 GHz. The supplied stackups stay lossless. The chapter also describes the loss of the silicon substrate.
- **Elmer FEM as an additional solver.** gds2palace can now generate simulation input not only for AWS Palace, but also for Elmer FEM, an open-source multiphysics solver. This adds two new workflows, both built from the same GDSII layout + XML stackup + Python model script approach already used for Palace:
  - **Elmer EM simulation** — S-parameter simulation, see chapter "Using gds2palace with Elmer FEM for EM simulation".
  - **Elmer thermal simulation** — steady-state heat conduction (temperature) simulation from user-defined heat sources and constant-temperature boundaries, see chapter "Using gds2palace with Elmer FEM for thermal simulation".
- **Derived layers in the XML stackup.** A layer's geometry can now be computed from other layers using boolean operations (AND/OR/XOR/NOT) and resizing, instead of only being read directly from GDSII. This is used, for example, to derive SG13G2 resistor geometry from existing process layers, see the example in folder `more_examples/derived_layers_and_resistors`.
- **Reference-relative layer positioning in the XML stackup.** A `<Layer>` or `<Dielectric>` can now be positioned as an offset from the top or bottom edge of another named layer/dielectric, instead of always requiring an absolute Zmin/Zmax. This removes the need to hand-recompute every dependent layer's z-position whenever a thickness changes.
- **Simplified handling of cutouts.** Mesh generation was redesigned so that the `preprocess_gds` option is no longer required for layouts with cutouts (holes) or other self-intersecting polygon boundaries.
- **setupThermal.** The companion desktop tool `setupEM` now also includes `setupThermal`, a GUI for building Elmer thermal models without writing Python code, and a GUI-driven XML stackup editor. See https://github.com/VolkerMuehlhaus/setupEM

The XML stackup file format used by both Palace and Elmer models has grown to support these new features. See chapter "Extended XML stackup format" in the Appendix, and the full attribute reference in `XML_stackup_format.md`.

2 About this workflow

Palace, for **PA** rallel **LA** rge-scale **C** omputational **E** lectromagnetics, is an open-source, parallel finite element code for full-wave 3D electromagnetic simulations. It can be scaled from single computer to large high performance simulation clusters and cloud-based computing.

https://awslabs.github.io/palace/stable/
https://aws.amazon.com/de/blogs/quantum-computing/aws-releases-open-source-software-palace-for-cloud-based-electromagnetics-simulations-of-quantum-computing-hardware/
The gds2palace workflow enables RFIC FEM simulation using Palace from GDSII layout files.

3 Workflow

In this document, the status of the IHP workflow developed for AWS Palace FEM solver is documented.
Two files must be provided by the user: the **layout in GDSII format** and a **simulation model script** in Python. This simulation script references the technology stackup file (XML format) and calls functions from the workflow code to create two output files for Palace: the **simulation mesh file** which is built from geometries and stackup, and the **simulation control file** in *.json format which defines material settings and simulation settings.

Having these two files, the user can now run the FEM simulation in Palace. The simulation model script can start an external command to start simulation, or the user can run Palace on a platform of his choice. This way, the files for Palace can be created on any desktop computer, and the Palace simulation can be done on the same computer or on a very different system. This gives many options to scale Palace simulation power as needed, using the exact same Palace input data from our workflow.

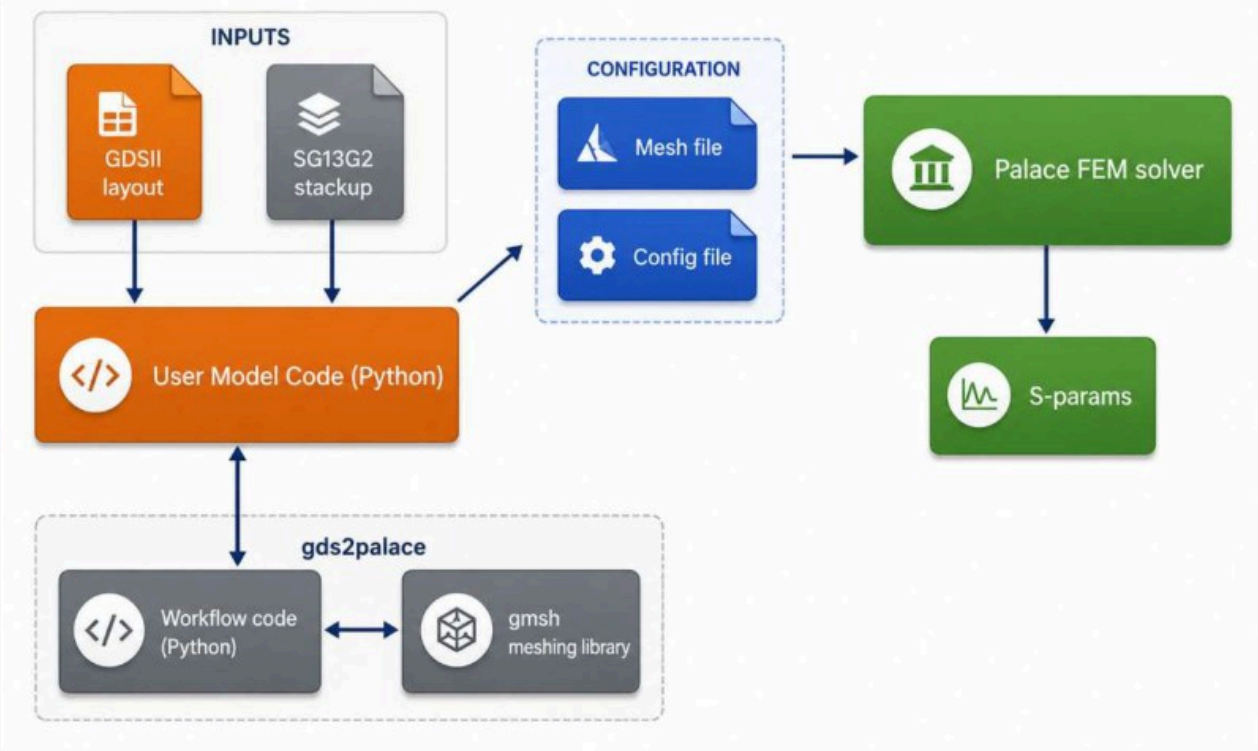


Figure 1: Workflow overview: from GDSII layout, XML stackup and model script to Palace input files

4 Required software and Python modules

The gds2palace Python workflow creates AWS Palace model files (mesh file + config file). To actually simulate these models, you need to have AWS Palace installed. Installing Palace is described below in chapter “Installing AWS Palace”.

4.1 All-in-one installation script for Linux

An installation script for Linux is available, which

- creates the Python venv for gds2palace and its user interface setupEM
- installs gds2palace, setupEM and all dependencies
- installs the AWS Palace solver as an Apptainer container image, and checks that it runs
- installs the scripts to run Palace models from gds2palace (run_palace, combine_snp)

See https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/tree/main/scripts/install_linux

4.2 Manual installation

It is recommended to install gds2palace into a Python virtual environment. In this document, we assume Linux environment and install to a virtual environment `palace` that is located in the user's home directory at `~/venv/palace`

This is how to create that virtual environment:

```
cd ~
python3 -m venv ~/venv/palace
```

4.3 Recommended installation of gds2palace as Python module

The easiest method to install the gds2palace Python workflow is to use the preconfigured package which automatically installs all dependencies (gdspy, gmsh, scikit-rf). We activate the virtual environment and install to there:

```
source ~/venv/palace/bin/activate
pip install --upgrade gds2palace
```

That's all we need to do for installing the gds2palace Python module. If you install gds2palace this way, you only need the gds2palace repository to download the examples and XML stackup files and documentation.

To run a model script using gds2palace, you then activate the venv first:

```
source ~/venv/palace/bin/activate
```

As an alternative, it would also be possible to work with a local copy of the gds2palace directory, as described below. However, this is more complicated and no longer recommended!

4.4 Alternative, no longer recommended: local gds2palace directory

It is possible to use a local directory with gds2palace module code. This directory can be downloaded from the gds2palace github repository and must always be copied to the directory where the *.py simulation model is located. The simulation model must then have a code line to include that local gds2palace directory in the Python path.

In this case, Python module dependencies must be installed manually:

```
source ~/venv/palace/bin/activate
pip install gdspy
pip install gmsh
pip install scikit-rf
```

This virtual environment with installed dependencies can be used to run gds2palace and the scripts that postprocess the Palace simulation results. To activate this Python environment on the command line:

```
source ~/venv/palace/bin/activate
```

4.5 External tools (not Python modules)

Two more tools are needed for parts of the workflow, but are not installed via pip:

- ParaView — to view field-dump output (Palace/Elmer EM) and Elmer thermal result files.
- An MPI implementation — only needed for multi-process Elmer runs (`settings['ELMER_MPI_THREADS']`, see chapter "Installing Elmer FEM"). Use OpenMPI or MPICH on Linux/macOS; on Windows, install Microsoft MPI.

5 Installing AWS Palace

To actually simulate the model created by gds2palace, you need to have AWS Palace installed. Different methods of installing Palace are described at https://awslabs.github.io/palace/stable/install/ but we have compiled the most relevant methods for gds2palace users below.

5.1 Installing the Palace solver using Apptainer

For development of gds2palace, Palace was installed using the Singularity/Apptainer installation method. This was rather simple and straightforward, even with no knowledge about container usage. A step-by-step guide can be found here: building-palace-apptainer.md

A pre-built container image for Palace 0.18.1 is also available, if you'd rather not build anything yourself. It runs on all Intel processors since 2013 and all AMD processors since 2017. To download it into your current directory:

```
$ apptainer pull palace_0181.sif oras://ghcr.io/volkermuehlhaus/palace_0181:latest
```

5.2 Installing the Palace solver using spack package manager

Palace can also be created from source with a few simple commands. All tools required by the build process will be downloaded and installed automatically by spack, so you can sit and watch while your system builds the software.
A step-by-step guide for compiling Palace using the **spack package manager for Linux** : building-palace-spack.md

5.3 Running Palace (after installation)

You can use any of the installation methods described on the AWS Palace web site. The gds2palace workflow does not change, it only creates the input files for Palace and does not care how you installed Palace, or on what platform you run the actual Palace simulation from these model files.

To start Palace from setupEM, a wrapper script **run_palace** is used, and this is where you point to your actual installation (even remote copy & remote simulation is possible). This script is available in the scripts folder in gds2palace repository, and needs to be adjusted for your path settings and number of processor cores.

6 Quick tour

This chapter will give a very brief overview of the workflow usage, without going into the details.

To get started, we need a directory that contains:

- GDSII layout file to be simulated
- XML stackup file
- Only if gds2palace is **not** installed as Python module: local copy of gds2palace directory

In addition, we need the simulation model code in Python, which brings the workflow to life. Below are the most important code sections that need to be configured by the user.

6.1 Simulation model: Input files

In this section, the user specifies the GDSII data source and the technology stackup file.

```
# ====================== input files and path settings ======================

gds_filename = "line_simple_viaport.gds"   # geometries
XML_filename = "SG13G2_nosub.xml"          # stackup

# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False
```

6.2 Simulation model: Simulation control

In this section, the user needs to specify frequency range and mesh settings. There are additional optional settings that can be applied, as discussed later in this document.

```
settings['unit']   = 1e-6  # geometry is in microns
settings['margin'] = 50    # distance in microns from GDSII geometry boundary to simulation boundary

settings['fstart']  = 0e9
settings['fstop']   = 100e9
settings['fstep']   = 2.5e9

settings['refined_cellsize'] = 2  # mesh cell size in conductor region
settings['cells_per_wavelength'] = 10   # how many mesh cells per wavelength, must be 10 or more

settings['meshsize_max'] = 70  # microns, override cells_per_wavelength
settings['adaptive_mesh_iterations'] = 0
```

Users who are familiar with the IHP openEMS workflow will notice many similarities, although some implementation details are different.

One important difference is that FEM simulates one frequency after another, so frequency settings have an effect on total simulation time. However, Palace is configured to use an adaptive frequency sweep, which will EM simulate only the minimum required number of frequencies across the band, and then create all other frequency points using interpolation. This adaptive sweep is enabled by default in our workflow.

Another important difference is the use of “refined_cellsize” to create the mesh: the gmsh meshing engine will always create a mesh based on all geometry detail, no matter how small it is, and then apply an additional mesh refinement based on “refined_cellsize” along the metal edges.

6.3 Simulation model: Ports

Additional important user input are the port definitions.

Similar to the IHP openEMS workflow, **ports are created based on polygons from the GDSII file** , located on special layers that are not part of the IHP layer table. Each port needs to use a different source layer in the GDSII file.

The port mapping defines the port number and port impedance, and then maps the port geometry from the special GDSII input layer (usually 201 and above) to actual IHP technology layers. For the vertical ports shown here, we have from_layername and to_layername. For in-plane ports, we would have target_layername instead. Finally, the port direction is required to specify vertical ports or in-plane ports and their direction/polarity.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
```

```
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=1, port_Z0=50, source_layernum=202,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
```

For this Palace workflow, the voltage parameter is not supported yet by the Palace solver. Palace only supports opposite polarity by reversing the port direction. **But here in our workflow, we use this voltage parameter to specify if a port is active: ports with voltage=0 will not be excited.**

It must be noted that Palace behaves differently from other FEM solvers: to get the full S- matrix, we need to run all port excitations, one after another. This only happens if all port voltages are not zero. Only then, the full S-parameter output file can be created.

ATTENTION : If any port excitation is zero, that "zero voltage" port will not be excited and that row in the S-parameter file will be padded with zeros. For example, if we simulate an 8-port circuit and only excite port 1, the S-parameter output will have valid results for S11, S21, … S81 but all other values will be zero. This can be useful to quickly simulate one specific path in the model.

To get the full S-parameter file, all ports must be defined with non-zero voltage, so that all port excitations are simulated, one after another.

PORT SHAPE IN GDSII : The workflow creates lumped ports in Palace, which need to be 2D sheets. This means that for vertical ports (via ports), you should draw a zero-width box in GDSII, resulting in a vertical 2D sheet with no width. If you define a vertical port from a box with finite xy area, the Palace port will be created as a vertical 2D sheet along the left edge (xmin) of that box if it is taller than wide, else along its bottom edge (ymin), not along its center line.

SEVERAL SHAPES ON ONE PORT LAYER : Each shape becomes one element of the same port, and the elements are in parallel (a Palace multielement lumped port; the port impedance stays the given Z0). This is useful for a terminal that is spread over several strips, e.g. the source fingers of a multi-finger transistor: draw one line per finger on the same port layer. Multielement ports are supported for Palace only, not for Elmer.

Composite ports (multiple EM ports grouped into one port in the final output file) are not yet supported by this workflow. All ports with non-zero voltage are simulated one after another, resulting in n-port S-parameters.

6.4 Running the model code to create Palace input files

The simulation model file (Python code) can be run on the command line. After reading and processing the input files, a 3D viewer comes up and shows the resulting 3D model. This viewer is the graphical interface of the gmsh meshing library, and provides many options for inspection of the model. At this point, the model is not meshed yet, so that we can see the raw geometries.

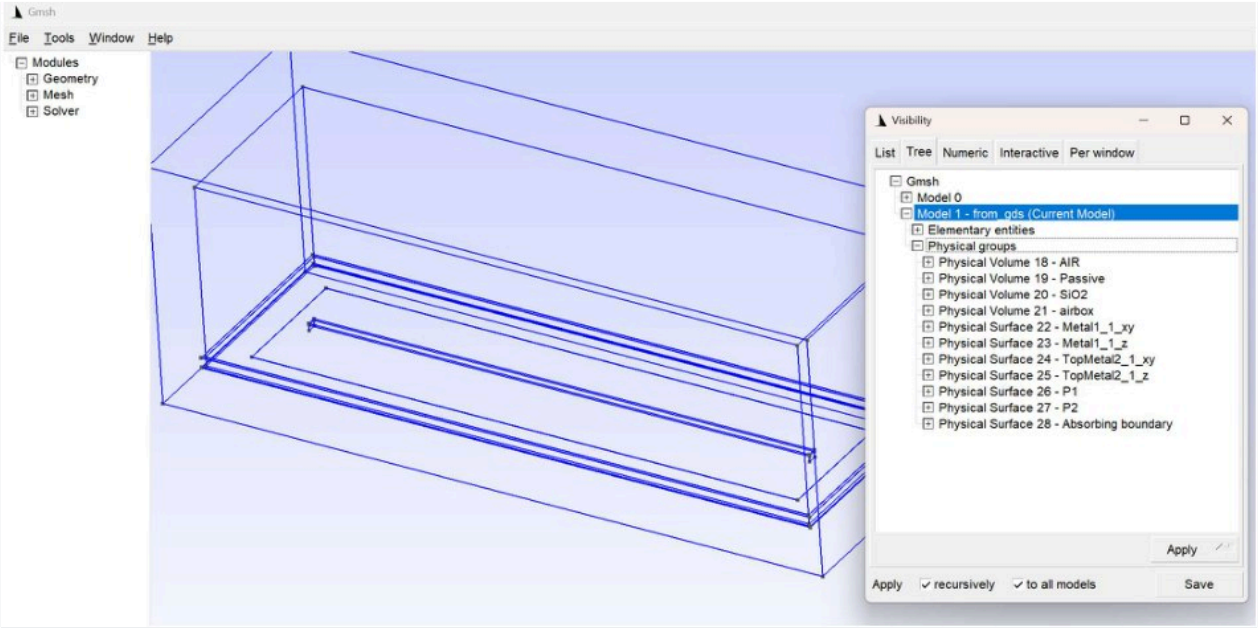


Figure 2: gmsh preview of the raw model geometry, before meshing

To see the structure of the 3D model, you can go to Tools > Visibility.

In the screenshot, you can see that dielectric boxes (SiO2, Passive, AIR etc) have been added around the GDSII polygons, with an offset value in xy direction from the “margins” parameter in the model file. Around that, we have another layer of air on all six sides, using the same margins value.

Metals have been created as surfaces. All polygons on each layer are merged, if possible, and then each of the resulting polygons is created as a separate surface. To be more precise, all faces of each resulting polygon body (top, bottom and side walls) form one surface group, named after the layer and a number, e.g. TopMetal2_1.

You can also see two ports P1 and P2 created as surfaces. The outer simulation boundary is this example is a surface “Absorbing boundary” that is defined as absorbing boundary in the Palace config file.

When we are done with inspecting the model (which is optional, no user action is required!), the gmsh viewer window can be closed to proceed with meshing. After closing the gmsh geometry preview, the simulation model script will mesh these geometries, which can take a while and will show lots of status information on the command line.

When meshing is completed, the gmsh 3D viewer will be displayed again, showing the overall mesh.

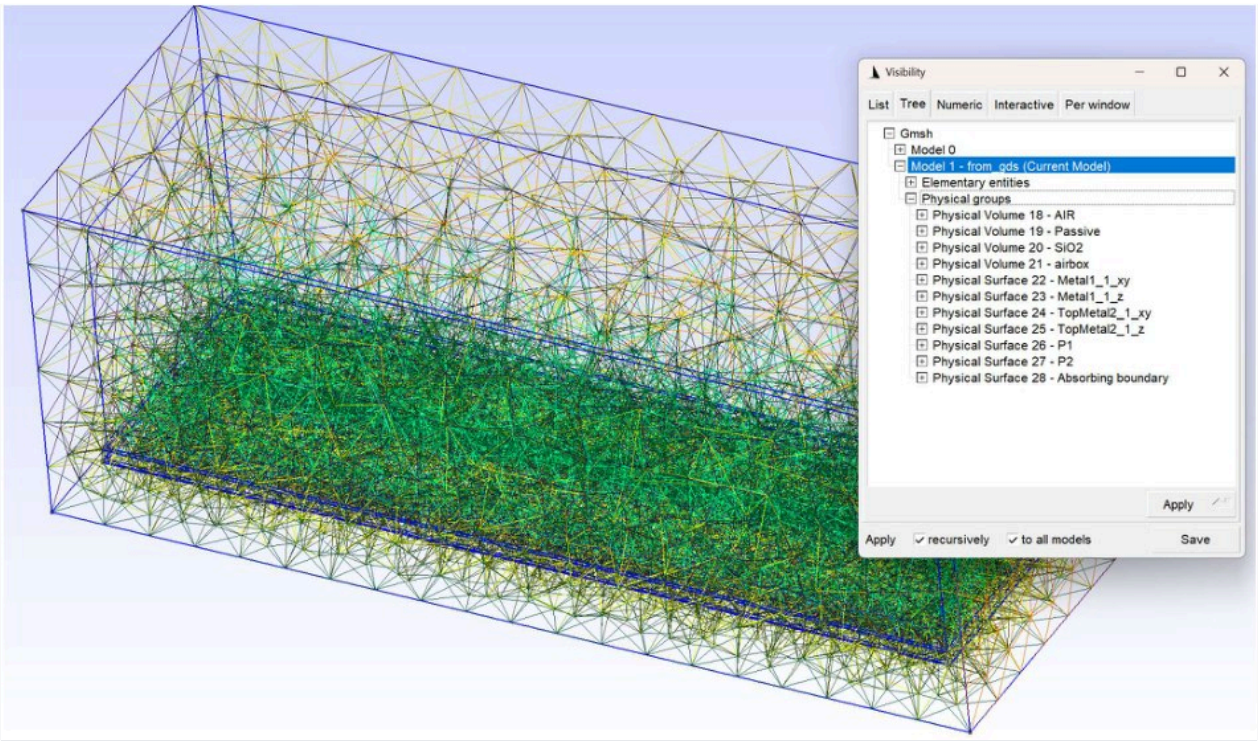


Figure 3: gmsh view of the complete mesh

This is too complex to see anything, but we can now go to Tools > Visibility and select one or more groups to be displayed:

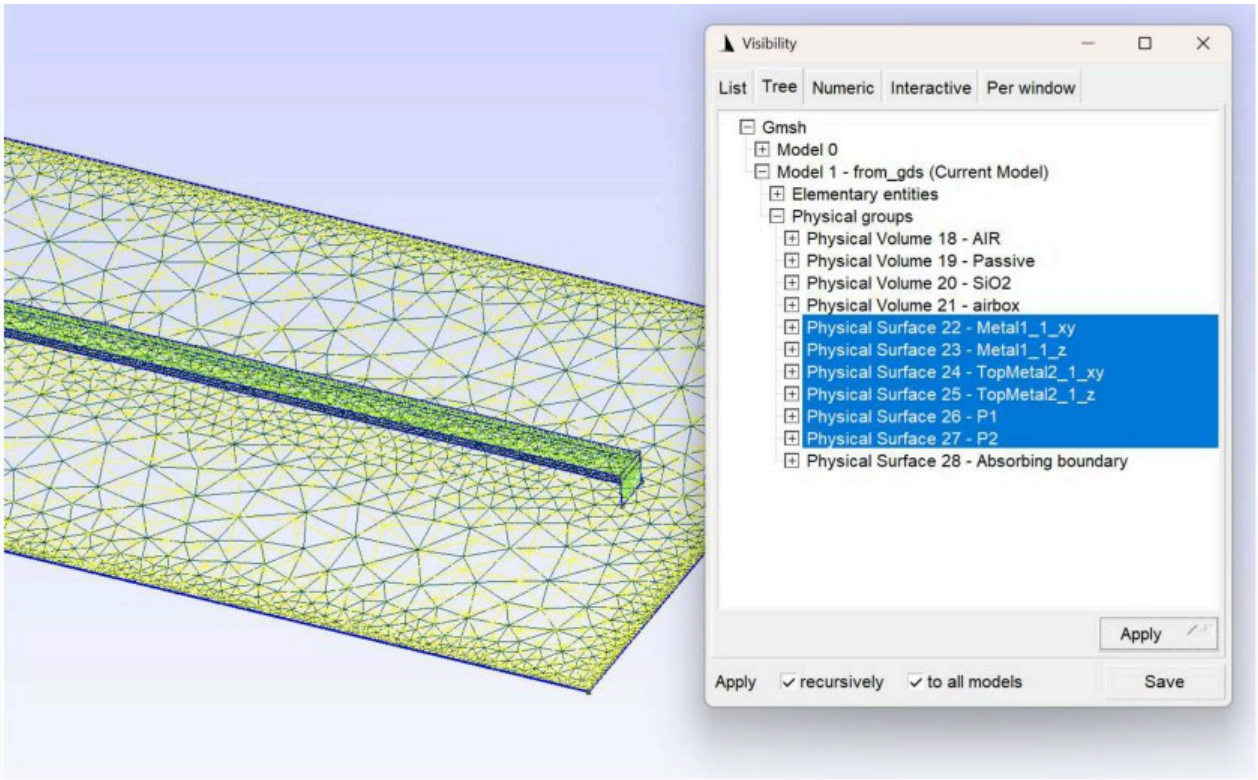


Figure 4: Mesh on the conductor surfaces and ports, selected in Tools > Visibility

This shows the mesh at the conductor surfaces of Metal1, TopMetal1 and the vertical ports. We can also display the meshed oxide, which has a lot of detail because the metal layers reside inside this volume. The metals have been “cut out” from the dielectric layer, and we see the mesh refinement around the conductor edges.

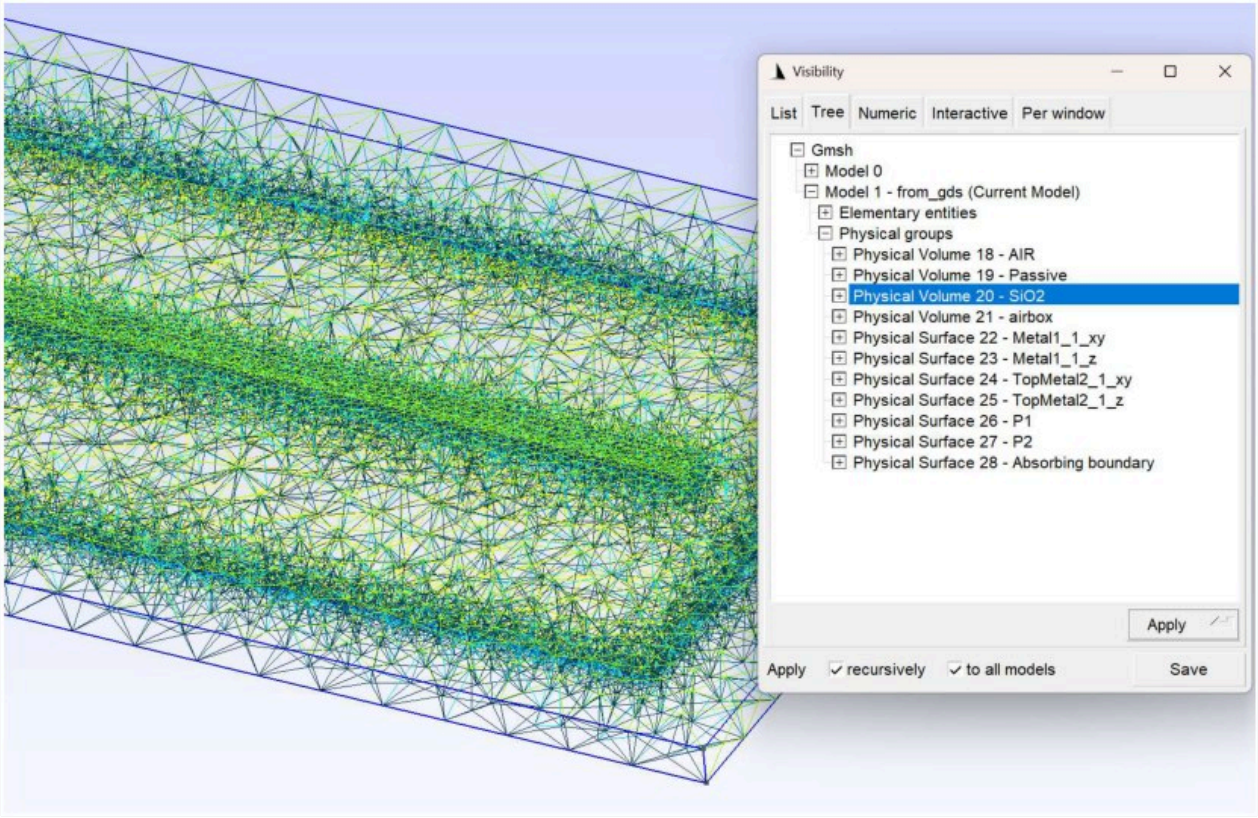


Figure 5: Mesh of the SiO2 volume, with the metals cut out and refined mesh at the conductor edges

When you are done with inspection of the model (which is optional, no user action is required!), the gmsh viewer window can be closed to finish model generation. The workflow code has now created the required file to start Palace: the simulation control file config.json and the mesh file.

```
{
    "Problem": {
        "Type": "Driven",
        "Verbose": 3,
        "Output": "output/palace_line_viaport"
    },
    "Model": {
        "Mesh": "palace_line_viaport.msh",
        "L0": 1e-06,
        "Refinement": {}
    },
    "Solver": {
        "Driven": {
            "MinFreq": 0.1,
            "MaxFreq": 100.0,
            "FreqStep": 2.5,
            "SaveStep": 0,
            "AdaptiveTol": 0.01
        },
        "Linear": {
            "Type": "Default",
            "KSPType": "GMRES",
            "Tol": 1e-06,
            "MaxIts": 400,
            ...
```

6.5 Running Palace FEM simulation from our input files

To simplify running the solver, in addition to mesh file and config file, a script file was also created named *run_sim*.

```
#!/bin/bash
run_palace config.json
combine_snp
```

(The `run_sim` of a model with port walls first solves the calibration standards, and since gds2palace 0.10.0 the `_data` folder also holds a copy of `combine_extend_snp.py`, the result script matching this gds2palace, which `combine_snp` runs when present.)

This starts a script *run_palace* where you can define in detail how to run Palace. It is recommended that you place this run_palace script in your PATH, configured for the actual machine where you want to run Palace.

For workflow development and test, Palace was installed into an apptainer container, as documented in the Palace documentation in chapter "**Build using Singularity/Apptainer**".

The script *run_palace* was then configured to start Palace inside the palace_014.sif container, which is located in the user's home directory, and passes one command line argument (the config.json file). The additional parameter *-np 16* tells palace to run using 16 threads. This will partition the simulation domain into 8 pieces, each running in a separate process, and the combine results into one output directory.

```
#!/bin/bash
apptainer exec ~/palace_014.sif palace -np 16 $1
```

Running Palace is not very spectacular, the simulation progress is shown in the terminal window.

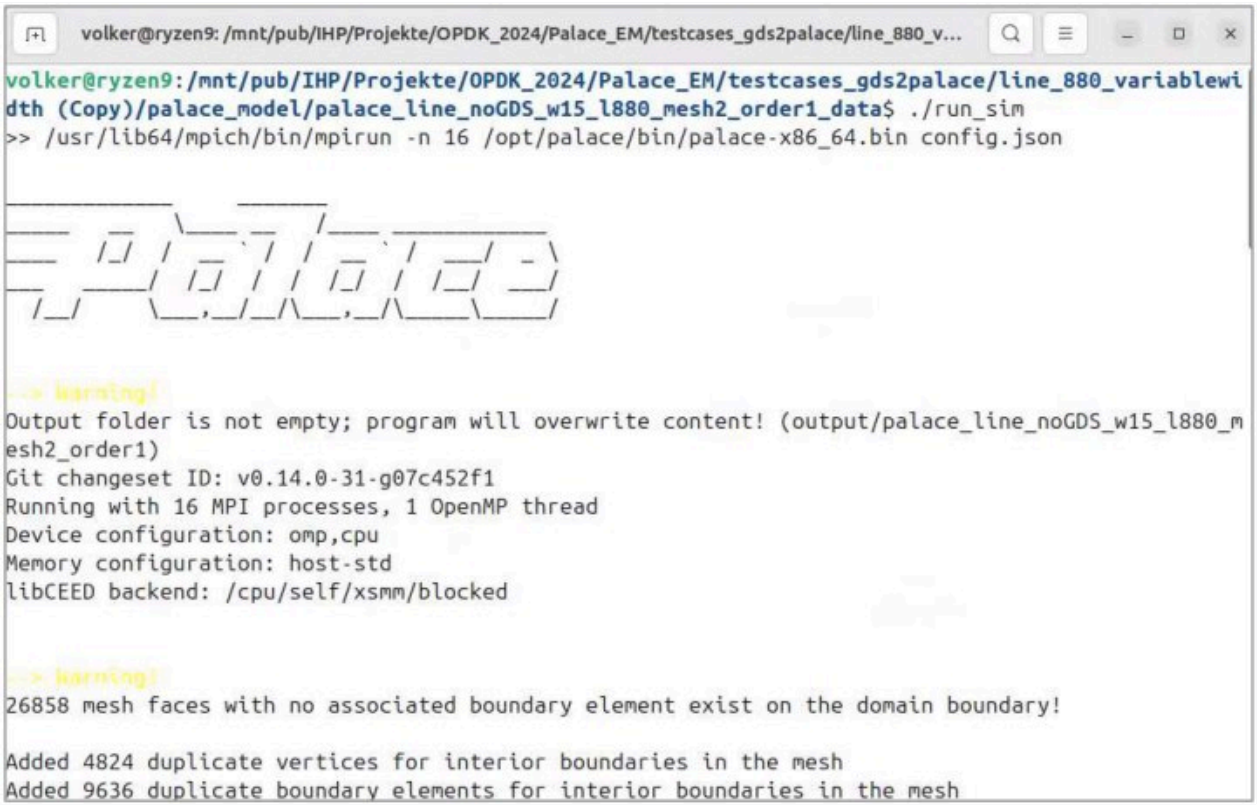


Figure 6: Palace simulation progress in the terminal window

When *run_palace* is finished, Palace output files are created in the *output* directory below the simulation model directory. S-parameters are in *.csv file format, which we need to convert now.



Figure 7: Palace output files

To do that conversion, the run_sim script where we started simulation with “run_palace” executes another command *combine_snp*.

combine_snp is a script that searches for Palace s-parameter files (port-S.csv) and converts them to Touchstone file format. If the model was simulated for selected port excitations only, missing rows in the S-parameter file will be padded with zeros.

It should be noted that Palace frequency domain simulation cannot go down to 0 Hz, and the workflow script will replace any 0 Hz start frequency by a low value like 1 GHz.

If simulation results include such low frequency data, combine_snp will create an additional S-Parameter file with suffix “_dc”, with a DC value extrapolated from the EM simulated data.[1] Always check that DC extrapolated dataset carefully before use!



Figure 8: Output files after combine_snp: Touchstone file and DC-extrapolated Touchstone file

This ends our quick tour of using the gds2palace workflow. There are many additional settings available, which will be described in detail in the next chapters.

[1] combine_snp is a Python script created for this workflow. DC extrapolation in this script is powered by Python library scikit-rf.

7 Simulation model file in detail

7.1 Input files

Typical simulation models require two input files: GDSII geometries and XML stackup.

```
# ====================== input files and path settings ======================

gds_filename = "BM_Ardavan_Rahimian_with_ports.gds"   # geometries
XML_filename = "SG13G2_nosub.xml"          # stackup
```

Two settings are available to control processing of the GDSII geometries:

```
# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False

# merge via polygons with distance less than .. microns, set to 0 to disable via merging.
merge_polygon_size = 0
```

preprocess_gds is no longer needed: layouts with cutouts (holes) or other self-intersecting polygon boundaries are now handled automatically. If an older model script still sets it to True, the GDSII reader ignores it and prints a note. Earlier versions needed it, otherwise meshing of layouts with holes failed with errors like "Exception: Curve loop is not closed".

merge_polygon_size applies to layers which are declared as *Type="via"* in the XML stackup file.

When enabled with a non-zero value, the code will merge polygons on these layers with the given distance. This is how it works: in the GDSII reader, polygons on this layer will be oversized by half the given value, then all overlapping polygons on that layer will be merged, then undersized by half the given value.

For via arrays that are oriented along the xy-axis, this will give the resulting bounding box if the maximum via spacing is no larger than the given value.

Merging turns the gaps between the vias into solid via material, so the merged polygon contains more via metal than the real via array. By default, it still gets the full via conductivity from the XML stackup, which overestimates the conductivity of the via array: in the IHP stackup files, via conductivity is calculated from the resistance per via in the process specification, so it describes a single via, not a via array. With settings['fill_factor_correction'] = True (see settings), the conductivity of each merged via polygon is multiplied by its fill factor (original via area / merged polygon area).

This correction only knows about merging done here, by merge_polygon_size. If the via arrays were already merged before, e.g. with gds_prepare_for_EM, they arrive as solid polygons with fill factor 1.0, and their conductivity is not corrected.

7.2 settings

settings for meshing and simulation control are implemented as a Python dictionary, which is passed to the simulation_setup.create_palace function. Some settings are required, other are optional with a meaningful default value.

These settings are always required:

settings['unit']: Unit of values in mesh, typically 1E-6
settings['margin']: Oversize of dielectrics from bounding box of drawing in xy plane. Can be a single value (all sides) or a list of 4 values [margin_xmin, margin_xmax, margin_ymin, margin_ymax]. All values must be 0 or more.
settings['fstart']: Start frequency in Hz
settings['fstop']: Stop frequency in Hz
settings['fstep']: Frequency step in Hz for output, not all of them need to be EM simulated due to adaptive

frequency sweep
settings['refined_cellsize']: Target mesh size at polygon edges

All other settings are optional, grouped here by what they control.

7.2.1 Frequencies

Discrete frequencies can be used in addition to fstart/fstop and fstep, or instead of fstart/fstop and fstep:

settings['fpoint']: Discrete frequency/frequencies, values enclosed in [ ]. Example: settings['fpoint']=[10e9, 15e9]
settings['fdump']: Same as fpoint, but Palace is configured to write a field dump for Paraview at this frequency/these frequencies. Example: settings['fdump']=[10e9]
settings['adaptive_sweep']: Enable adaptive frequency sweep, default is True

7.2.2 Mesh size and accuracy

settings['cells_per_wavelength']: Calculated at highest frequency,value must be 10 or more, default is 10
settings['meshsize_max']: Maximum mesh size limit, in addition to cells, default is 70
settings['refined_cellsize_override']: Optional per-layer override of refined_cellsize,example: settings['refined_cellsize_override']=[['Metal3',10],['Metal2',2]]
settings['substrate_refinement']: Extra mesh refinement into substrate, usually not required, default is False
settings['order']: Order of basis function for FEM solver (order 2 is more accurate,order 1 is only for quick & dirty results). Default is 2

7.2.3 Adaptive mesh refinement

settings['adaptive_mesh_iterations']: Iterations for adaptive mesh refinement, often not required when using fine initial mesh, default is 0
settings['adaptive_mesh_conformal']: Use conformal AMR refinement instead of Palace's default nonconformal (hanging-node) refinement, default is False
settings['amr_tol']: Target relative error for adaptive mesh refinement: refinement stops once the error estimate is below this value. Only used with adaptive_mesh_iterations > 0, default is 0.01
settings['amr_max_dof']: Maximum number of unknowns for adaptive mesh refinement: refinement stops once the model reaches this size. Only used with adaptive_mesh_iterations > 0, default is 2e6
settings['save_adaptive_mesh']: Save mesh file from adaptive iteration for possible re-use, default is False

7.2.4 Linear solver (Palace)

settings['complex_coarse_solve']: Let Palace's sparse direct coarse solver factorize the full complex system, instead of only its real part (Palace's own default). **Default is True, for every model**, whether metals are meshed as surface impedance or filled. The real-part-only preconditioner ignores absorbing boundaries, port resistances and losses. Some models then need far more iterations or don't converge at all, and a frequency that doesn't converge gets **wrong S-parameters**, not just slightly inaccurate ones. Which models are affected can't be predicted from the settings: it happened at high frequency (a D-band grounded CPW line failed at 170 GHz) and at a single low frequency (an inductor failed at 2.4 GHz and gave 1.1 nH instead of 2.5 nH, while the same model converged at 0.1 and 10 GHz). With True, both converged in 1-31 iterations and ran 3.5-6x faster. Models that already converge without it get identical results, in about the same or less time. The cost is peak RAM, which depends on the model: 1.0-1.6x at order 2 and 1.5-1.9x at order 1 in our test cases. Set it to False only if a model doesn't fit into memory, and then check the Palace output for `did NOT converge`
settings['solver_maxits']: Iteration limit of Palace's linear solver (GMRES) per frequency. If it is reached, Palace only logs "solver did NOT converge" and carries on, and the S-parameters at that frequency are unreliable. Default is 400
settings['solver_tol']: Relative residual tolerance of Palace's linear solver. Default is 1e-6

7.2.5 Simulation boundary

settings['boundary']: List with 6 values for boundary at xmin,xmax,ymin,ymax,zmin,zmax. Values can be ABC/PML, PEC or PMC, and PORTWALL on xmin, xmax, ymin or ymax for calibrated ports at the layout edge

(see “Calibrated de-embedding at port walls”). Default: [‘ABC’,‘ABC’,‘ABC’,‘ABC’,‘ABC’,‘ABC’]
settings[‘air_around’]: Other spacing of air around dielectrics, default is same as margin (with a list of 4 margin values: the margin of each side, and the largest margin for zmin and zmax). Can be a single value or a list of 6 values [air_xmin, air_xmax, air_ymin, air_ymax, air_zmin, air_zmax]. A value of 0 is allowed on any side, placing the simulation boundary flush with the dielectric/metal stack on that side (e.g. a backside ground plane serving directly as the PEC/ABC boundary, with no wasted air layer below it).

7.2.6 Conductor modelling

settings[‘conductor_model’]: The conductor model. Errors, run times and what they mean for typical designs: see “Choosing the conductor model” in chapter “Conductor loss modelling”. Default is ‘surface’

- `'surface'`: a surface impedance on all conductor faces, the conductors are hollow. The fastest model, and accurate for inductance, coupling, resonances, match and balance. Its loss is less accurate: the peak Q of coils on TopMetal1/TopMetal2 with 2-8 µm traces is 18-30 % too high, and thin layers with current on one face get up to twice their loss.
- `'solve_inside'`: the conductors are meshed as volumes with their bulk conductivity. This gives the right loss at low frequency, where the skin depth is not small compared to the conductor cross section. At high frequency, the loss is a few percent too high at a 2 µm mesh. Use it for Q and loss while designing. Needs more RAM and simulation time than the surface model.
- `'solve_inside_skin'`: like `'solve_inside'`, plus thin layers along all faces of each conductor (top, bottom and side walls). The mesh then has thin cells next to the conductor surface, where the skin effect puts the current, while the cell size along the surface stays at `refined_cellsize`. Use it for final numbers of Q and loss.

There is nothing to set for the skin layers. Their depths are about 0.4 and 1 skin depth at the highest simulation frequency (the largest of `fstop`, `fpoint`, `fdump`), from each layer’s conductivity. A depth is only used if a core of at least 1/3 of the layer thickness remains; the model script prints the depths used per layer. Some layers get no skin layers:

- Vias, sheet layers and PEC layers.
- Layers thinner than 4 skin depths at the highest frequency. In SG13G2, these are Metal1 below about 1 THz, Metal2-Metal5 below about 700 GHz, TopMetal1 below about 35 GHz and TopMetal2 below about 15 GHz. They gain little, while their very thin skin layers cost much solver effort. For the 0.42 µm Metal1 ground plane of `palace_line_viaport.py`, skin layers changed the line resistance by about 1 % and made the simulation 2.6 times slower. In the D-band balun of the mesh convergence studies, skin layers in Metal3-Metal5 left the result unchanged and made it 1.7 times slower.
- Layers with a conductivity above 1e8 S/m. No real metal conducts better than silver (6.3e7 S/m), so such a material stands in for a perfect conductor, e.g. `LOWLOSS` (1e10 S/m) for SUBGND in older stackups, and its skin layers would be a few nanometres thin.

So in a low GHz simulation, `'solve_inside_skin'` builds the same model as `'solve_inside'`. Skin layers were tested on TopMetal2 inductors and lines up to 150 GHz with `refined_cellsize` 2, see chapter “Conductor loss modelling”. Elmer thermal models always use solid conductor volumes, whatever this setting is.

Scripts from gds2palace before 0.9.0 use **settings[‘filled_metals’]** instead. It still works: `True` is the same as `'solve_inside'`, `False` the same as `'surface'`, and the model script prints a note. If both settings are given with different meanings, the script stops with an error.

settings[‘fill_factor_correction’]: Only has an effect with merge_polygon_size > 0. Via array merging fills the gaps between vias with via material, so each merged via polygon is scaled down to its real via conductivity: its conductivity is multiplied by its fill factor (original via area / merged polygon area). Merged vias on the same layer are grouped into separate materials, starting a new group when the fill factor is more than 20% below the group’s highest value, and each group uses its mean fill factor. Vias with the reserved PEC material are grouped but not scaled. Applies to Palace and Elmer EM (electrical conductivity) and to Elmer thermal (heat conductivity, including temperature table values). Default is False

7.2.7 Script control and output files

settings['no_gui']: Run script without showing gmsh user interface, useful for automated processing, default is False
settings['no_preview']: don't show unmeshed geometry, immediately show meshed model, default is False
settings['preview_only']: Only show the unmeshed geometry in gmsh, then stop without generating a mesh, default is False
settings['save_gmsh_unrolled']: Also save gmsh geometry file without meshing, for later inspection, default is False
settings['config_suffix']: Text appended to the names of the generated config.json and port_information.json files, e.g. '_fine' gives config_fine.json, default is '' (no suffix)

7.2.8 Model data set by the script

The model script also passes the stackup and layout data in the settings dictionary. These are not options: the script sets them after reading the XML stackup and the GDSII file.

settings['materials_list'], settings['dielectrics_list'], settings['metals_list']: Stackup data, as returned by stackup_reader.read_substrate()
settings['allpolygons']: Layout polygons, as returned by gds_reader.read_gds()
settings['simulation_ports']: Port definitions, created with simulation_setup.all_simulation_ports()
settings['sim_path']: Output directory for the generated model files
settings['model_basename']: Base name of the generated model files, e.g. the mesh file model_basename.msh

7.3 Port configuration

Similar to the IHP openEMS workflow, **ports are created based on polygons from the GDSII file** , located on special layers that are not part of the IHP layer table. Each port needs to use a different source layer in the GDSII file.

The port mapping defines the port number and port impedance, and then maps the port geometry from the special GDSII input layer (usually 201 and above) to actual IHP technology layers. For the vertical ports shown here, we have from_layername and to_layername. For in-plane ports, we would have target_layername instead. Finally, the port direction is required to specify vertical ports or in-plane ports and their direction/ polarity.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=1, port_Z0=50, source_layernum=202,
    from_layername='Metal1', to_layername='TopMetal2', direction='z'))
```

For this Palace workflow, the voltage parameter is not supported yet by the Palace solver. Palace only supports opposite polarity by reversing the port direction. **But here in our workflow, we use this voltage parameter to specify if a port is active: ports with voltage=0 will not be excited.**

It must be noted that Palace behaves differently from other FEM solvers: to get the full S- matrix, we need to run all port excitations, one after another. This only happens if all port voltages are not zero. Only then, the full S-parameter output file can be created.

ATTENTION : If any port excitation is zero, that "zero voltage" port will not be excited and that row in the S-parameter file will be padded with zeros. For example, if we simulate an 8-port circuit and only excite port 1, the S-parameter output will have valid results for S11, S21, … S81 but all other values will be zero. This can be useful to quickly simulate one specific path in the model.

To get the full S-parameter file, all ports must be defined with non-zero voltage, so that all port excitations are simulated, one after another.

PORT SHAPE IN GDSII : The workflow creates lumped ports in Palace, which need to be 2D sheets. This means that for vertical ports (via ports), you should draw a zero-width box in GDSII, resulting in a vertical 2D sheet with no width. If you define a vertical port from a box with finite xy area, the Palace port will be created as a vertical 2D sheet along the left edge (xmin) of that box if it is taller than wide, else along its bottom edge (ymin), not along its center line.

SEVERAL SHAPES ON ONE PORT LAYER : Each shape becomes one element of the same port, and the elements are in parallel (a Palace multielement lumped port; the port impedance stays the given Z0). This is useful for a terminal that is spread over several strips, e.g. the source fingers of a multi-finger transistor: draw one line per finger on the same port layer. Multielement ports are supported for Palace only, not for Elmer.

Composite ports (multiple EM ports grouped into one port in the final output file) are not yet supported by this workflow. All ports with non-zero voltage are simulated one after another, resulting in n-port S-parameters.

Port parasitics in results: A lumped port is a small sheet with a physical size, so it adds a little capacitance and inductance to the result. Other 3D volume meshing EM tools behave the same for lumped ports. For ports at the layout edge, gds2palace removes them by calibration, see “Calibrated de-embedding at port walls” below. For other ports, `combine_snp` removes an estimated port inductance, calculated from the port’s size.

7.4 Calibrated de-embedding at port walls

7.4.1 Why a port needs calibration

A lumped port drives the line through a small sheet. Its field fringes around the sheet, so the port adds a capacitance to ground and a short piece of inductance. Both depend on the line and the stackup. A formula can only estimate them: for a via port in the de-embedding study, the estimated inductance was about 45 % too low.

A calibration measures the port instead. The same port drives short lines of known length, the standards. Two of them, of length L and 2 L, differ only by a piece of line, and comparing them leaves the port alone. This is how planar EM solvers remove their box wall ports (J. C. Rautio, “double delay”). In the study, it removed the port 30 to 100 times better than the estimate.

7.4.2 Port walls

gds2palace calibrates the ports on a port wall. A port wall is a box side set to `'PORTWALL'` in `settings['boundary']`: a PEC wall right at the layout edge, where the lines leave the layout. The PEC wall is the ports’ ground.

```
settings['boundary'] = ['PORTWALL', 'ABC', 'ABC', 'ABC', 'ABC', 'ABC']   # xmin is a port wall
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    target_layername='TopMetal2', direction='x'))
```

A port wall has no dielectric oversize and no air, so the tool sets both to 0 on that side, whatever `settings['margin']` and `settings['air_around']` say there, and logs it. It also sets the direction of each port on the wall to point into the layout, so that both ports of a pair have the same polarity. Every port on a port wall must be excited (voltage 1), because the calibration needs the full S-matrix.

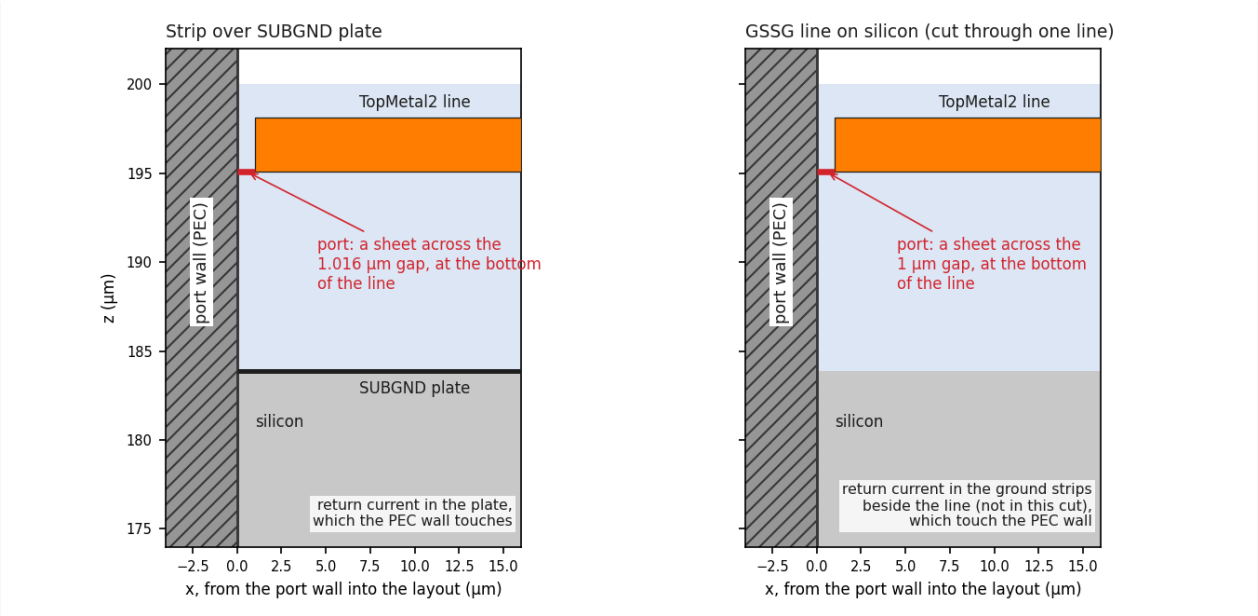


Figure 9: Gap port at the port wall, side view

The port in side view, cut through a line. The line stops a small gap before the PEC wall, and the port is a sheet across that gap at the bottom of the line, so the PEC wall is the port's ground. Left: a strip over the SUBGND plate, which touches the wall. Right: a GSSG line on silicon without a plate; its ground strips run beside the line and touch the wall.

7.4.3 What to draw

1. Draw the line to the layout edge.
2. Draw the port polygon on the layout edge, over the end of the line. Its depth into the layout is the gap between the PEC wall and the line, and the tool cuts it out of the line. A zero-width line along the edge is fine too: the tool then cuts a gap of D/32 (D is defined below). The polygon's width doesn't matter, the port takes the line's width. One port per line: a polygon that also covers a ground strip or a second line is an error.
3. Ground metal must reach the layout edge next to the lines: ground strips beside them, or a ground plane under them. Without it, the model is not written. Over bare silicon, the return current has no fixed path near the wall, and the port's capacitance then grows with the length of the standards instead of settling.

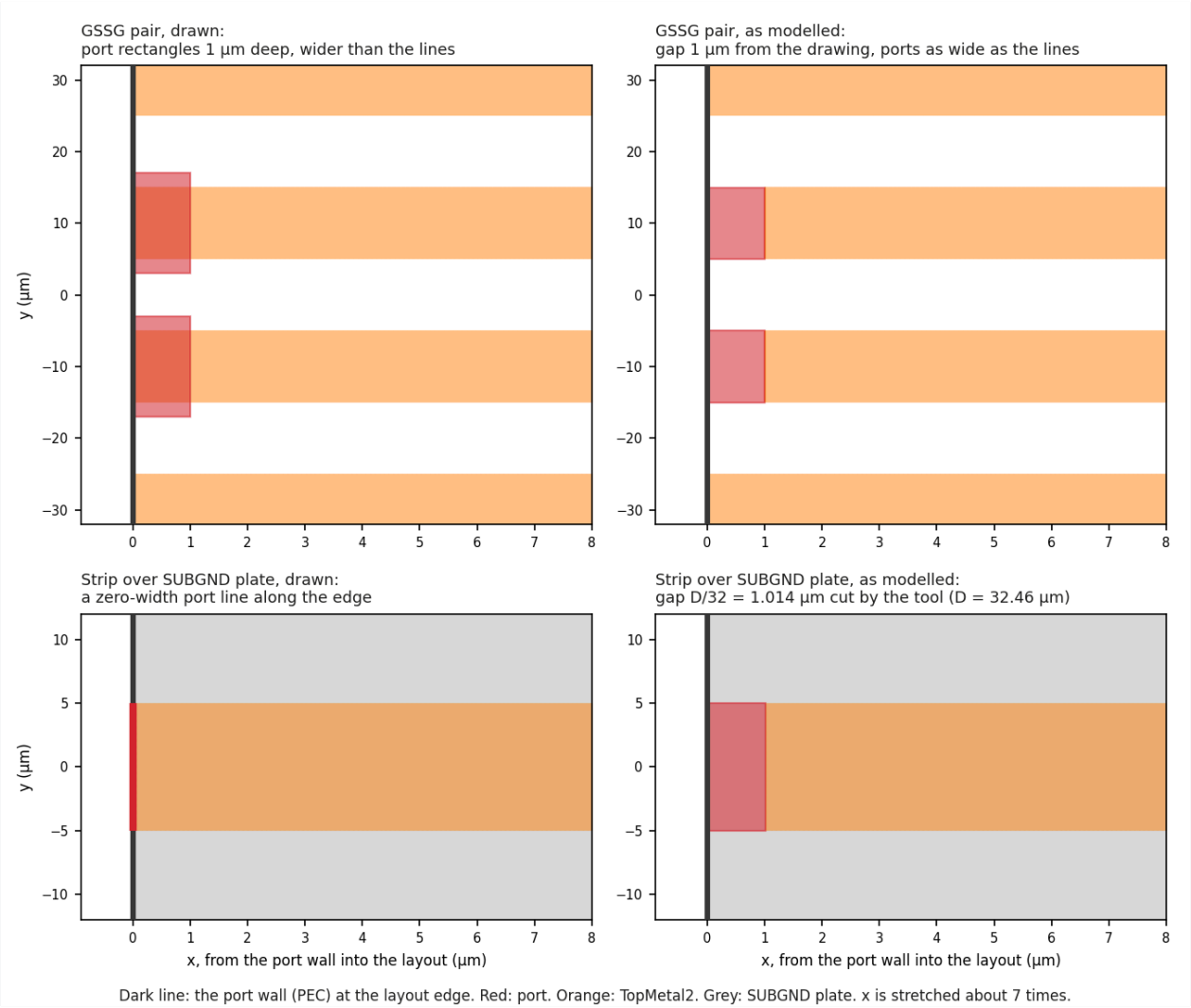


Figure 10: What to draw and what gds2palace makes of it, top view

Top view at the port wall, as drawn (left) and as modelled (right). Top: a GSSG pair with port rectangles 1 µm deep and wider than the lines; the model keeps the 1 µm gap and makes each port as wide as its line. Bottom: a strip over the SUBGND plate, drawn to the edge with a zero-width port line along the edge; the tool cuts a gap of D/32. The x axis is stretched to show the gap.

All ports on one port wall are calibrated together, so the coupling between the two ports of a pair is removed too. The lines must cross the wall at a right angle and keep their cross-section for a while: the tool stops with an error when the cross-section changes within D/4 of the wall, unless the port wall is moved outward (`port_wall_extension`) far enough that the extended feed gives the standards those D/4 of straight line: then metal close to the layout edge, e.g. the outer turn of a coil, belongs to the device next to the reference plane and is accepted. That looks at the metal within D of the lines, the port's region; a pad frame or a slot in a ground plane farther away that ends or closes near the wall is only noted, since the standards extrude the cross-section at the wall there while the layout differs. A stacked line, e.g. TopMetal1 and TopMetal2 joined by a via array, is cut on all its layers, and the via array becomes a solid via bar in the standards.

A via port on a port wall, e.g. from a layout tool's export, can't stay a via port: its sheet would lie in the PEC wall. It is used as a gap port instead, when one end of the via port is the line at the port (one piece of metal about as wide as the port) and the other end is not (a ground plane, or no metal at all). The port then sits on the line's metal and points from ground to the line, like every wall port, and the log says so for each port, also when this reverses the via port's drawn polarity. If both ends or neither end is a line, the model is not written: draw an in-plane port on the line instead.

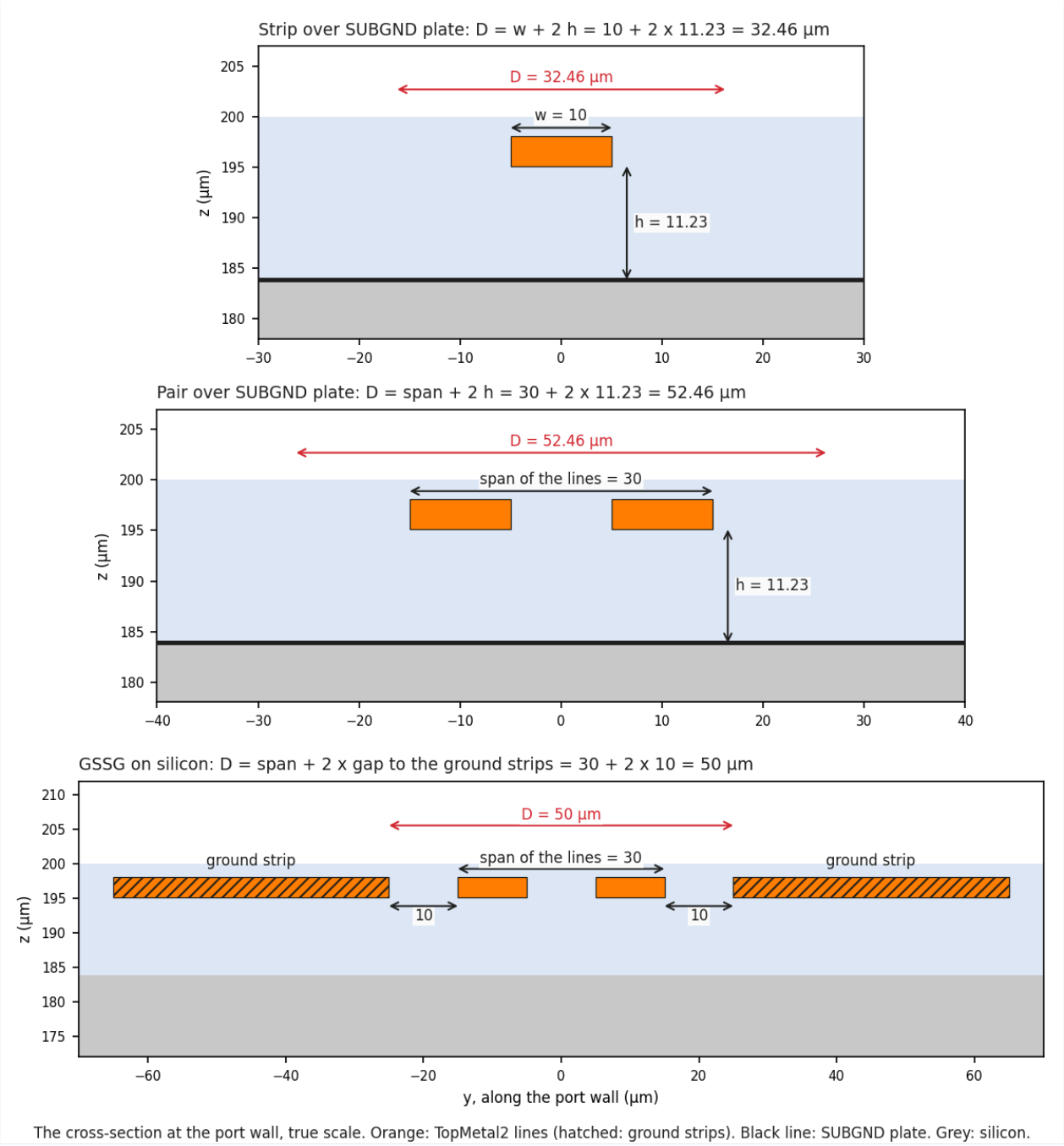
7.4.4 The standards

For each port wall, the tool writes two standards into the model's `_data` folder: the layout's cross-section at the wall, extruded to thru lines of length L and 2 L. They use the model's stackup, conductor model and mesh settings, because each mesh has its own small line error, and the calibration only removes it when model and standards share it. They are solved at three frequencies (fstop/20, fstop/5 and fstop). Both ends of a line have the same cross-section, so two such port walls share one pair of standards.

The length follows from the size of the cross-section at the cut:

D = the span of the lines + 2 × the distance to the nearest ground metal

That is w + 2h for a strip over a plate, and the inner ground-to-ground spacing for coplanar lines. The port's fringing field settles over a few D, so L = 4 D. Longer standards are worse, not better: L and 2 L are meshed separately, and their small difference in phase grows with length. So L is also kept below a fifth of the guided wavelength at the highest frequency, but not below 2 D. The log then gives the frequency up to which the calibration is fully valid. With L = 4 D, this limits D to about a twentieth of the guided wavelength, for lines with an effective index of about 2, like TopMetal2 lines on SG13G2, about 75 µm at 100 GHz. A cut where the ground is close to the lines, like a GSG pad feed, is always fine.



The cross-section at the port wall, true scale. Orange: TopMetal2 lines (hatched: ground strips). Black line: SUBGND plate. Grey: silicon.

Figure 11: How D is calculated

Several lines on one port wall. All ports on a wall are calibrated together, and the standards always carry the whole cross-section, so the coupling between the lines is removed in any case. D only sets the gap and the length of the standards, and for that only lines that are coupled count together: two neighbouring lines whose spacing is less than twice their distance to ground form one group, and D is that of the largest group. A differential pair is one group, as before. Lines far apart against their distance to ground, such as the resonator arms of a filter cut by a split region, each have their own D, so a wide cut no longer gets long standards and a low frequency limit. The log names the groups when there is more than one.

D in the cross-section at the port wall, as gds2palace calculates it. With a plate under the lines, the distance to ground is the height h above the plate: D = w + 2 h for one strip, the span of both lines + 2 h for a pair. With ground strips beside the lines, it is the gap to the ground strips, so D is the inner ground-to-ground spacing. gds2palace takes the nearest ground metal that reaches the wall, in any direction, and prints D in the log and in the setupEM Ports tab.

These rules come from a study with nine cross-sections from 14 µm to 2 mm, on chip and on a PCB: `test_data/port_deembedding_study`.

`run_sim` solves the standards before the model, and skips a standard whose result is still there for the same mesh and settings.

7.4.5 Results

`combine_snp` writes `<model>_deembedded.sNp` as before. Its first lines say what was done at each port, e.g. `! port 1: calibrated at port wall xmin, reference plane at the wall`, or `!port 3: estimated port inductance 2.1 pH` for a port inside the layout. The file `<model>_deembedding.json` next to it lists the port's elements, the line at each port wall (Z0, n_eff and loss per mm per mode), the frequency up to which the calibration is valid, and warnings.

Check the line: if its Z0 and n_eff are not what you expect for the line you drew, the calibration isn't right either. The script warns "calibration doubtful" when the port's capacitance is negative, the line's n_eff is outside the possible range, or the port loses more power than it stores.

7.4.6 Reference plane shift

`settings['reference_plane_shift'] = {'xmin': 60}` moves the reference plane 60 µm into the layout along the feed, using the line from the standards. The shift must stay within the part of the feed that has the wall's cross-section.

The shift is exact only if the feed is the standards' line, including how its ground is connected at the far end. In the standards, the ground strips reach a PEC wall at both ends, so a current can circulate through them. Ground strips that end open before the device make the feed a different line, by an amount that depends on the geometry. One example: in the de-embedding study's GSSG feed (TopMetal2 lines 10 µm wide and 10 µm apart, ground strips 40 µm wide at 10 µm), the feed's inductance per length was about 8 % higher than the standards' line.

7.4.7 Moving the port wall outward

A device close to the port wall sits in the port's fringe field, and the PEC wall mirrors its field. When the layout can't be cut further out, `settings['port_wall_extension'] = {'ymin': 'auto'}` moves the port wall outward instead: the feed is extended by the cross-section at the wall, the port moves to the new wall, and the reference plane is shifted back to the layout edge. The result is at the layout edge as before, but the device is further from the wall. `'auto'` extends until the device is at least D and half its own size away from the wall; a number extends by that length. A reference plane shift still counts from the layout edge.

The extended feed is a copy of the cross-section at the wall, so it is exactly the standards' line where the return current flows the same way, e.g. in a plane under the lines. With ground strips beside the lines that end open before the device, the extended feed is not the standards' line, and the error grows with the length of the extension; how large it is depends on the geometry (see the example under "Reference plane shift"). The setupEM preview shades the extension, which is not in the drawing.

7.4.8 Warnings while the model is written

- **"the device is inside the port's fringe field"**: keep the device at least D away from the port wall, or move the port wall outward (above).
- **"the wall mirrors its field"**: a PEC wall acts as a mirror for the device's field. Keep the device at least half its own size away from the wall, or move the port wall outward. The device's size is measured along the wall, without the ground metal that reaches the wall.
- **"standards shortened"**: D is large compared to the wavelength. Cut where the ground is closer to the lines.
- **"vias … are extruded as solid bars"**: the standards replace a via array at the cut by a solid via bar. This was not tested in the study.

7.4.9 Settings

settings['boundary']: `'PORTWALL'` on a side makes it a port wall (xmin, xmax, ymin or ymax). **settings['reference_plane_shift']:** Dictionary with a distance per port wall, e.g. `{'xmin': 60}`, in drawing units, positive into the layout. Default: no shift **settings['port_wall_extension']:** Dictionary per port wall, e.g.

`{'ymin': 'auto'}`: `'auto'` or a length in drawing units by which the port wall is moved outward, the feed extended by the cross-section at the wall. Default: none
settings['wall_extension']: Dictionary per side that is not a port wall, e.g. `{'xmax': 2000}`: a length in drawing units by which the layout's cross-section at that edge (the metal layers that reach it) is continued outward. For the pieces of a split region, where another piece has a port wall with an extension on that side. Default: none
settings['deembedding_length']: The length L of the shorter standard, in drawing units. Default: 4 D, shortened to a fifth of the guided wavelength at the highest frequency, at least 2 D **settings['deembedding']:** `'none'` writes no de-embedded file. Default: calibrated at port walls, estimated port inductance at the other ports

Limits: coupling between different ports on different walls is not removed. Port walls are for Palace; Elmer doesn't support them yet.

7.5 Filenames and flow control

If you want to have additional control over the names of files and directories that are created by the workflow, have a look at this section at the beginning of the simulation model file:

```
# get path for this simulation file
script_path = utilities.get_script_path(__file__)

# use script filename as model basename
model_basename = utilities.get_basename(__file__)

# set and create directory for simulation output
sim_path = utilities.create_sim_path (script_path,model_basename)
print('Simulation data directory: ', sim_path)

# change path to models script path
modelDir = os.path.dirname(os.path.abspath(__file__))
os.chdir(modelDir)
```

For example, if you want the customize the model name, to reflect settings used for meshing, you can define these values as variables and append that to the model_basename variable.

```
mesh  = 5  # define here, so that we can add this information to output filenames
order = 3  # define here, so that we can add this information to output filenames

XML_filename = "SG13G2_nosub.xml"          # stackup

# preprocess GDSII for safe handling of cutouts/holes?
preprocess_gds = False

# merge via polygons with distance less than .. microns, set to 0 to disable via merging.
merge_polygon_size = 0

# get path for this simulation file
script_path = utilities.get_script_path(__file__)

# use script filename as model basename, with parameters for suffix
model_basename = utilities.get_basename(__file__) + '_mesh' + str(mesh) + '_order' + str(order)

# set and create directory for simulation output
sim_path = utilities.create_sim_path (script_path,model_basename)
print('Simulation data directory: ', sim_path)

# change path to models script path
modelDir = os.path.dirname(os.path.abspath(__file__))
os.chdir(modelDir)

# ======================== simulation settings ================================
```

```
settings = {}

settings['unit']   = 1e-6  # geometry is in microns
settings['margin'] = 50    # distance in microns from GDSII geometry boundary to simulation boundary

settings['fstart']  = 0e9
settings['fstop']   = 100e9
settings['fstep']   = 2.5e9

settings['refined_cellsize'] = mesh  # mesh cell size in conductor region
settings['cells_per_wavelength'] = 10   # how many mesh cells per wavelength, must be 10 or more

settings['meshsize_max'] = 70  # microns, override cells_per_wavelength
settings['adaptive_mesh_iterations'] = 0
settings['order'] = order
```

If you want the script to create the Palace files without showing the mesh in gmsh 3D viewer, this is also possible.

You can use the optional **settings[‘no_gui’]** and either set this to *True* (always run without graphical user interface) or you can check for an optional command line parameter passed to the simulation model script.

```
settings['no_gui'] = ('nogui' in sys.argv)  # check if nogui specified on command line
```

In this case, running the model as

```
python mymodel.py
```

would run with 3D mesh viewer, or running the model as

```
python mymodel.py nogui
```

would run without the 3D viewer. Maybe this gives some idea what customization is possible.

8 Examples

This chapter gives examples on configuration settings, especially mesh control, and their results.

8.1 Single microstrip line

The first testcase for meshing is a microstrip line with TopMetal2 over Metal1 ground. Width is 15 µm to achieve 50 Ohm line impedance, length is 880µm.
This model is simulated with mesh configurations that vary 2 mesh parameters:

- **‘refined_cellsize’** defines the mesh resolution along the conductor edges
- **‘order** ‘ defines the order of basis functions, i.e. the degrees of freedom for field variation within one cell

By default, we want to simulate with an order=2, which is a good trade-off between accuracy and speed. However, in this simple example, we will check how these two values can be configured for accurate results, and what this does to simulation time.

The basic idea is this: if we have a higher order of basis functions, the fields can be modelled more accurately within each mesh cell, so we need less mesh cells ? larger value of refined_cellsize.

If we use first order basis functions, we need higher mesh density to accurately model the field distribution. Mesh on conductors at **‘refined_cellsize’** = 5:

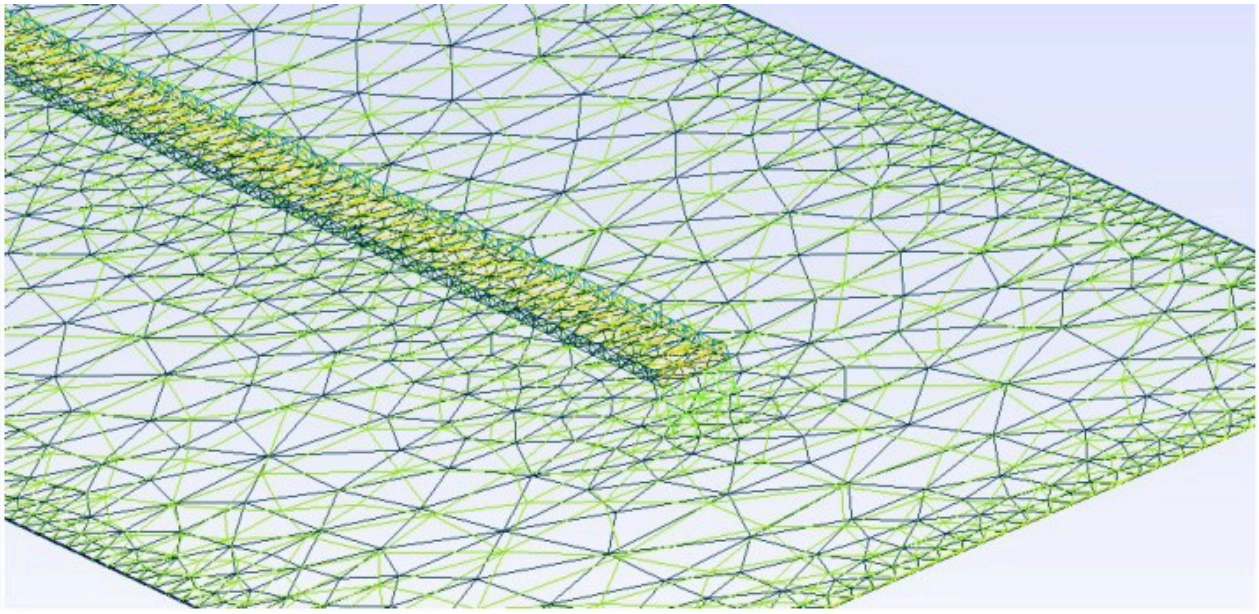


Figure 12: Mesh on the conductors with refined_cellsize = 5

Mesh on conductors at **'refined_cellsize'** = 2:

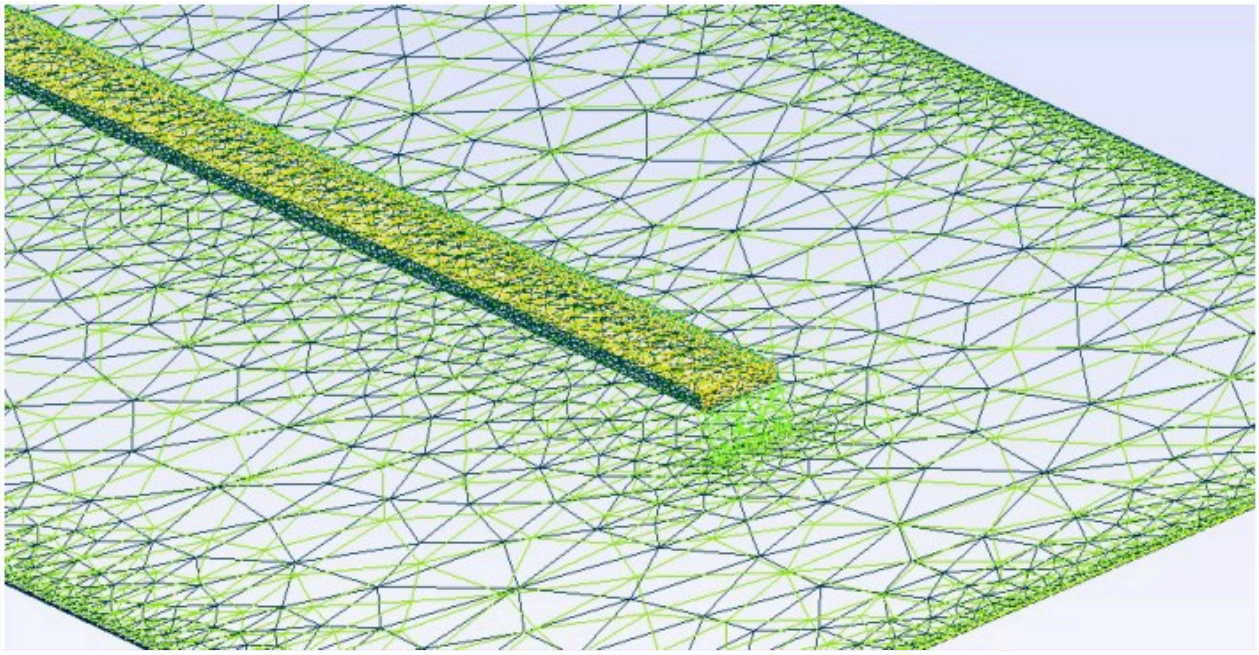


Figure 13: Mesh on the conductors with refined_cellsize = 2

These two mesh densities are now simulated with different order of basis function.

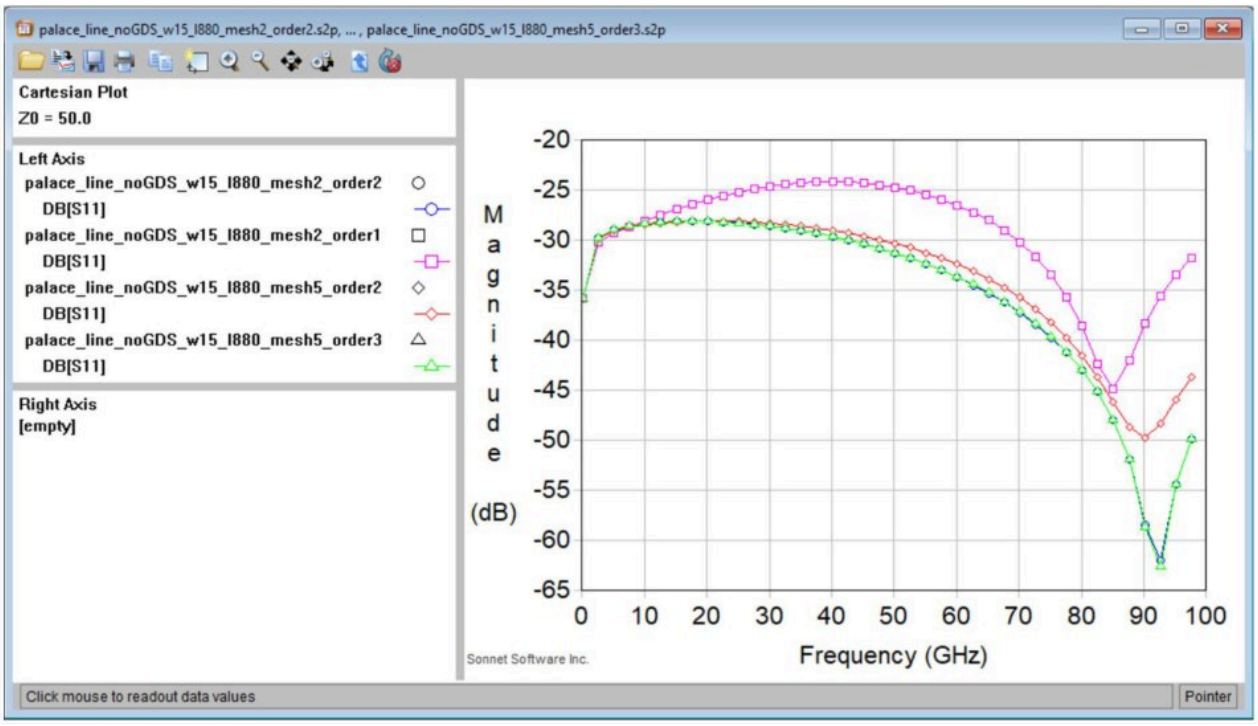


Figure 14: Microstrip line: S11 for different mesh size and basis function order

The clear outlier in this return loss plot is the pink curve, simulated with mesh order = 1 at 2µm refined cellsize. This result is different from the other curves obtained with higher mesh order. Our default simulation would be the blue curve, obtained with refined_cellsize = 2µm and order = 2.
Going to a larger value of refined_cellsize = 5µm (less mesh density at the edges) at order = 2 gives slightly different results, shown by the red curve.

If we combine the larger refined_cellsize = 5µm with higher order = 3, to give each cell more degrees of freedom, we get similar results as our baseline (default) simulation: the green and blue curve are visually identical for S11. For S21 transmission, we get a similar result: these two are almost identical.

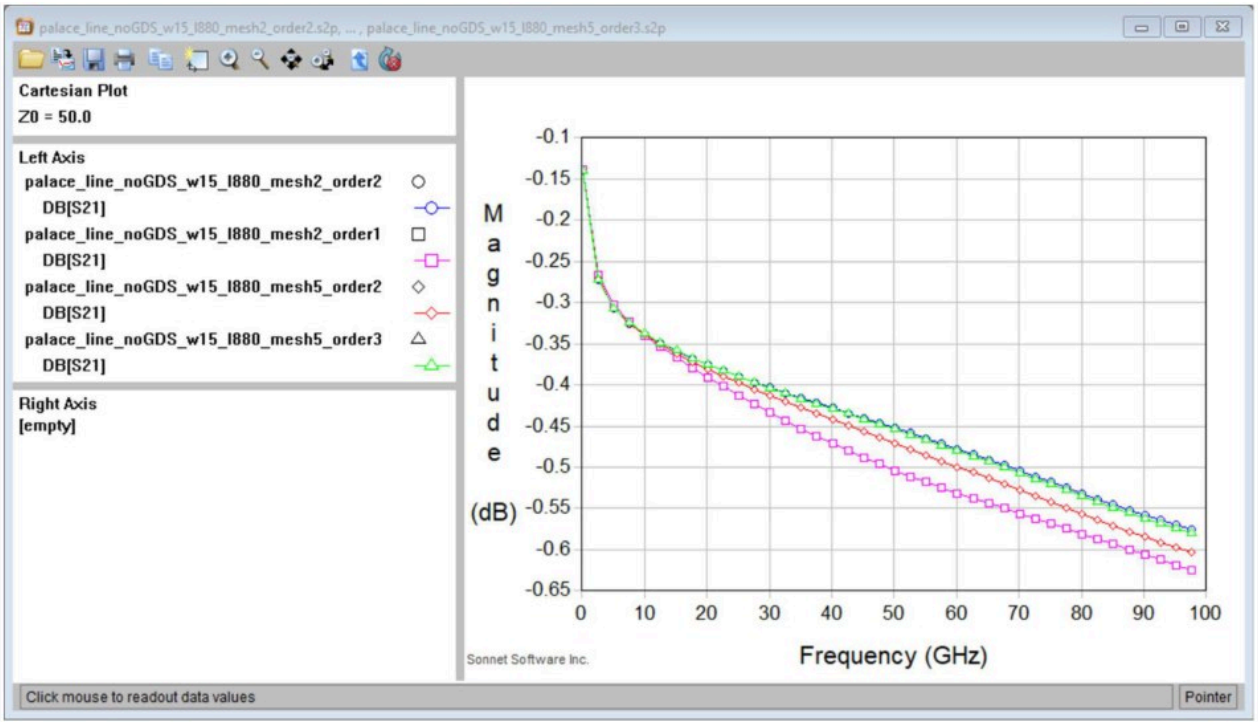


Figure 15: Microstrip line: S21 magnitude for different mesh size and basis function order

For phase of S21 we also have similar results: the order = 1 result is a clear outlier, but all other results are very close.

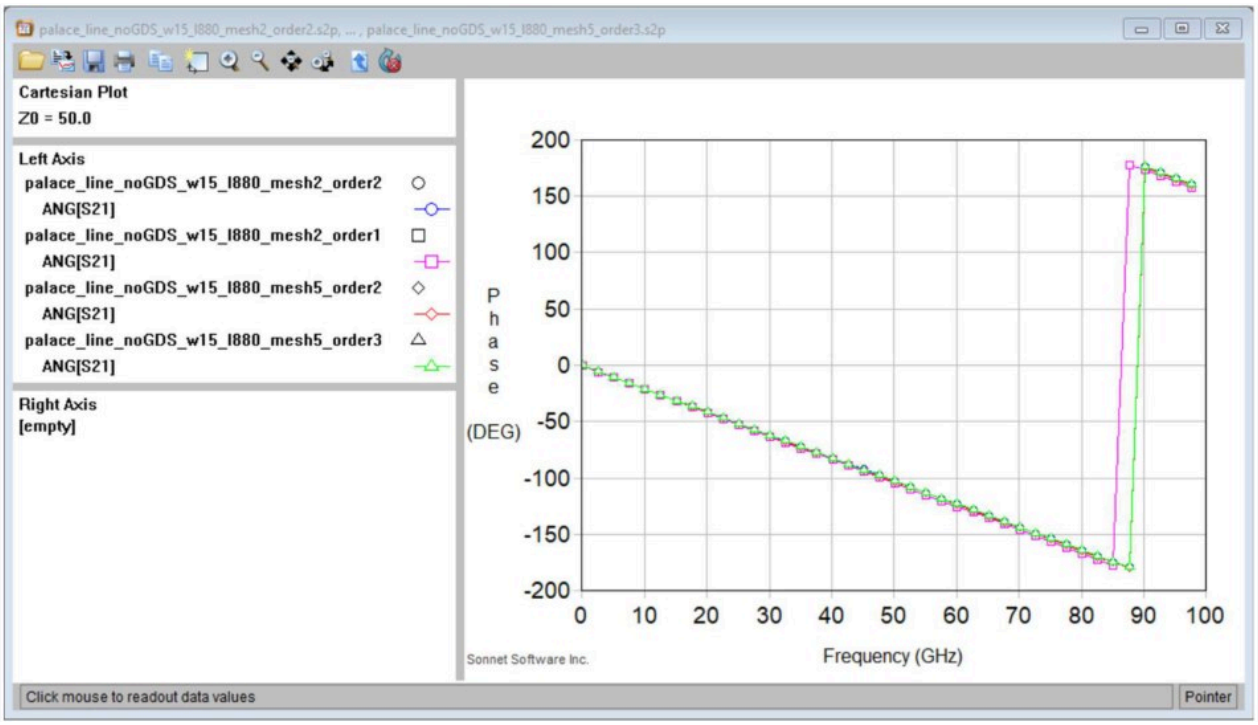


Figure 16: Microstrip line: S21 phase for different mesh size and basis function order

Conclusion: refined_cellsize=2 at order=2 is a good starting point, and for this example we don't need to go any further. Decreasing the mesh resolution to refined_cellsize=5 at order=2 is possible if we can accept some small change in results. Let's look at simulation times for these 4 different meshing approaches:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2, order=2: 733s, DOF 1640126
refined_cellsize=2, order=1: 64s, DOF 321999
refined_cellsize=5, order=2: 282s, DOF 699658
refined_cellsize=5, order=3: 1967s, DOF 1982661

All simulations were done using adaptive frequency sweep and required 10 discrete frequencies to be EM simulated. Total time in the table also includes interpolation to the specified sweep range. It can be seen that refined_cellsize=2, order=2 is the best choice for accurate data, we get results that are visually identical to refined_cellsize=5, order=3 at less simulation time (733 s total vs. 1967 s total).

In the next chapter, we will go from that baseline setting, and investigate the use of adaptive mesh refinement.

8.2 Balun 140-170 GHz

This testcase is a balun from the phase shifter project by Rupok Das, published for the OPDK July 2025 tapeout:

- https://github.com/IHP-GmbH/TO_July2025/tree/main/FMD_QNC_D_Band_Phase_Shifter



Figure 17: Balun 140-170 GHz: layout

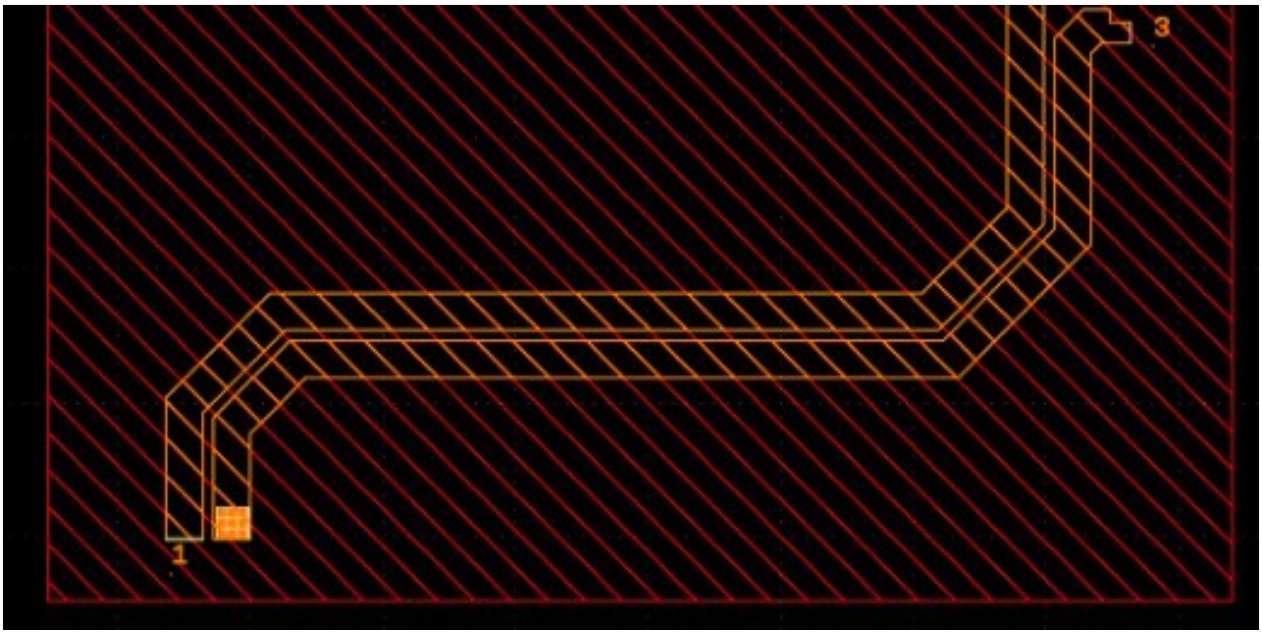


Figure 18: Balun 140-170 GHz: layout with port numbers

Line width is 7µm with TopMetal2 over Metal3 ground, gap width is 2µm.

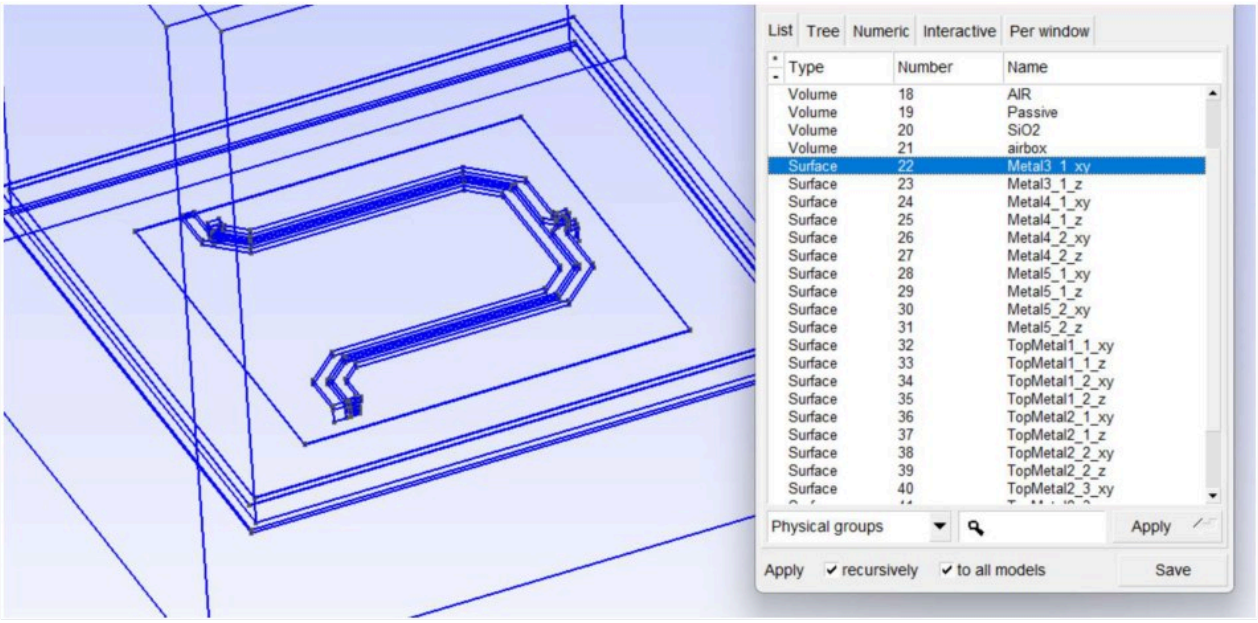


Figure 19: Balun 140-170 GHz: model geometry in gmsh

The critical dimension in this design is the 2µm wide gap between the coupler traces, the solver needs to model the field distribution in that gap with sufficient accuracy.

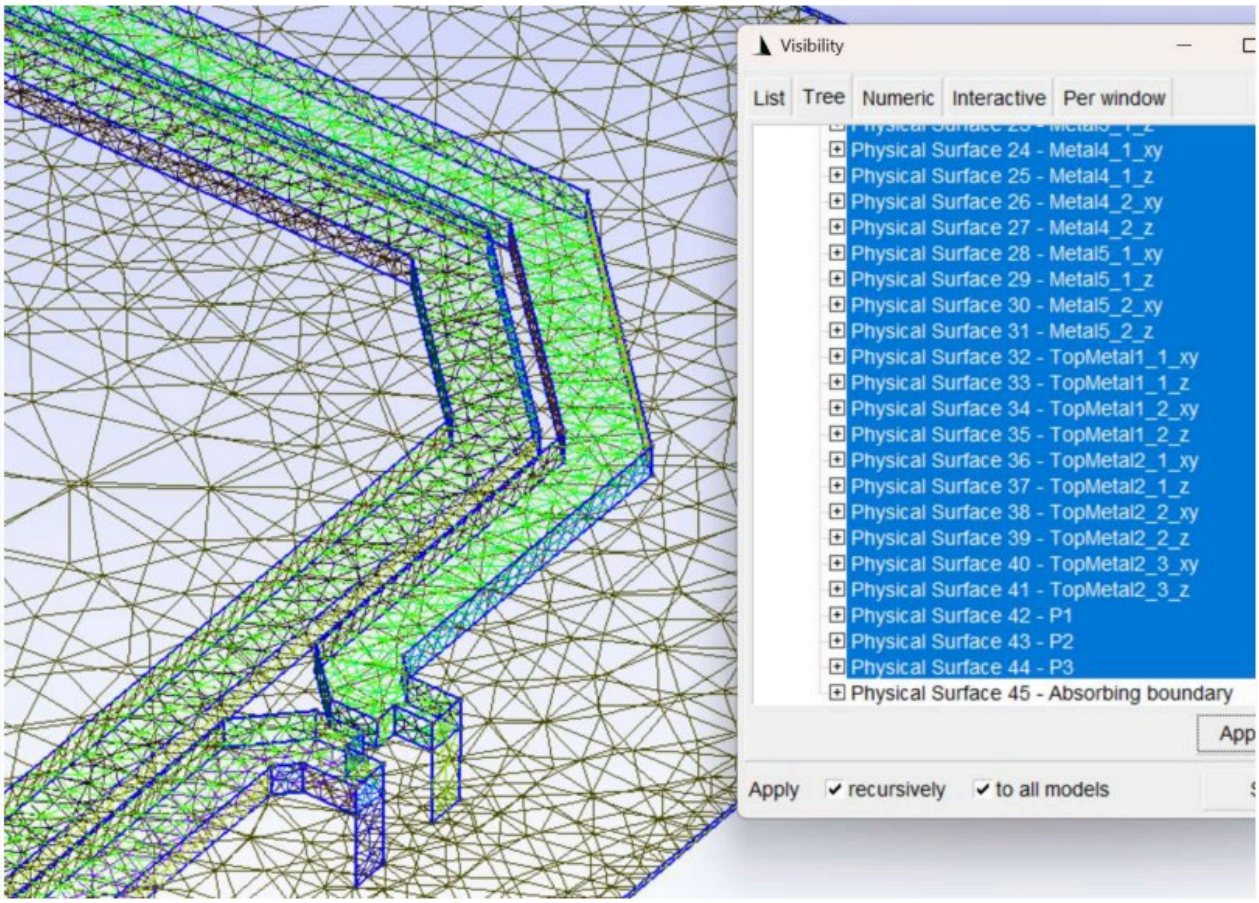


Figure 20: Balun 140-170 GHz: mesh on the conductor surfaces

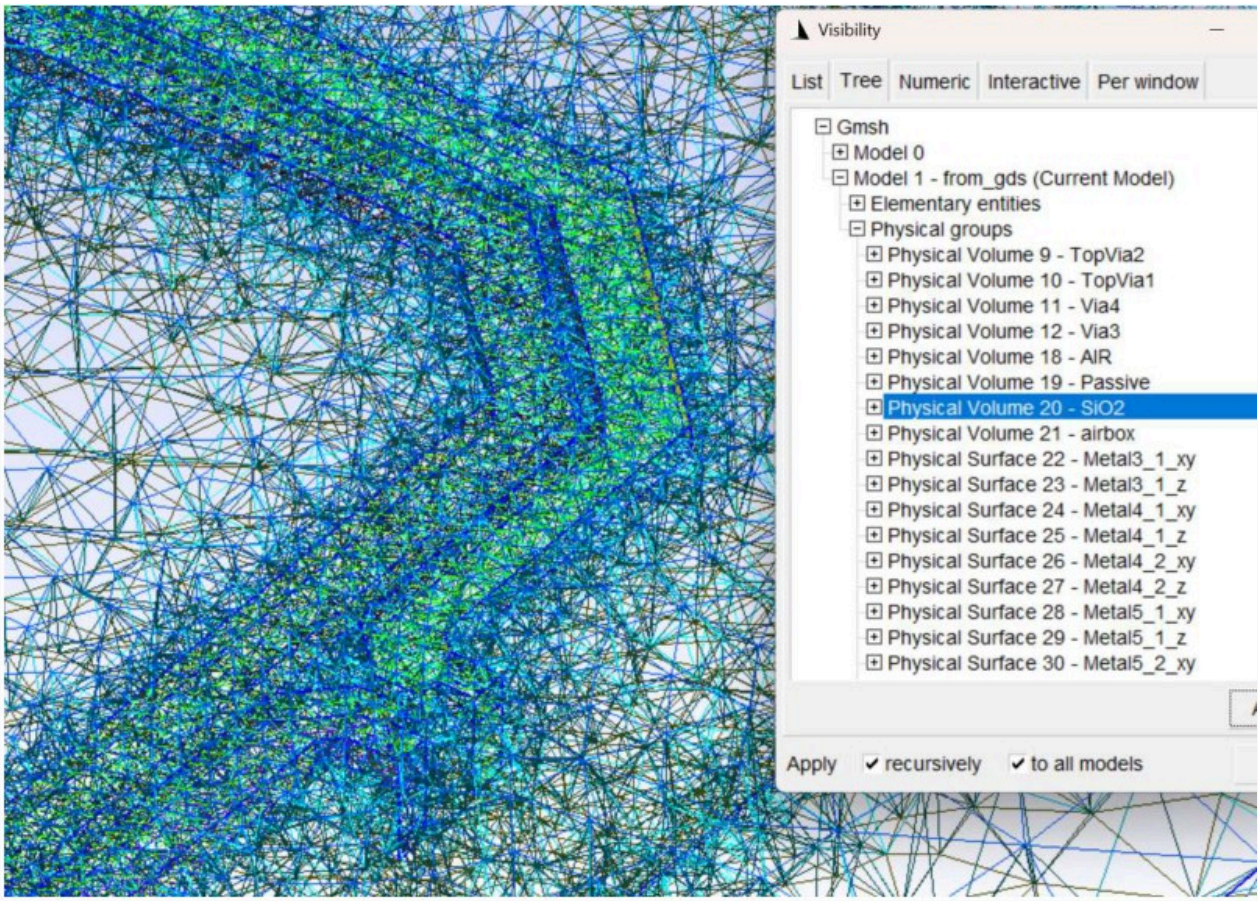


Figure 21: Balun 140-170 GHz: mesh of the SiO2 volume

We will first look at results from refined_cellsize=5 and refined_cellsize=2, both with order=2, over the frequency band 100 to 200 GHz, with all 3 port excitations simulated for full S- parameter data.

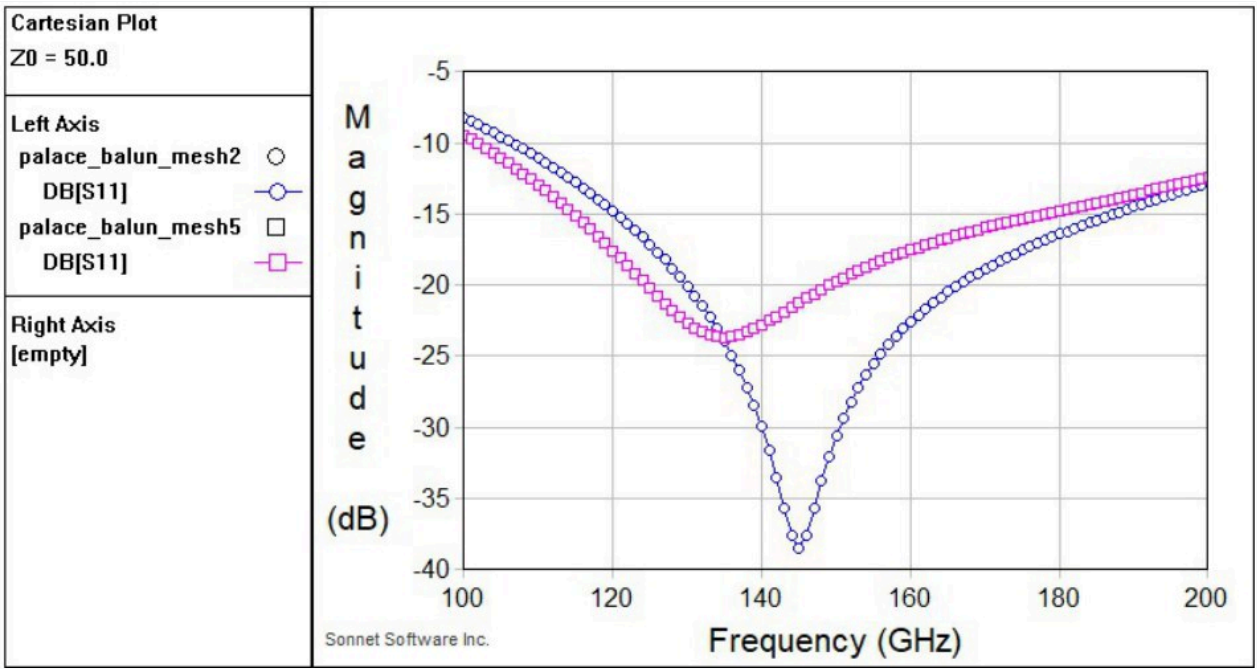


Figure 22: Balun: S11 for refined_cellsize = 5 and 2

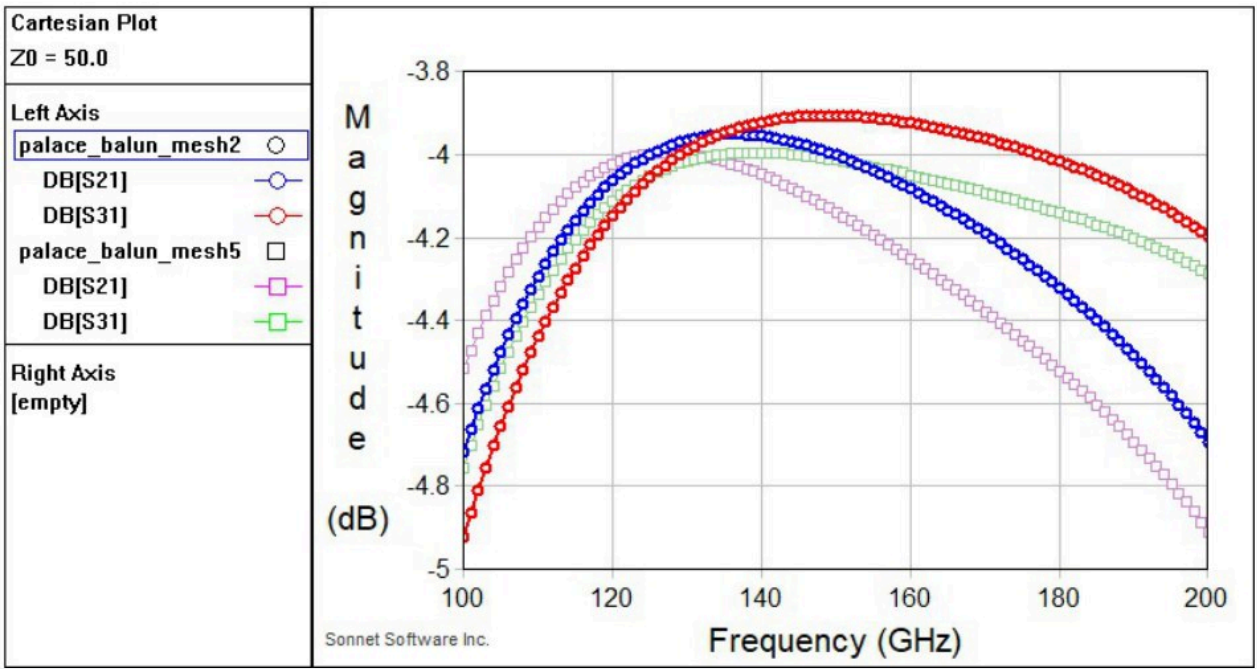


Figure 23: Balun: S21 and S31 for refined_cellsize = 5 and 2

There is quite a difference in S11 and S21,S31 results, and we can assume that the finer mesh gives more accurate results … but how far are we off from convergence? We could now simulate with an ever smaller refined_meshsize = 1, or we can use adaptive mesh refinement now. This will run the simulation starting with an initial mesh size, and then refine the mesh in the most relevant regions.

By default, adaptive mesh refinement in Palace uses data from all simulation frequencies and all port excitations, which means a significant increase in overall simulation time because a full simulation is used during mesh refinement. The excerpt below shows settings from the Palace configuration file for an adaptive sweep with maximum of 2 adaptive mesh refinements, with up to 2 million degrees of freedom.

```
"Model": {
    "Mesh": "palace_balun_mesh5_AMR.msh",
    "L0": 1e-06,
    "Refinement": {
        "UniformLevels": 0,
        "Tol": 0.01,
        "MaxIts": 2,
        "MaxSize": 2000000.0,
        "Nonconformal": true,
        "UpdateFraction": 0.7,
        "SaveAdaptMesh": false
    }
},
"Solver": {
    "Driven": {
        "MinFreq": 100.0,
        "MaxFreq": 200.0,
        "FreqStep": 1.0,
        "SaveStep": 0,
        "AdaptiveTol": 0.01
    },
```

Note for experts: We could decide to create an adaptive mesh refinement at one target frequency, or a few selected target frequencies, and then store that refined mesh for the final full frequency sweep. This is the approach chosen by many commercial FEM solvers, where the user needs to decide what frequencies are used for AMR.

However, that is not implemented in this gds2gmsh workflow. The present implementation of adaptive mesh refinement in gds2gmsh workflow does each refinement step at all frequencies and for all port excitations, to provide a safe and reliable mesh refinement. In this case, the refined mesh exists in memory only and is not stored to disk.

Below is the result from an adaptive mesh refinement with 2 refinement steps, based on an initial mesh with refined_cellsize=5. This result is very close to the results obtained with a (fixed) mesh created from refined_cellsize=2, so both models come to similar results from different (initial) mesh.

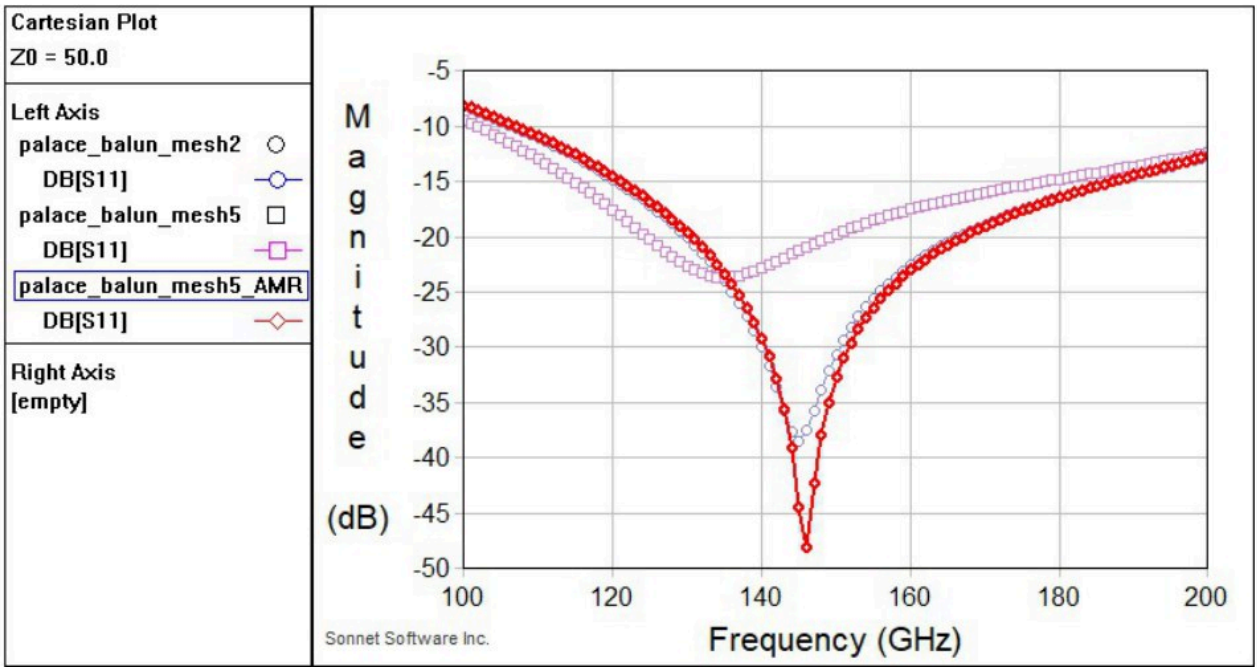


Figure 24: Balun: S11 with adaptive mesh refinement, compared to the fixed meshes

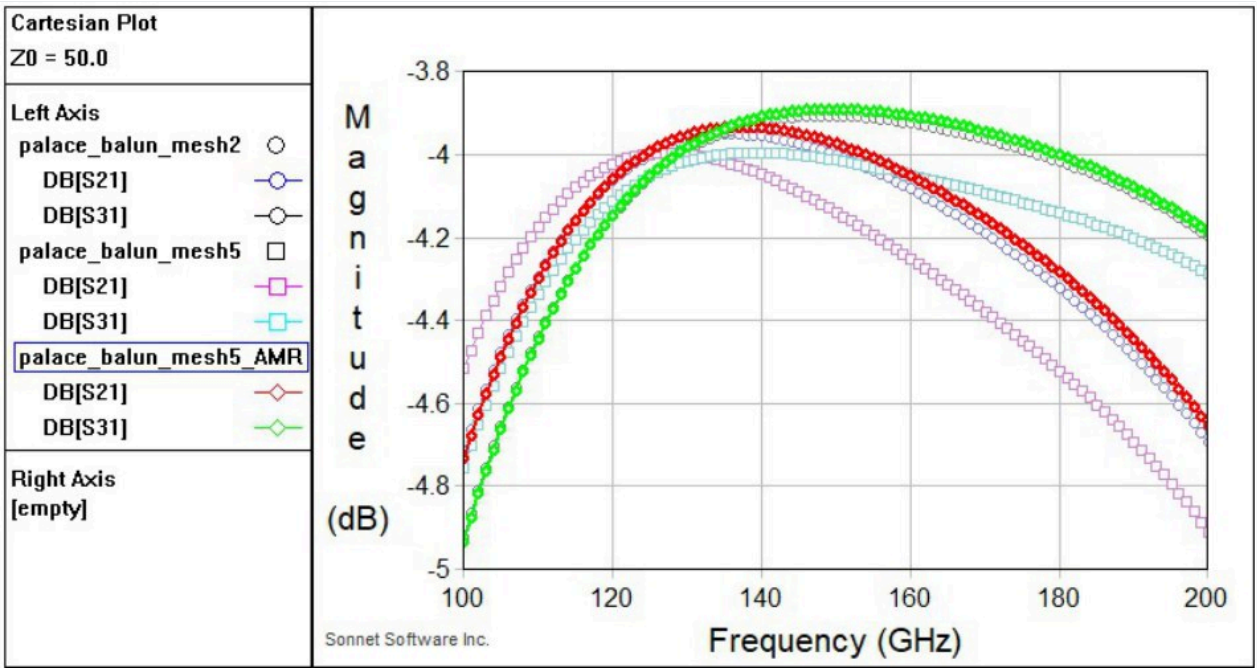


Figure 25: Balun: S21 and S31 with adaptive mesh refinement, compared to the fixed meshes

Here is the overview of simulation times and mesh size for full sweep over all 3 port excitations:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2: 580s (13 freq), DOF 836878

refined_cellsize=5: 218s (13 freq), DOF 353056
refined_cellsize=5 + AMR 2x: 1132s (13 freq), DOF 353056, 402138,710236

We could now spend more time to double check result with an adaptive mesh refinement starting from an initial mesh that is even finer, and to save time, that could be done at one or few frequencies only. Below is the result for 3 selected frequencies with 2 mesh refinements starting from 2µm:

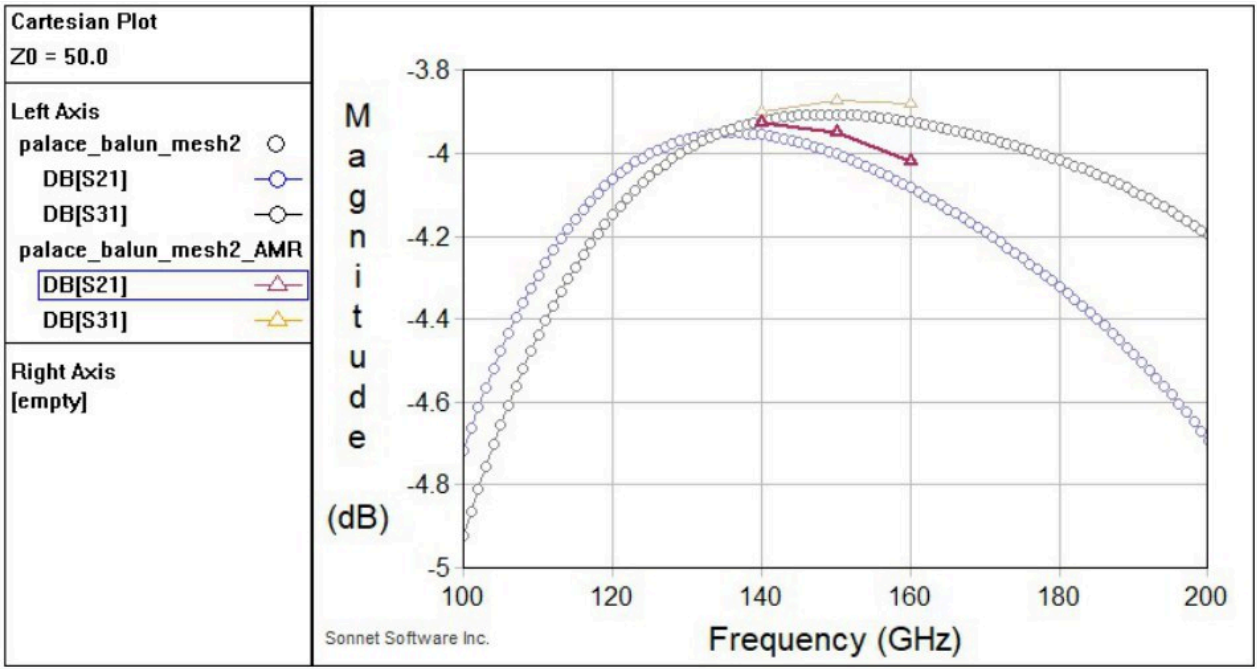


Figure 26: Balun: S21 and S31 with adaptive mesh refinement from the finer initial mesh, at selected frequencies

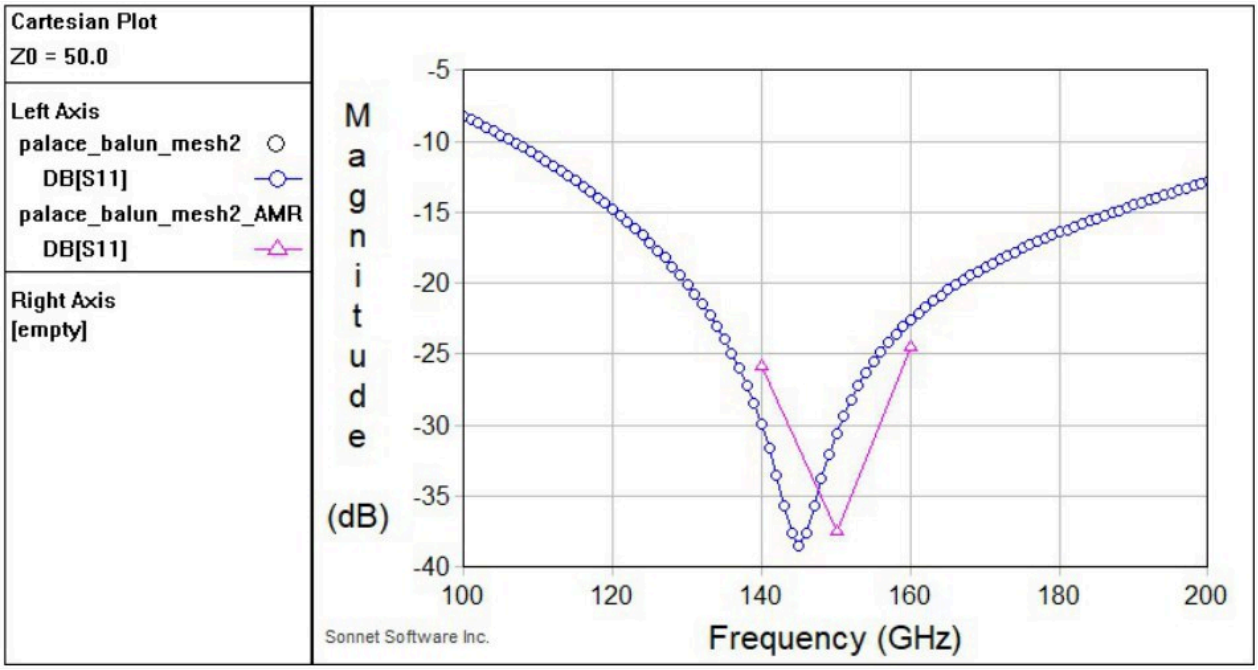


Figure 27: Balun: S11 with adaptive mesh refinement from the finer initial mesh, at selected frequencies

This gives an idea where a converged result would be located in frequency, at approximately 150 GHz resonance instead of 145 GHz calculated with coarser mesh. Here is the overview of simulation times and mesh size for 3 frequencies (not full sweep!) over all 3 port excitations:

Simulation time total, Degrees of freedom (from log)
refined_cellsize=2 + AMR 2x: 1346s (3 freq), DOF 804166, 961866, 1742956

9 Conductor loss modelling

gds2palace has three conductor models, chosen with `settings['conductor_model']`:

- `'surface'`, the default: the conductors are hollow bodies with a surface impedance on all faces. Like many RF FEM solvers, this model doesn't mesh the inside of the conductors, which at high frequency would need sub-micron cells near the surface for the skin effect. It is the fastest model.
- `'solve_inside'`: the conductors are meshed as volumes with their bulk conductivity.
- `'solve_inside_skin'`: volumes, plus thin layers along all faces for the skin effect.

The surface model has two errors in the conductor loss: one at low frequency, described first, and one at high frequency, described in "Accuracy at high frequency". The volume models avoid both. "Choosing the conductor model" below sums up the errors and run times of all three models.

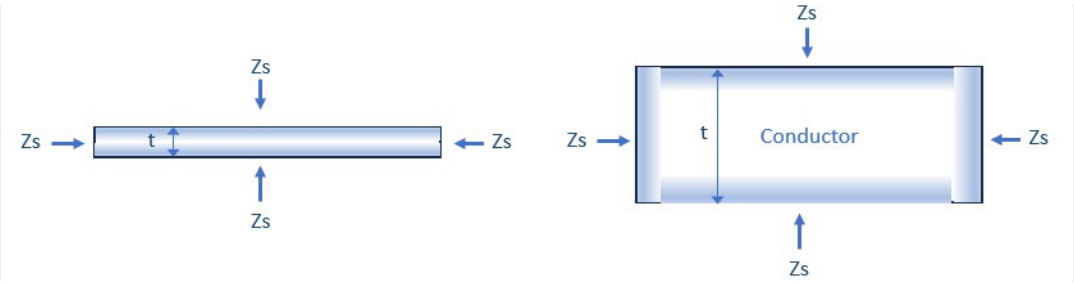


Figure 28: Metals modelled as hollow bodies with surface impedance Zs on all sides

The surface impedance can be calculated easily when physical metal dimensions are much larger than skin depth. However, at low frequency and in the transition between skin effect regime and low frequency, we also need to consider conductor dimensions.

9.1 Limits of conductor loss calculation

With the surface model, gds2palace creates the faces of each conductor and defines them as a conductor sheet with conductivity and thickness in the Palace config.json file. Thickness in this case is the value defined for that metal in the XML stackup file, e.g. 3µm thickness for TopMetal2.

The code in Palace does not know anything about the aspect ratio of the conductor, and uses a surface impedance calculation that is accurate for wide sheets, where we have a conductor that is much wider than the conductor height. Summing up the surface impedance mapped to top and bottom side of a thin sheet, the resulting impedance would be accurate down to DC where it is limited by physical conductor cross section, as long as both faces carry the same current. A thin layer with current on one face only, such as a ground plane under a line, gets up to twice its loss, see "Accuracy at high frequency".

However, in our RFIC use case, we might have conductor aspect ratio that reaches almost square shape, and we need to account for side wall currents, not only for closely spaced coupled lines.

Mapping the surface impedance (as calculated internally by Palace from conductor thickness) onto all surfaces, i.e. side walls as well as top and bottom, will lead to an overestimate of available conductor cross section at low frequency:

The Palace calculation[2] of Zs is exact when applied to top and bottom side only, with the same current on both. If we apply this value to all sides, the extreme case at DC will over-estimate the conductor cross section by factor (width+thickness)/width.

[2] https://github.com/awslabs/palace/blob/main/palace/models/surfaceconductivityoperator.cpp

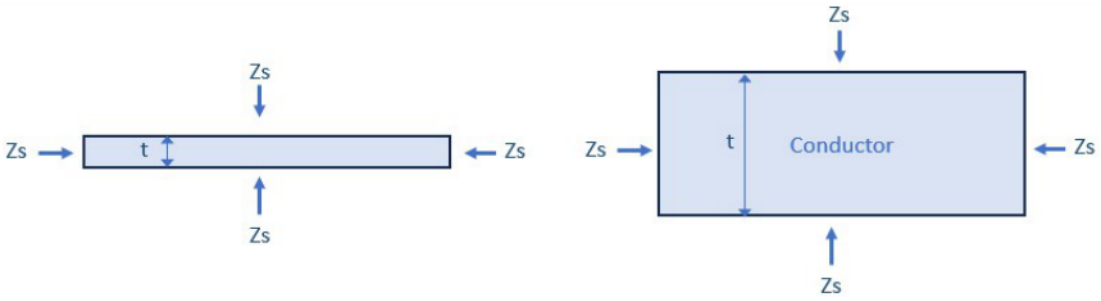


Figure 29: Surface impedance for a thin sheet (left) and a conductor with near-square cross section (right)

The volume models of `settings['conductor_model']('solve_inside','solve_inside_skin')` do not have this error, see “Choosing the conductor model” below.

9.2 Accuracy at high frequency

At high frequency, the current flows in a thin layer at the surface of a conductor. This is the skin effect. In TopMetal2, the skin depth is about 0.9 µm at 10 GHz and 0.3 µm at 100 GHz, much less than the 3 µm metal thickness. The three conductor models of `settings['conductor_model']` handle the skin effect in different ways, and each has its own error. This section explains these errors, so that you can choose the right model for your design.

9.2.1 Surface impedance (`'surface'`, the default)

The surface model does not mesh the inside of a conductor. Instead, each face gets a surface impedance that describes the skin effect. This impedance is exact for a wide, flat sheet. Real conductors are not wide sheets, and this causes two errors.

Too little loss in thick conductors. At the edges and corners of a trace, the current crowds into the corners. A surface impedance per face cannot describe this. So the loss of TopMetal1 and TopMetal2 traces is too low, not only at low frequency (see “Limits of conductor loss calculation”), but also at mm-wave frequencies. For TopMetal2 lines, the resistance is still about 10-20 % too low at 50-150 GHz. For coils with 2-8 µm traces on the top metals, the peak Q is about 20-30 % too high, and below the peak even more. Coils with wider traces were not tested. Their error is probably smaller, because the side walls and edges carry a smaller part of the current.

Too much loss in thin layers with current on one face. A ground plane under a line carries its current only on the face toward the line. The same is true for the plates of a MIM capacitor. The surface model assumes that both faces of a conductor share the current. The face that carries all the current then gets twice the real sheet resistance. This matters for layers thinner than about two skin depths: in SG13G2, Metal1-Metal5 and the MIM plates up to 150 GHz. The loss of such a layer can double. For the line example `palace_line_viaport.py` below, the line resistance is 20-40 % too high, and the 880 µm line shows up to 0.08 dB too much insertion loss. The subsection “Palace’s External option” explains the details.

The two errors have opposite signs. On a given structure, they can add up or cancel. So a surface model result does not tell you whether the loss is too high or too low.

9.2.2 Conductors meshed as volumes (`'solve_inside'`)

With `'solve_inside'`, gds2palace meshes the conductors as volumes with their bulk conductivity. The solver then computes the current inside the metal, including the current crowding at edges and corners. At low frequency, this model is exact.

At high frequency, the mesh must resolve the skin effect. What matters is the cell size normal to the conductor surface. With a 2 µm mesh, the 3 µm thick TopMetal2 gets about two cells through its thickness. The second order elements follow the current decay into the metal quite well, but they leave some current too deep in the metal. The loss is then a few percent too high, up to about 10 % at 100 GHz. A finer mesh reduces this error. But it also refines the mesh along the whole conductor surface, which makes the model much larger.

9.2.3 Volumes with skin layers (`'solve_inside_skin'`)

`'solve_inside_skin'` adds thin layers along all faces of the conductors. The mesh cells in these layers are thin normal to the surface, where the current changes fast. Along the surface, they keep the normal cell size. This is the same idea as a skin depth based mesh operation in HFSS.

The depths of the skin layers follow from the skin depth at the highest simulation frequency. You don't need to set them. Thin layers like Metal1-Metal5 get no skin layers: they would gain little, but their very thin skin layers would cost much solver time.

For a single-turn TopMetal2 inductor up to 100 GHz, skin layers with a 2 µm mesh came within about 1 % of a fully converged model, in a ninth of its run time. Without skin layers, the 2 µm mesh gave a Q about 8 % too low. The surface model gave a Q about 22 % too high.

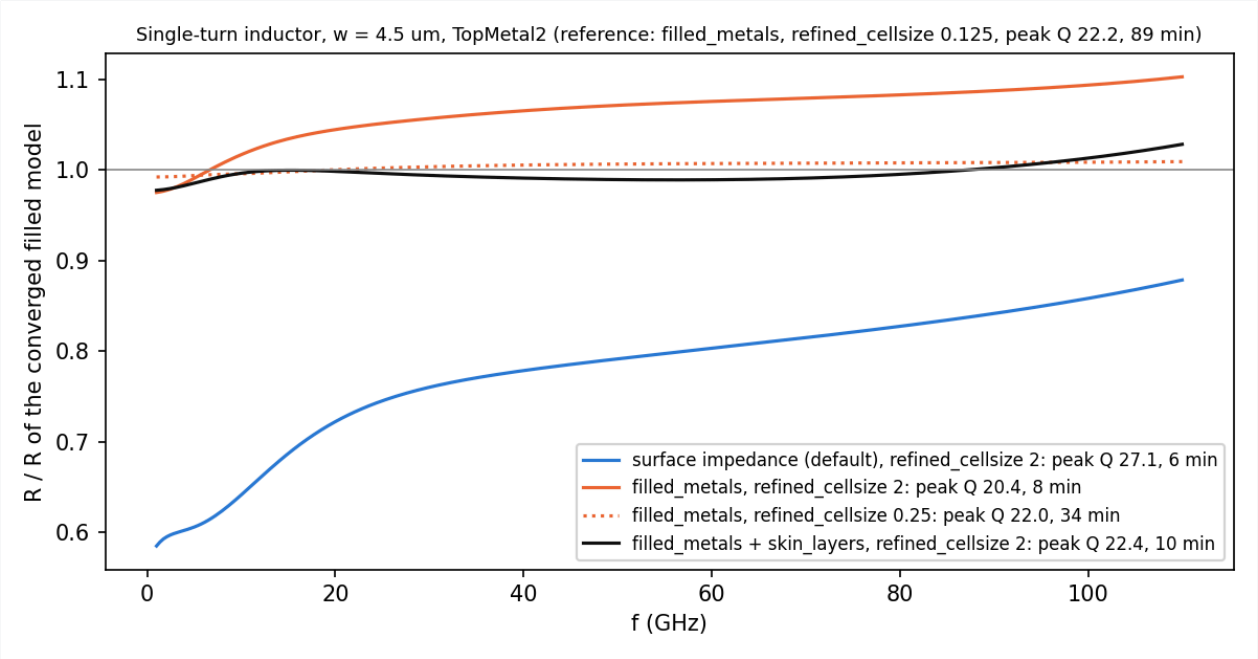


Figure 30: Single-turn inductor: differential resistance of the conductor models, relative to 'solve_inside' at a 0.125 µm mesh

The line example `palace_line_viaport.py` is a 15 µm wide TopMetal2 line over a Metal1 ground plane. Its line resistance, from two line lengths, is compared here with a 2D calculation. With skin layers, the model is within a few percent of the 2D result up to 100 GHz. The surface model is 20-40 % too high, because of the thin Metal1 ground plane.

In S21, the 880 µm example line has an insertion loss of 0.1-0.5 dB between 1 and 100 GHz. Compared with the most accurate run, the surface model adds up to 0.08 dB to that, and `'solve_inside'` up to 0.03 dB. With skin layers, the difference is below 0.01 dB.

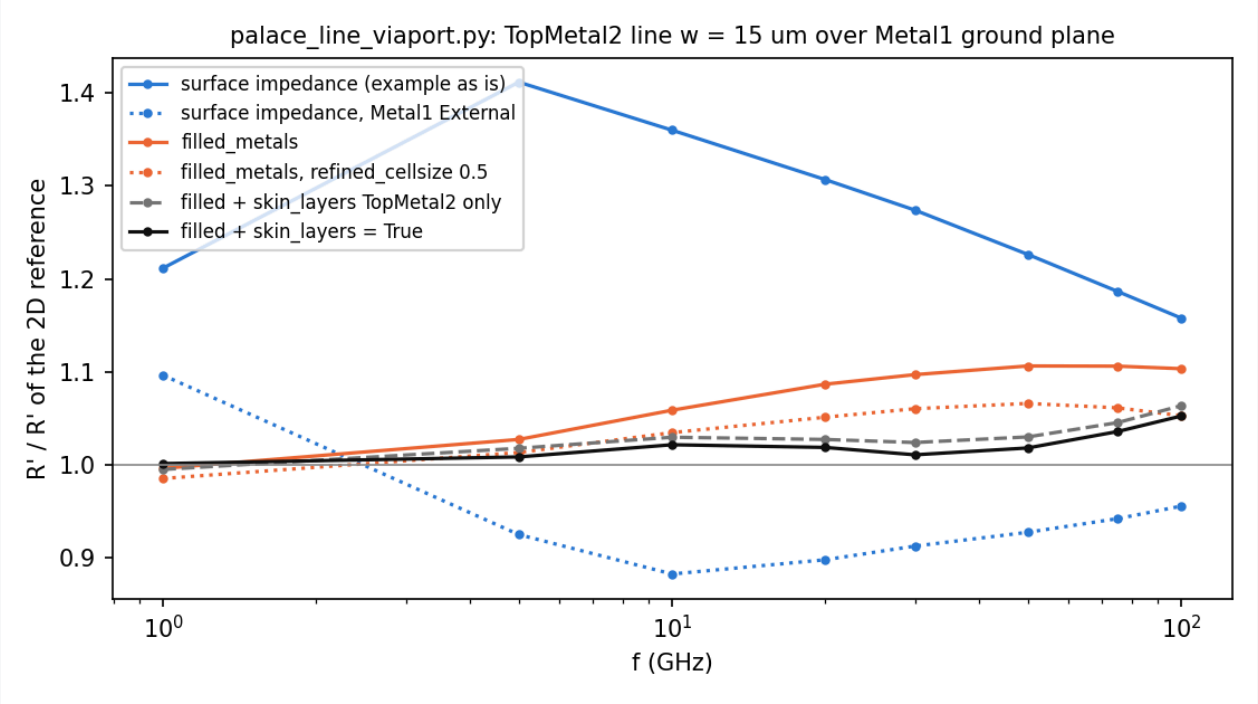


Figure 31: palace_line_viaport.py: line resistance of the conductor models, relative to a 2D calculation

9.2.4 Choosing the conductor model

For most designs, the choice is simple:

purpose	`settings['conductor_model']`
resonance frequencies, inductance, coupling, match and balance	`'surface'`, the fastest
Q and insertion loss during design	`'solve_inside'`, a few percent pessimistic at a 2 µm mesh
Q and insertion loss, final numbers	`'solve_inside_skin'`

The table below sums up the three models. The errors are for a 2 µm mesh, compared with `'solve_inside_skin'` at 1 µm. They come from step 4 of the mesh convergence studies in `more_examples/mesh_convergence`, with five SG13G2 structures. The run times are for the coils and baluns, from the conductor model study in `more_examples/conductor_models`, which ran all models on one machine, one run at a time. On straight lines, the volume models cost less: 1.0-1.1x and 1.4-2.0x.

	`'surface'` (default)	`'solve_inside'`	`'solve_inside_skin'`
Model	surface impedance on all conductor faces	conductors meshed as volumes	volumes with thin layers along the faces for the skin effect
In HFSS	impedance boundary (Solve Inside off)	Solve Inside	Solve Inside with a skin depth based mesh operation
Before gds2palace 0.9.0	`filled_metals = False`	`filled_metals = True`	not available
Peak Q of coils on TopMetal1/2	18-30 % too high	4-8 % too low	2-3 % too low
Insertion loss (baluns)	0.02-0.14 dB too low	0.03-0.15 dB too high	up to 0.05 dB too high
Unknowns at the same mesh	1x	about 1.2x	1.75-2.5x
Run time at the same mesh	1x	1.25-1.7x	1.5-3.5x

The Q errors are for the peak Q of coils with 2-8 µm traces. Below the peak, the surface model is even more optimistic: 24-37 % at half the peak frequency. Coils with wider traces were not tested. Their error is probably smaller, because the side walls and edges carry a smaller part of the current.

What the errors mean for typical designs, with the same mesh and reference. For the line, the line resistance R’ is compared with the 2D calculation, and the attenuation in dB/mm with the most accurate run (skin layers also in the Metal1 ground plane).

Design	Figure of merit	`surface`	`solve_inside`	`solve_inside_skin`
LC VCO tank, coil on TopMetal1/ TopMetal2 with 2-8 µm traces	coil Q at its peak; phase noise ~ $1/Q^2$ where the coil sets the tank Q	Q +18…+30 %: phase noise 1.4-2.3 dB too optimistic	Q −4…-8 %: phase noise 0.35-0.7 dB pessimistic	Q −2…-3 %
LNA input match, baluns, matching networks	insertion loss, adds 1:1 to the noise figure of an LNA	0.02-0.14 dB too optimistic	0.03-0.15 dB pessimistic	up to 0.05 dB pessimistic
MIM capacitor in a tank or matching network	series resistance, capacitor Q	series R about 2x (`palace_rfcmim.py`, 10 MHz: 1.88 instead of 0.87 Ω), capacitor Q about half	right	right
TopMetal2 line over a Metal1 ground plane (`palace_line_viaport.py`)	attenuation (dB/ mm), line resistance R’	up to 0.09 dB/mm too lossy (+9…+39 %); R’ +21…+41 %	up to 0.03 dB/mm too lossy (+8 %); R’ up to +11 %	within 0.01 dB/mm (+1 %); R’ −0.5… +6.4 %
any structure	inductance, coupling, SRF, match, amplitude and phase balance	all three models within about 1 % of each other		

Some notes on these numbers:

- **Loss errors in dB hide the relative error.** 0.05 dB on a balun with 1.2 dB insertion loss is 4 % of its loss. At a PA output, 0.1 dB is 2.3 % of the output power.
- **The reference is a converged simulation, not a measurement.** Compared with a measurement, the stackup matters as much as the conductor model. In the L6n2 inductor study (`more_examples/`

`measured_vs_simulated/more_accurate_models_L6n2_v3`), a volume model with the planar stackup had a Q 8 % below the measured one. With the conformal passivation over TopMetal2, it came within 3-5 % in Q and within 2 % in SRF.

- **Sweeping far beyond your band is fine.** The skin layers are sized for the highest frequency. On a 30 GHz transformer, skin layers sized for 200 GHz changed Q by only 1 %.
- **Skin layers don't replace a fine enough mesh.** The cell size must not exceed the smallest gap or trace width, for every conductor model. With traces 2 µm apart, `'solve_inside_skin'` at a 4-5 µm mesh was worse than `'solve_inside'` at 2 µm.

9.2.5 Palace's External option

This subsection explains why the surface model doubles the loss of thin layers with current on one face.

Palace's `Conductivity` boundary uses the surface impedance $Z_s = (\gamma/\sigma)\cdot\coth(\gamma h/2)$ for a conductor of thickness h. At DC, this is $2/(\sigma h)$, twice the sheet resistance of the layer. This is meant for a conductor with two faces in the mesh. A thin sheet inside the model is an example: Palace splits it into two faces, each face gets this Z_s, and the two faces in parallel give the right sheet resistance $1/(\sigma h)$.

The option `"External": true` doubles h. Then Z_s tends to $1/(\sigma h)$, the full sheet resistance on one face. This is meant for a conductor with only one face in the mesh, such as a metal wall that forms the outer boundary of the model.

gds2palace models each conductor as a hollow body with all its faces in the mesh, and does not set `External`, except for a conductor at the simulation boundary (see below). This is exact when both faces carry the same current, for example a Metal1 line at low frequency. It makes no difference for conductors much thicker than the skin depth: with or without `External`, Z_s is then nearly the same. For a ground plane with current on one face, it doubles the loss.

Setting `External` by hand on a ground plane would fix that case. But it would halve the loss of conductors that carry current on both faces, and gds2palace puts all faces of a conductor in one group. So no setting of the surface model is right for all conductors. Meshing the thin layers as volumes is the better fix: a thin layer is only one or two cells thick, so this costs little, and the current in it is then right. The volume models `'solve_inside'` and `'solve_inside_skin'` do this.

9.2.6 Conductors at the simulation boundary

With `margin` and `air_around` both 0 on a side, a conductor can reach the simulation boundary. In the `'surface'` model the conductor is hollow, so its face on the boundary has no mesh on either side. gds2palace leaves this face out of the model. What this means for the loss depends on which face it is.

A side wall or end face (x and y sides). The top and bottom faces remain and share the current as before, so Z_s stays the same. For a line 5 µm wide and 0.49 µm thick along the boundary, R at low frequency is 4 % below `'solve_inside'`. A line away from the boundary is 8 % below it, because it has one more side wall.

A top or bottom face (z sides). One main face is left, and it carries all the current. gds2palace sets `External` on this conductor. For a thin ground plane as the outer wall of the model, the loss is then about 15 % too high, compared with about 2 times too high without `External`.

A sheet resistor. Rs is the sheet resistance itself, for one face or two, so `External` plays no role. With a `PMC` boundary, the resistance is correct when an edge of the sheet is on the boundary, and when the sheet lies in the top or bottom boundary plane. With a `PEC` boundary, a sheet that ends on the boundary is a correct resistor to that wall. An absorbing boundary is different, see below.

An absorbing boundary right at a conductor changes the result, in every conductor model. At low frequency it acts as a resistive sheet between the conductor and its surroundings. Keep some air between a conductor and an absorbing boundary, unless you have checked that this doesn't matter for your structure.

10 Dielectric loss modelling

10.1 Silicon substrate

The silicon below the oxide is not an insulator. In the XML stackup files it is a material with `Type="Semiconductor"` and a conductivity: in the latest SG13G2 files 2 S/m for the substrate and 5 S/m for the EPI layer on top of it, both with a permittivity of 11.9. gds2palace passes this conductivity to Palace, see “Mapping of Volumes and Surfaces to Palace materials” in the Appendix.

The silicon adds loss wherever the fields of a structure reach into it. The electric field drives a current through the conducting silicon, and the changing magnetic field of a coil induces eddy currents in it. A ground plane between the structure and the silicon, e.g. on Metal1 or the SUBGND sheet, keeps the fields out of the silicon. The line below has such a ground plane, so its models leave the substrate out.

10.2 The line measurement

The 15 µm wide TopMetal2 line over a Metal1 ground plane (`palace_line_viaport.py`, `palace_line_noGDS.py`) was measured by IHP in 2014 with two lengths, 880 µm and 2580 µm, up to 110 GHz. The GSG pads were de-embedded from the measurement, which leaves some ripple in the data.

The simulations use `conductor_model = 'solve_inside_skin'`, the most accurate conductor model (see “Choosing the conductor model”), with `refined_cellsize` 2 and `order` 2, and the stackup `SG13G2_nosub.xml` of the examples. The models have ideal via ports at the line ends, where the measurement has pads.

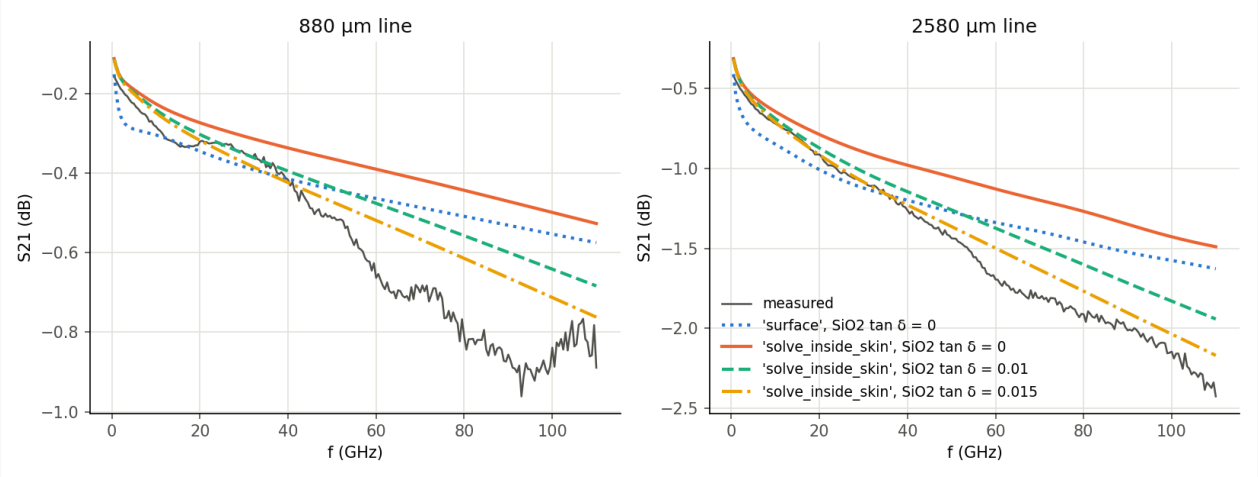


Figure 32: S21 of the 880 µm and 2580 µm lines, measured and simulated

Pads and ports affect both lengths in the same way. The difference between the two lengths is 1700 µm of line, and its loss is the loss of the line itself. It follows from the transfer matrices of both lengths, as in a multiline calibration, for the measurement and for the simulation alike. The figure below shows this attenuation per mm. It compares the line without pads and ports, so it is the better comparison.

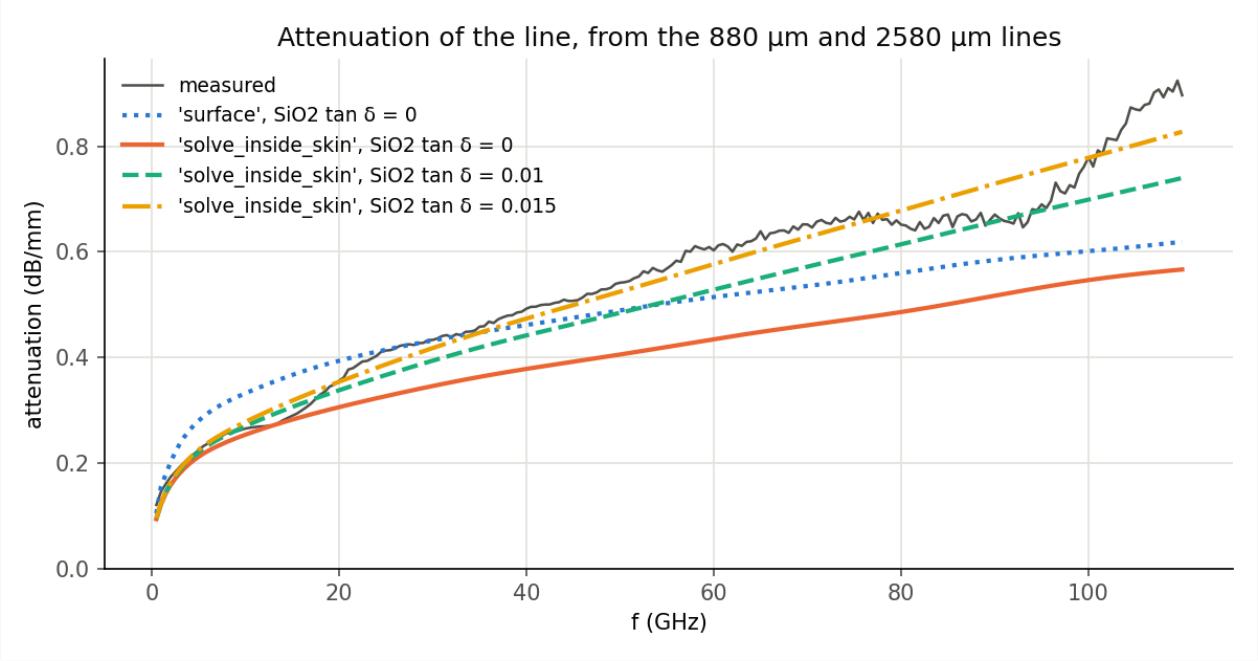


Figure 33: Attenuation of the line from the 880 µm and 2580 µm lines, measured and simulated

10.3 Lossless SiO2

In the supplied XML stackup files, SiO2 is lossless. With this stackup, the simulated attenuation agrees with the measurement from 3 to 10 GHz, within about 5 %. Below 3 GHz, the measured attenuation is higher, by up to 0.03 dB/mm (17 % at 1 GHz). There, all loss is conductor loss, so the metal of the measured wafer may differ in thickness or conductivity from the stackup; this measurement can't tell. Above 10 GHz, the measured loss grows faster with frequency. At 20 GHz, the simulation is 14 % low, at 100 GHz 30 % (0.55 instead of 0.78 dB/mm).

The surface model (`'surface'`, dotted) seems to agree better between 20 and 40 GHz. This is an error cancellation. The surface model doubles the loss of the thin Metal1 ground plane (see "Accuracy at high frequency"), and in this range the extra loss happens to replace the missing loss. Earlier editions of this guide used the surface model and found agreement up to 40 GHz.

A loss that grows faster with frequency than the conductor loss points to dielectric loss. With the skin effect, conductor loss grows about with the square root of frequency. The loss of a dielectric with a constant loss tangent grows in proportion to frequency.

10.4 SiO2 with a loss tangent

The loss tangent of the SiO2 in this process is not known. A paper by IHP authors on the SG25H technologies found values around 0.01:
Korndörfer F, F Sischka Optimization of the Substrate Parameters for EM-simulators MOSAK/ESSDERC/ESSCIRC Workshop (Munich) 14 Sep., 2007

- https://www.mos-ak.org/munich_2007/posters/P05_MOS-AK_Korndoerfer.pdf

With a loss tangent of 0.01 in all SiO2 layers, the simulation closes about 60 % of the gap. With 0.015, it agrees with the measured attenuation within about 5 % from 20 to 80 GHz. Between 80 and 100 GHz, the measured attenuation stays flat for a while, and the simulation is up to 12 % above it; at 100-110 GHz, it is within about 10 % again. In S21, 0.015 follows the 2580 µm line up to 40 GHz. Above, both measured lines have more loss, the 880 µm line most; S21 includes the pads and their de-embedding.

10.5 Conclusion regarding dielectric losses

Some dielectric loss seems to be real: a loss tangent of about 0.015 in SiO2 explains most of the measured line loss from 20 to 110 GHz. This value is fitted to one line on one wafer, measured in 2014. It is not a process value,

and the measurement can't tell dielectric loss from other losses that grow with frequency. So the supplied XML stackup files stay lossless.

To see how much dielectric loss matters for your design, set `DielectricLossTangent` of the SiO2 material in a copy of the XML stackup file, e.g. to 0.015, and compare the result with the lossless one. Palace uses the loss tangent; the Elmer EM output does not.

11 Advanced topics

11.1 Adaptive mesh refinement at selected frequencies only

By default, adaptive mesh refinement in Palace uses data from all simulation frequencies and all port excitations, which means a significant increase in overall simulation time because a full simulation is used during mesh refinement. To do a two-step approach where the adaptive mesh refinement used only a limited set of frequencies, you could run the gds2palace workflow with those frequencies only, and change the config.json file to **store the resulting mesh data to disk** , by setting SaveAdaptMesh to true.

```
"Model": {
    "Mesh": "palace_balun_mesh2_AMR.msh",
    "L0": 1e-06,
    "Refinement": {
        "UniformLevels": 0,
        "Tol": 0.01,
        "MaxIts": 2,
        "MaxSize": 2000000.0,
        "Nonconformal": true,
        "UpdateFraction": 0.7,
        "SaveAdaptMesh": false
    }
},
```

Adaptive frequency sweep would be **disabled** by removing the "AdaptiveTol" line in the frequency block.

After running Palace with this config file, you will find a refined mesh file in the output folder. Note that the final mesh refinement result is **not** located in the "iteration_nn" folder, the results of adaptive sweep can be found one level above (!) in the parent folder. You would then create another config.json file with the full frequency range and adaptive frequency sweep enabled, with the Mesh setting pointing to that adaptive mesh file created by the AMR before.

11.2 Using wave ports instead of lumped ports not supported

This gds2palace workflow creates lumped ports. The Palace solver also supports wave ports, but they are not implemented here: in Palace 0.18, two wave ports can't share one face, which a differential pair needs, and the line's mode index changes with frequency on silicon. To remove the port parasitics, use calibrated ports at a port wall instead, see "Calibrated de-embedding at port walls".

12 Using S-Parameter output, model extraction

S-Parameters are an accurate wideband description of the simulation results, but can be difficult to use in time domain circuits simulators. In this chapter, we will look at some possible solutions.

12.1 Lumped circuit model extraction

For a limited range of devices, we can use straightforward calculation of the equivalent lumped circuit model. Such a calculation is available here, based on simple narrowband calculation:

https://github.com/VolkerMuehlhaus/lumpedmodel

Currently, these devices are supported:

- Untapped inductors (2-port)
- MIM capacitor (2-port)
- Transmission line model (RLGC, 2-port data)

These tools are provided as Python scripts. The user needs to specify a target frequency, the fit result is then plotted along with the input data to show the quality of the model fit. For sure, there are other extraction tools available elsewhere that might cover more use cases.

12.2 Mathematical “black box” vector fit

Another approach is to do vector fitting of arbitrary n-port data. One possible implementation is available here, using the vector fit in scikit-rf library: https://github.com/VolkerMuehlhaus/lumpedmodel/tree/main/vector_fit

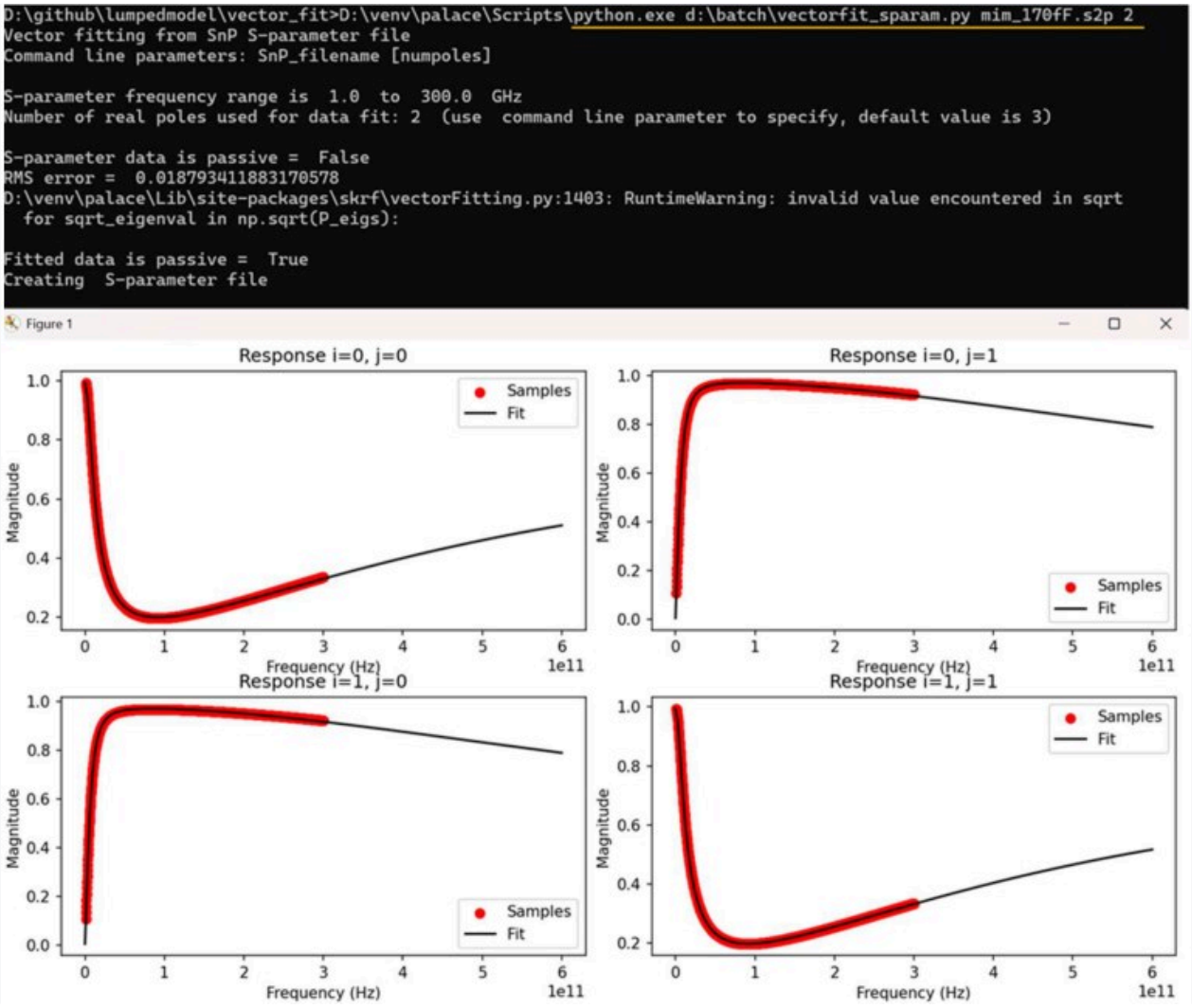


Figure 34: Vector fit of S-parameter data

13 Using gds2palace with Elmer FEM for EM simulation

Elmer FEM is an open-source multiphysics FEM solver. In addition to the AWS Palace output described so far in this guide, gds2palace can create input files for Elmer's electromagnetic (`VectorHelmholtz`) solver, producing S-parameters from the same layout, stackup and port model already used for Palace.

13.1 Installing Elmer FEM

Elmer FEM is not distributed with gds2palace and must be installed separately, see https://www.elmerfem.org/. gds2palace needs to find two Elmer command-line tools: `ElmerGrid` (mesh conversion) and `ElmerSolver` (the solver itself).

- **Windows:** set the environment variable `ELMER_HOME` to your Elmer install directory. gds2palace looks for `%ELMER_HOME%\bin\ElmerGrid.exe`.
- **Linux/macOS:** make sure `ElmerGrid` and `ElmerSolver` are available on your `PATH`.

If you use `settings['ELMER_MPI_THREADS']` to run Elmer across multiple processes, you also need an MPI implementation installed (see "External tools (not Python modules)" above) — on Windows this is Microsoft MPI, providing the `mpiexec`/`mpirun` launcher the generated `run_elmer` script uses.

13.2 From a Palace model to an Elmer model

An Elmer EM model uses the same **GDSII layout**, **XML stackup file**, **settings dictionary** and **port definitions** already described in chapter "Simulation model file in detail". The only change needed at the end of the model script is which output function is called:

```
# Palace output
config_name, data_dir = simulation_setup.create_palace (excite_ports, settings)
utilities.create_run_script(sim_path)
```

becomes

```
# Elmer FEM output
config_name, data_dir = simulation_setup.create_elmer (excite_ports, settings)
utilities.create_elmer_run_script(sim_path, settings)
```

Everything else in the model script — `gds_filename`, `XML_filename`, `settings['unit']`, `settings['margin']`, `settings['refined_cellsize']`, and port definitions via `simulation_setup.all_simulation_ports()` / `simulation_setup.simulation_port(...)` — stays exactly the same as for a Palace model.

Important difference in how ports are excited: Palace runs one solver pass per active port excitation (controlled by each port's `voltage` and the `excite_ports` list) to build up the full S-matrix, and ports with `voltage=0` are skipped to save simulation time. Elmer's EM solver instead uses a *constraint modes* ("scanning") analysis that solves for **all** ports defined in the model in a single run. This means:

- Every port you define in `simulation_ports` appears in the resulting Elmer S-matrix, regardless of its `voltage` setting.
- There is no equivalent to the Palace "excite only port 1" trick to save time — with Elmer, you get the full n-port S-matrix or nothing.
- Depending on port count and mesh size, a full n-port Elmer run can be faster or slower than n separate Palace excitation runs; this depends on the specific model.

13.3 Frequency sweep

`settings['fstart']`, `settings['fstop']`, `settings['fstep']`, `settings['fpoint']` and `settings['fdump']` are read the same way as for Palace and written to a plain frequency list (`frequencies.dat`) for Elmer. Note that Palace's adaptive frequency sweep interpolation

(`settings['adaptive_sweep']`) has no Elmer equivalent — Elmer solves at every frequency you list, so simulation time scales directly with the number of frequency points.

13.4 Mesh order and solver method

```
settings['order'] = 2          # 1 = linear, 2 = quadratic (recommended), default 2
settings['iterative'] = False  # True = iterative linear solver, False = direct solver (default)
```

- `settings['order']` selects the basis function order, same setting as used for Palace. Elmer supports order 1 (linear) and order 2 (quadratic); other values fall back to the order-1 solver recipe.
- `settings['iterative']` selects between Elmer's direct solver (MUMPS, default, robust for small/medium models) and an iterative solver, which can be more memory-efficient for large models.
- `settings['ELMER_MPI_THREADS']` (optional) enables MPI-parallel solving: gds2palace partitions the mesh with `ElmerGrid` for the given number of MPI processes, and the generated `run_elmer` script (see below) uses `mpirun` to start `ElmerSolver`.

```
settings['ELMER_MPI_THREADS'] = 8   # partition mesh and solve using 8 MPI processes
```

13.5 Boundary conditions and current limitations

`settings['boundary']` (the six-sided outer simulation boundary, values ABC/PML/PEC/PMC) works the same way as for Palace, including PMC.

A few other Palace features are not yet available in the Elmer EM flow:

- **Sheet resistor layers** (`Type="sheet"` metal paired with a `Resistor` material, as used for on-chip resistors) are not yet supported for Elmer EM output.
- **Composite ports** (grouping several EM ports into one output port) are not supported, same limitation as for Palace.

13.6 Running the Elmer EM simulation and getting S-parameters

`utilities.create_elmer_run_script(sim_path, settings)` writes a `run_elmer` script into the model output directory, next to the generated Elmer input files (`physics.sif`, `case.sif`, `frequencies.dat`, `mesh/`, `ELMERSOLVER_STARTINFO`):

```
#!/bin/bash
ElmerSolver
combine_snp
```

or, if `settings['ELMER_MPI_THREADS']` is set to more than 1:

```
#!/bin/bash
mpirun -np 8 ElmerSolver case.sif
combine_snp
```

Run it from the model output directory:

```
cd <model>_data
./run_elmer
```

`ElmerSolver` finds `case.sif` automatically via `ELMERSOLVER_STARTINFO`. The same `combine_snp` postprocessing script already used for Palace results also recognizes Elmer's `scalar_results` output and converts it to Touchstone (`.sNp`) format, so downstream use of Elmer S-parameters is identical to Palace results.

14 Using gds2palace with Elmer FEM for thermal simulation

In addition to EM simulation, gds2palace can generate input for Elmer's steady-state Heat Equation solver, producing a 3D temperature field from user-defined heat sources and constant-temperature boundaries. This shares the same GDSII + XML stackup pipeline as the EM flows, but does not use frequencies or ports.

Elmer FEM must be installed as described above (`ElmerGrid` and `ElmerSolver` on `PATH`, or `ELMER_HOME` set on Windows).

A complete, extensively documented worked example (model script, stackup XML, and GDSII layout) is available at `more_examples/thermal_simulation_using_Elmer/Elmer_Thermal_Workflow.md` — this chapter summarizes the key points.

14.1 Stackup XML: thermal material properties

For a thermal run, every `<Material>` needs a thermal conductivity. Only steady-state conductivity is evaluated so far; electrical parameters (used for S-parameter simulation) are not needed for a thermal-only model, though the same stackup file can be reused for both EM and thermal simulation.

```
<Material Name="TopMetal2" Type="Conductor" Conductivity="30300000.0"
          Density="2700" ThermalConductivity="237" Color="ff8000"/>
```

- `Density` (kg/m^3) is written through to Elmer but not used by the steady-state solver.
- `ThermalConductivity` is a single value in W/(m·K), for materials whose conductivity doesn't change meaningfully with temperature.
- `ThermalConductivityTable` points to a `<Table>` instead, for temperature-dependent conductivity (e.g. silicon substrate):

```
<Material Name="HighResSubstrate" Type="Semiconductor" Conductivity="0.025"
          ThermalConductivityTable="SiliconThermalCond" Density="2329"/>
...
<Tables>
  <Table Name="SiliconThermalCond">
    <Point Temperature="280" Value="163.00"/>
    <Point Temperature="290" Value="155.20"/>
  </Table>
</Tables>
```

gds2palace writes this straight through as an Elmer temperature-dependent material table — no manual `.sif` editing needed. See chapter "Extended XML stackup format" in the Appendix and `XML_stackup_format.md` for the full attribute reference.

14.2 Heat sources and constant-temperature boundaries

Instead of ports, a thermal model declares where heat enters and leaves through `thermal_objects`, using ordinary marker polygons drawn on dedicated GDSII layers:

```
thermal_objects = simulation_setup.all_thermal_objects()
thermal_objects.add_heatsource(simulation_setup.heatsource(
    power=0.65, source_layernum=201, target_layername='TFR'))
thermal_objects.add_consttemp(simulation_setup.constanttemp(
    temp=298, source_layernum=202, target_layername='BACKSIDEGND'))
```

- `source_layernum` is a GDSII layer number you draw purely as an xy footprint marker; it does not set a z-position.
- `target_layername` is the name of an existing `<Layer>` from the stackup XML — its z-range becomes the 3D volume the source/boundary is applied to.

- `heatsource(power, ...)` dissipates `power` Watts as a volumetric heat source over that volume. In the example above, 0.65 W is applied to the `TFR` resistor layer via marker layer 201.
- `constanttemp(temp, ...)` fixes that volume's temperature (Kelvin), acting as a heat sink/reference boundary. In the example above, layer 202 pins the `BACKSIDEGND` layer to 298 K (25 °C), a typical backside heatsink boundary.

The target layer for a heat source must be a volume (`Zmax` larger than `Zmin`). The target layer for a constant-temperature boundary is usually a zero-thickness sheet layer (`Zmax = Zmin`) — if it has finite thickness instead, two boundaries are created, one at each face.

14.3 Model script settings

```
settings['unit']              = 1e-6   # geometry is in microns
settings['margin']            = 100    # air margin around the layout, in microns
settings['refined_cellsize']  = 5      # extra-fine mesh near heat sources
settings['meshsize_max']      = 100    # coarsest mesh size, in microns
settings['elmer_thermal']     = True   # selects the thermal flow instead of EM/Palace
settings['thermal_objects']   = thermal_objects
```

`settings['no_gui'] = True` skips the interactive gmsh mesh preview, useful for unattended/batch runs. Then create the model:

```
config_name, data_dir = simulation_setup.create_elmer_thermal (settings)
```

Note there is no `excite_ports` argument here — a thermal run has no ports.

14.4 Running ElmerSolver and viewing results

Output goes to an `elmer_model/` folder next to the script, containing the gmsh mesh, the Elmer-native `mesh/` folder (from `ElmerGrid`), `case.sif` (materials, body/volume definitions, the heat source as a volumetric Body Force, and the constant-temperature Boundary Condition), and `ELMERSOLVER_STARTINFO`. It's worth a quick look at `case.sif` before solving, especially the first time you use a new stackup — it's plain text and easy to sanity-check.

```
cd elmer_model
ElmerSolver
```

Results appear in the same directory:

- `thermal_results.vtu` — full 3D temperature field, open in ParaView.
- `thermal_results.dat` — quick min/max temperature summary (plain text).

If `settings['ELMER_MPI_THREADS']` is set to more than 1, gds2palace also partitions the mesh for you; run `ElmerSolver_mpi` with your MPI launcher of choice instead of the single-threaded `ElmerSolver`.

14.5 Using setupThermal instead of writing Python code

setupThermal is a desktop GUI (part of the `setupEM` project) that drives this same flow without writing a Python model script by hand: pick the layout/stackup files, add heat sources and constant-temperature boundaries in a table (GDSII source layer, target stackup layer name, and power in W or temperature in K), then generate and run the model, or launch `ElmerSolver` directly from the app with output streamed to an on-screen log. The **Model** tab shows the generated Python code, so you can inspect or export it if you'd rather take over from there manually.

15 Appendix

15.1 Understanding volumes and surfaces created from GDSII

The workflow creates different types of geometry: volumes and surfaces. This depends on the mapping in the XML stackup file.

Geometries from GDSII layers are defined in the “Layers” section of the XML file :

- Type=“conductor” is used for all regular metal layers. With the default `conductor_model = 'surface'`, they are represented in the mesh by **“Surface”** objects on all boundaries (top, bottom, side walls). In other words, these metals are represented by hollow shells and not by solid volumes. With `'solve_inside'` or `'solve_inside_skin'`, they are **“Volume”** objects instead, see “Conductor loss modelling”.
- Type=“sheet” is used for metal layers that have no physical height in the model, they are represented as one flat **“Surface** “ only. A sheet uses a Resistor material or the reserved material PEC, see “Mapping of Volumes and Surfaces to Palace materials”.
- Type=“via” is used for via layers, these are represented in the mesh by **“Volume”** objects.

Important: Conductor layers must never be stacked directly, they must be separated by a via layer!

Extract from an XML stackup file, showing layer mappings with conductor, via and sheet types. It uses absolute z positions, as the legacy files do; the latest files position layers relative to each other (see “Extended XML stackup format”). The ground layers SUBGND and BACKSIDEGND are sheets of the reserved material PEC, as in the latest files:

```
  </Dielectrics>
  <Layers>
    <Substrate Offset="183.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    <Layer Name="Metal1" Type="conductor" Zmin="1.0400" Zmax="1.4600" Material="Metal1" Layer="8"/>
    <Layer Name="Metal2" Type="conductor" Zmin="2.0000" Zmax="2.4900" Material="Metal2" Layer="10"/>
    <Layer Name="Metal3" Type="conductor" Zmin="3.0300" Zmax="3.5200" Material="Metal3" Layer="30"/>
    <Layer Name="Metal4" Type="conductor" Zmin="4.0600" Zmax="4.5500" Material="Metal4" Layer="50"/>
    <Layer Name="Metal5" Type="conductor" Zmin="5.0900" Zmax="5.5800" Material="Metal5" Layer="67"/>
    <Layer Name="TopMetal1" Type="conductor" Zmin="6.4303" Zmax="8.4303" Material="TopMetal1" Layer="126"/>
    <Layer Name="TopMetal2" Type="conductor" Zmin="11.2303" Zmax="14.2303" Material="TopMetal2" Layer="134"/>
    <Layer Name="TopVia2" Type="via" Zmin="8.4303" Zmax="11.2303" Material="TopVia2" Layer="133"/>
    <Layer Name="TopVia1" Type="via" Zmin="5.5800" Zmax="6.4303" Material="TopVia1" Layer="125"/>
    <Layer Name="Via4" Type="via" Zmin="4.5500" Zmax="5.0900" Material="Via4" Layer="66"/>
    <Layer Name="Via3" Type="via" Zmin="3.5200" Zmax="4.0600" Material="Via3" Layer="49"/>
    <Layer Name="Via2" Type="via" Zmin="2.4900" Zmax="3.0300" Material="Via2" Layer="29"/>
    <Layer Name="Via1" Type="via" Zmin="1.4600" Zmax="2.0000" Material="Via1" Layer="19"/>
    <Layer Name="Cont" Type="via" Zmin="0.4000" Zmax="1.0400" Material="Cont" Layer="6"/>
    <Layer Name="SUBGND" Type="sheet" Zmin="0" Zmax="0" Material="PEC" Layer="250"/>
    <Layer Name="BACKSIDEGND" Type="sheet" Zmin="-183.75" Zmax="-183.75" Material="PEC" Layer="251"/>
    <Layer Name="MIM" Type="conductor" Zmin="5.6043" Zmax="5.7540" Material="MIM" Layer="36"/>
    <Layer Name="Vmim" Type="via" Zmin="5.7540" Zmax="6.4303" Material="Vmim" Layer="129"/>
    <Layer Name="LBE" Type="dielectric" Zmin="-183.75" Zmax="0" Material="Air" Layer="157"/>
  </Layers>
</ELayers>
</Stackup>
```

Resulting mesh with surface and volume objects:

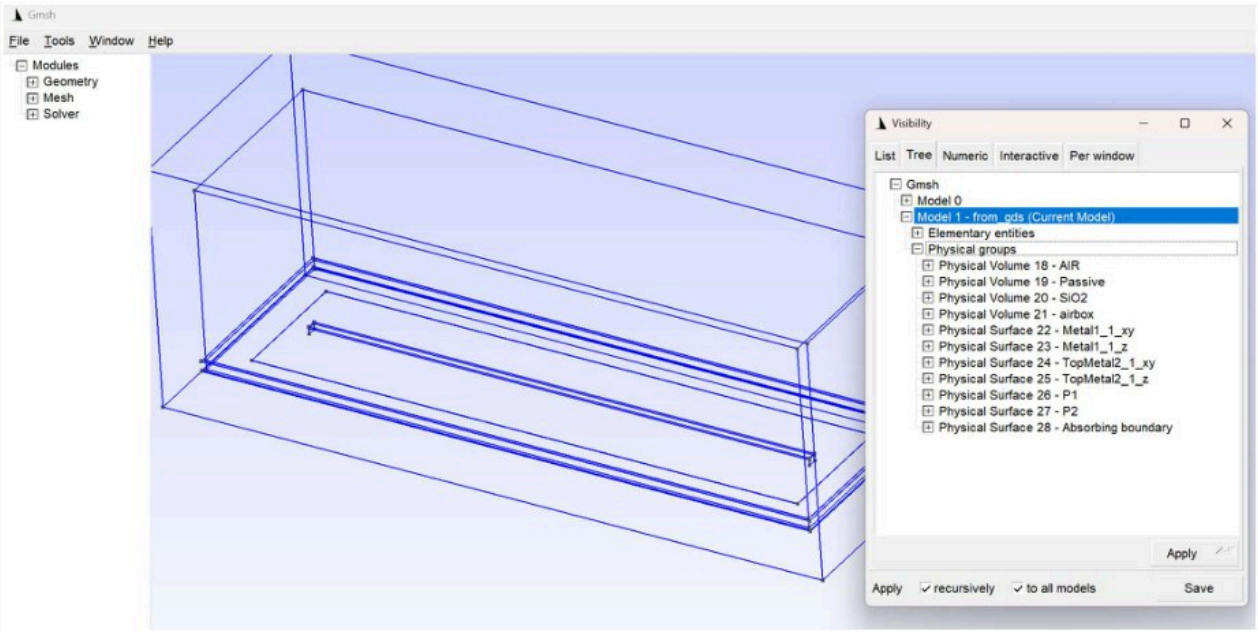


Figure 35: Resulting mesh with surface and volume objects

Dielectrics: There are additional materials in the stackup that are not drawn in the GDSII file, such as substrate or oxide or passivation. These are created as **“Volume”** objects in the mesh.
Dielectrics are defined in the section of the XML file, and usually cover the entire drawing area (bounding box of GDSII drawing) plus an additional “margin” that is defined in the simulation model code.

```
</Materials>
<ELayers LengthUnit="um">
  <Dielectrics>
    <Dielectric Name="AIR" Material="AIR" Thickness="200.0000"/>
    <Dielectric Name="Passive" Material="Passive" Thickness="0.4000"/>
    <Dielectric Name="SiO2" Material="SiO2" Thickness="15.7303"/>
    <Dielectric Name="EPI" Material="EPI" Thickness="3.7500"/>
    <Dielectric Name="Substrate" Material="Substrate" Thickness="180.0000"/>
  </Dielectrics>
  <Layers>
    <Substrate Offset="183.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    ...
```

As an option, the size of these dielectric layers can be defined by an optional parameter **Boundary** which sets the size of the dielectric from the bounding box of that layer in the GDSII file.

```
</Materials>
<ELayers LengthUnit="um">
  <Dielectrics>
    <Dielectric Name="iAIR" Material="AIR" Thickness="50.0000" Boundary="1039"/>
    <Dielectric Name="iSubstrate" Material="Substrate" Thickness="80.0000" Boundary="1039"/>
    <Dielectric Name="iSiO2" Material="SiO2" Thickness="12.27" Boundary="1039"/>
    <Dielectric Name="iPassive" Material="Passive" Thickness="0.4000" Boundary="1039"/>
    <Dielectric Name="Underfill" Material="Underfill" Thickness="46.2" Boundary="1039"/>
    <Dielectric Name="Passive" Material="Passive" Thickness="0.4000" Boundary="39"/>
    <Dielectric Name="SiO2" Material="SiO2" Thickness="15.7303" Boundary="39"/>
    <Dielectric Name="EPI" Material="EPI" Thickness="3.7500" Boundary="39"/>
    <Dielectric Name="Substrate" Material="Substrate" Thickness="180.0000" Boundary="39"/>
    <Dielectric Name="AIR" Material="AIR" Thickness="50.0000" Boundary="39"/>
  </Dielectrics>
  <Layers>
    <Substrate Offset="233.75"/>
    <Layer Name="Activ" Type="conductor" Zmin="0.0000" Zmax="0.4000" Material="Activ" Layer="1"/>
    ...
```

15.2 Mapping of Volumes and Surfaces to Palace materials

Palace offers a wide range of options to assign material properties to volumes and surfaces. Our workflow creates that mapping in the Palace control file (config.json), as described below:

Via layers (Type="via") are created as volumes, and the conductivity defined in the XML stackup is assigned to these volumes. In the Palace configuration file excerpts below, "Attributes" is the number of the physical volume or surface in the mesh.

```
"Domains": {
    "Materials": [
        {
            "Attributes": [
                9
            ],
            "Permittivity": 1.0,
            "Conductivity": [
                314300.0,
                314300.0,
                3143000.0
            ]
        },
```

To account for the z-directed nature of via arrays, which allows current flow predominantly in z direction, the conductivity from XML is only assigned to z-direction, and the value in xydirection is reduced by a factor of 10. This avoids issues with "unreal" currents flowing on the side walls of merged via polygons after via array merging.

With settings['fill_factor_correction'] = True, the conductivity of merged via polygons is additionally multiplied by their fill factor, in all directions. Via polygons on the same layer with different fill factors then get separate materials, in physical groups named after the layer and the fill factor, e.g. TopVia2_x0.49. The same applies to Elmer output: Elmer EM scales the (scalar) electrical conductivity, Elmer thermal the heat conductivity, and the Elmer bodies are named after these physical groups.

Metal layers (Type="conductor") are created as hollow elements surrounded by surfaces, with surface impedance to define metal loss (default `conductor_model = 'surface'`). The conductivity and thickness are obtained from the XML file. With `'solve_inside'` or `'solve_inside_skin'`, they are volumes with the conductivity of the XML file instead, like the vias.

```
"Boundaries": {
    "Conductivity": [
        {
            "Attributes": [
                27
            ],
            "Conductivity": 21640000.0,
            "Thickness": 0.41999999999999993
        },
```

All faces, top, bottom and side walls, get the same conductivity and thickness. This is exact for top and bottom of wide conductors with the same current on both faces, down to low frequency where the skin depth is larger than the physical conductor height. On the side walls, it over-estimates the conductor cross section at low frequency, see "Limits of conductor loss calculation". A thin layer with current on one face only gets up to twice its loss, see "Palace's External option". A conductor whose top or bottom face is on the simulation boundary gets `"External": true`, see "Conductors at the simulation boundary".

Metal sheet layers (Type="sheet") are created as a single, flat surface. Their position in the stackup should be defined with Zmin=Zmax for better readability, but actually the Zmax value is ignored here.

```
<Layer Name="MIM" Type="conductor" Zmin="5.6043" Zmax="5.7540" Material="MIM" Layer="36"/>
<Layer Name="Vmim" Type="via" Zmin="5.7540" Zmax="6.4303" Material="Vmim" Layer="129"/>
<Layer Name="LBE" Type="dielectric" Zmin="-183.75" Zmax="0" Material="Brick" Layer="157"/>
<Layer Name="ResistorA" Type="sheet" Zmin="22.0" Zmax="22.0" Material="ResA" Layer="220"/>
<Layer Name="ResistorB" Type="sheet" Zmin="15.0" Zmax="15.0" Material="ResB" Layer="221"/>
```

A sheet layer uses a Type="Resistor" material definition in the XML file, with fixed sheet resistance in Ohm/square, or the reserved material PEC (see below). A sheet with a Type="Conductor" material stops the model script with an error.

```
<Material Name="Vmim" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="2191000.0"
Color="ffe6bf"/>
<Material Name="MIM" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="500000.0"
Color="e6ffbf"/>
<Material Name="LBE" Type="Dielectric" Permittivity="1.0" DielectricLossTangent="0.0" Conductivity="0"
Color="d0d0d0"/>
<Material Name="ResA" Type="Resistor" Rs="3.5e-1" Color="d0d0d0"/>
<Material Name="ResB" Type="Resistor" Rs="1.1" Color="d0d0d0"/>
```

In the Palace config file, this results in an Impedance boundary with Rs value mapped to the surface.

```
"Impedance": [
    {
        "Attributes": [
            37,
            38
        ],
        "Rs": 0.35
    },
    {
        "Attributes": [
            39
        ],
        "Rs": 1.1
    }
],
```

Accuracy of IHP SG13G2 PDK resistors as sheets. The SG13G2 stackups build the resistor sheets `RSIL`, `RPPD` and `RHIGH` from the drawn poly, salicide block and implant layers. The sheet has the drawn size and the nominal sheet resistance. The PDK computes the resistance differently. Its pcells and SPICE models use $R = Rsh \cdot l / (w + \Delta w) + 2 \cdot rz / w$, with two corrections:

- **A width offset Δw.** The PDK uses an effective width that differs from the drawn one: rhigh 0.04 µm narrower, rsil and rppd 0.01 µm and 0.006 µm wider. This changes the width only, not the length.
- **An end resistance rz** at each end, between the silicided contact head and the resistor body: rsil 4.5 Ω·µm, rppd 35 Ω·µm, rhigh 80 Ω·µm. No drawn sheet carries it. In the EM model, only the contacts (`Cont`) add some resistance there.

Both corrections are fixed amounts, so physically small resistors are affected most: narrow ones by the width offset, short ones by the end resistance. Palace simulations of PDK resistor cells as drawn, each between two via ports, at 10 MHz:

Resistor (w × l)	PDK value	Palace	Error
rhigh 0.655 µm × 1.245 µm	2997 Ω	2599 Ω	−13 %
rppd 1 µm × 1.665 µm	500 Ω	442 Ω	−12 %
rhigh 0.5 µm × 4.29 µm	13.0 kΩ	11.7 kΩ	−10 %
rhigh 2 µm × 1.325 µm	999 Ω	906 Ω	−9 %
rsil 0.5 µm × 2.33 µm	50.0 Ω	48.9 Ω	−2 %
rsil 2 µm × 13.06 µm	50.0 Ω	49.9 Ω	−0.2 %

rhigh and rppd come out about 10 % low. rsil is close, because its contacts in the EM model happen to make up for most of its end resistance. If the exact value of an rhigh or rppd resistor matters, leave the resistor out of the EM model: put ports at its two Metal1 heads and connect the PDK resistor model there in the circuit simulator.

The reserved material `PEC` (`Material="PEC"`, no `<Material>` entry needed) is an ideal conductor. In the latest IHP stackups, the sheets SUBGND, BACKSIDEGND and REF_FOR_TRANSISTOR use it. A PEC sheet, and with the surface model a PEC conductor, goes into Palace's `PEC` boundary, together with the PEC sides of the simulation box. Palace has no ideal conductor volume, so a PEC via layer, and a PEC conductor with `'solve_inside'` or `'solve_inside_skin'`, is a volume with a conductivity of 1e10 S/m instead (a via layer again with a tenth of it in x and y).

Dielectrics and semiconductors (the `<Dielectrics>` stack and Type="dielectric" layers) are volumes with the permittivity of their material. A material with `DielectricLossTangent` above 0 gets that loss tangent. Otherwise it gets its `Conductivity`, which is how the silicon is lossy: in the latest SG13G2 stackups, the substrate (`Type="Semiconductor"`) has 2 S/m and the EPI layer 5 S/m. A material with both a loss tangent and a conductivity gets only the loss tangent.

```
"Materials": [
    {
        "Attributes": [
            3
        ],
        "Permittivity": 11.9,
        "Conductivity": 2.0
    },
    {
        "Attributes": [
            5
        ],
        "Permittivity": 4.1,
        "LossTan": 0.01
    },
```

15.3 Extended XML stackup format

The XML stackup format has grown beyond the basic dielectric stack and conductor/via layer mapping described above. The current major additions, all optional and backward-compatible with older stackup files, are:

- **Derived layers** — a layer's geometry can be computed from other layers via boolean operations (`AND`/`OR`/`XOR`/`NOT`) and resizing, instead of only being read directly from GDSII. Used, for example, to derive on-chip resistor geometry from existing process layers.
- **Reference-relative positioning** — a `<Layer>` or `<Dielectric>` can be positioned as an offset from the top or bottom edge of another named layer/dielectric, instead of requiring an absolute Zmin/Zmax.
- **Thermal material properties** — `Density`, `ThermalConductivity` and `ThermalConductivityTable` on `<Material>`, plus a `<Tables>` block for temperature-dependent conductivity, used by the Elmer thermal flow described in chapter "Using gds2palace with Elmer FEM for thermal simulation".

Using derived layers, reference-relative positioning, or thermal conductivity tables requires `schemaVersion="3.0"` in the `<Stackup>` root element. The stackup reader prints a warning if a file declares a newer `schemaVersion` than the installed gds2palace version supports, so an outdated installation is easier to notice.

The full attribute reference, with examples, is in `XML_stackup_format.md`, and derived-layer details specifically are in `derived_layers.md`.

15.4 Software versions used in this document

The results in this document come from different dates:

- Unless noted otherwise: Palace v0.14, gds2palace Python scripts as of October 13, 2025, in an Apptainer container on Ubuntu 24.04.02 LTS, AMD Ryzen 9-7950X (16 cores) with 128 GB RAM.
- Conductor loss modelling: Palace v0.18.0, gds2palace 0.9.0 (October 2026).
- Dielectric loss modelling: Palace v0.19.0, gds2palace 0.10.4 (November 2026).

16 List of examples

Here is an overview of model examples.

16.1 palace_line_viaport.py

Microstrip line on TopMetal2 over Metal1 ground plane, with via ports on both ends. Geometry is read from GDSII. The stackup does not include bulk silicon because that is shield by the ground plane anyway.

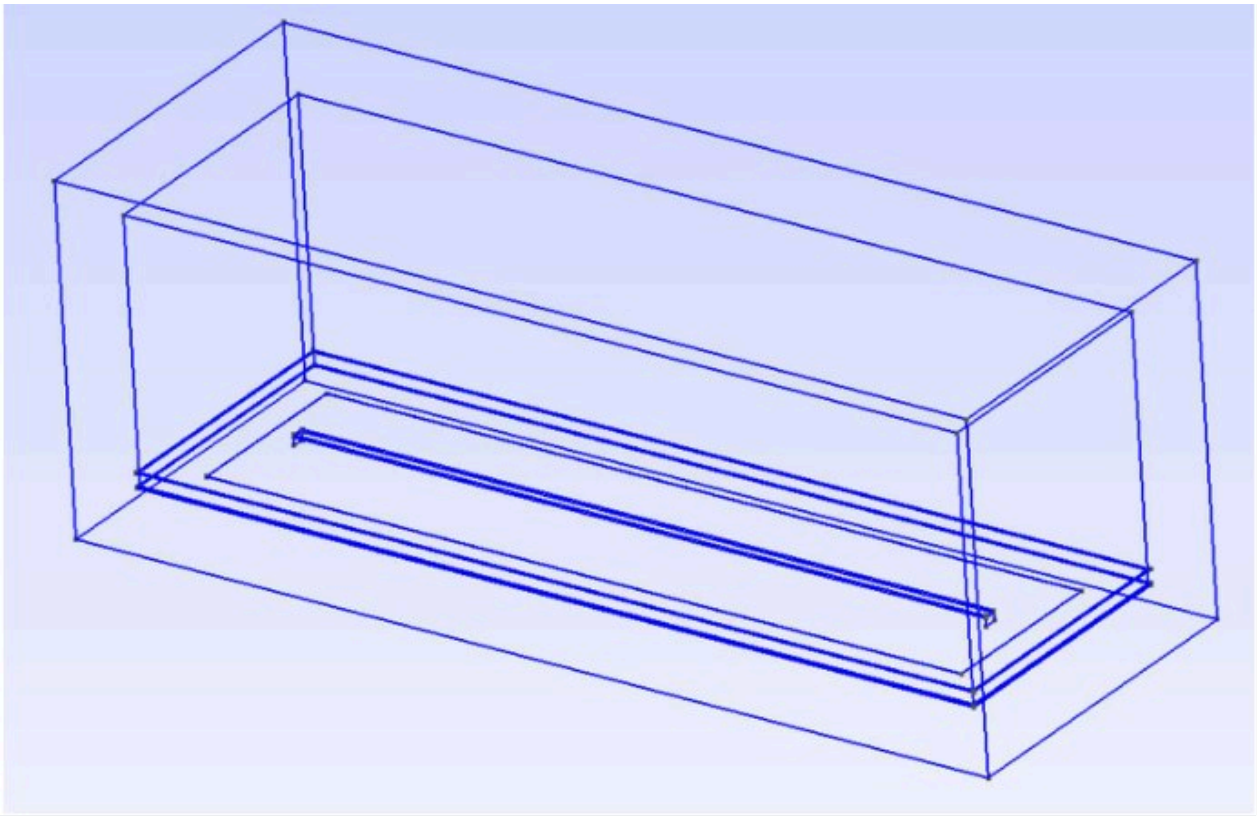


Figure 36: Example palace_line_viaport.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_line_viaport.py

16.2 palace_line_noGDS.py

Similar to palace_line_viaport.py but geometry is created by code instead of reading it from GDSII. https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_line_noGDS.py

16.3 palace_ind_frame.py

This is a 2-port inductor embedded into a metal frame. Geometry from GDSII.

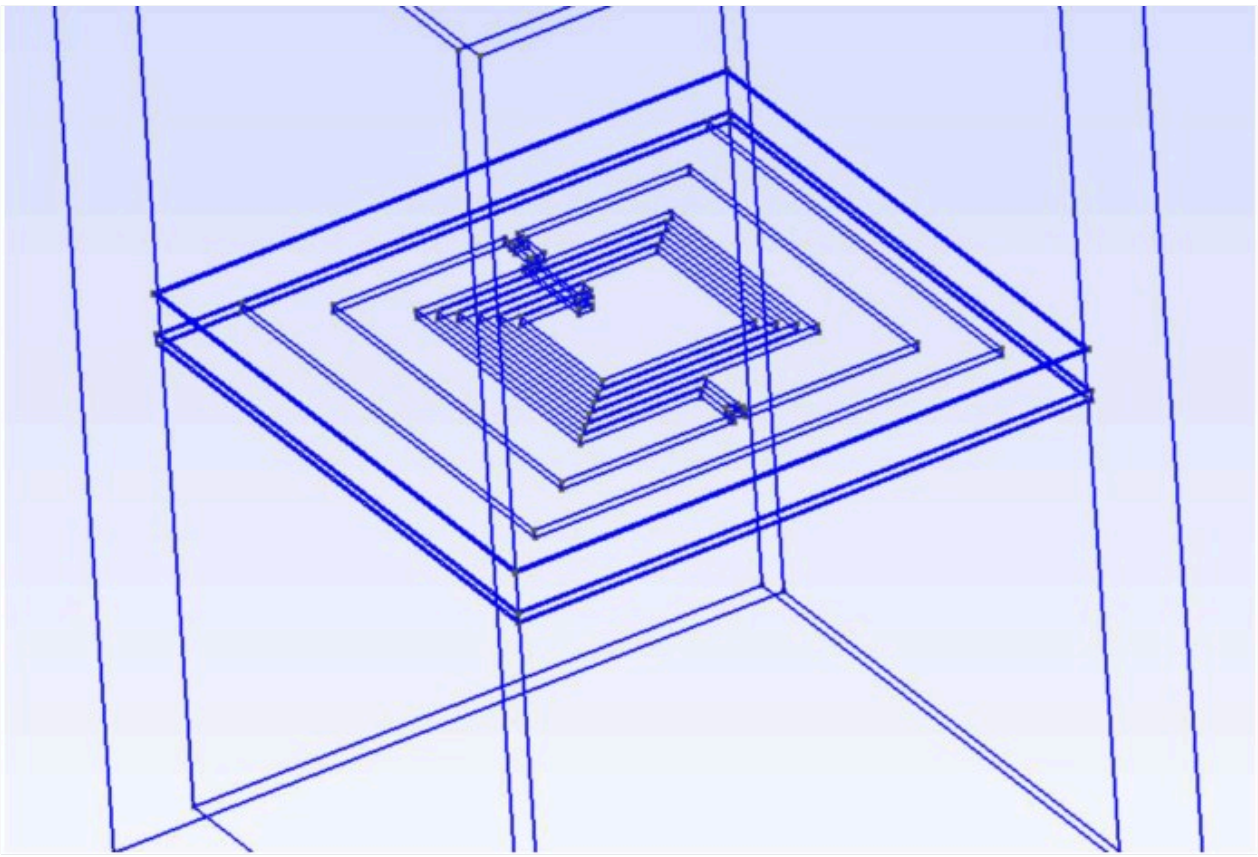


Figure 37: Example palace_ind_frame.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_ind_frame.py

16.4 palace_L2n0.py

2-port octagon inductor with via ports down to an artificial metal plane that sits on the surface of bulk silicon.

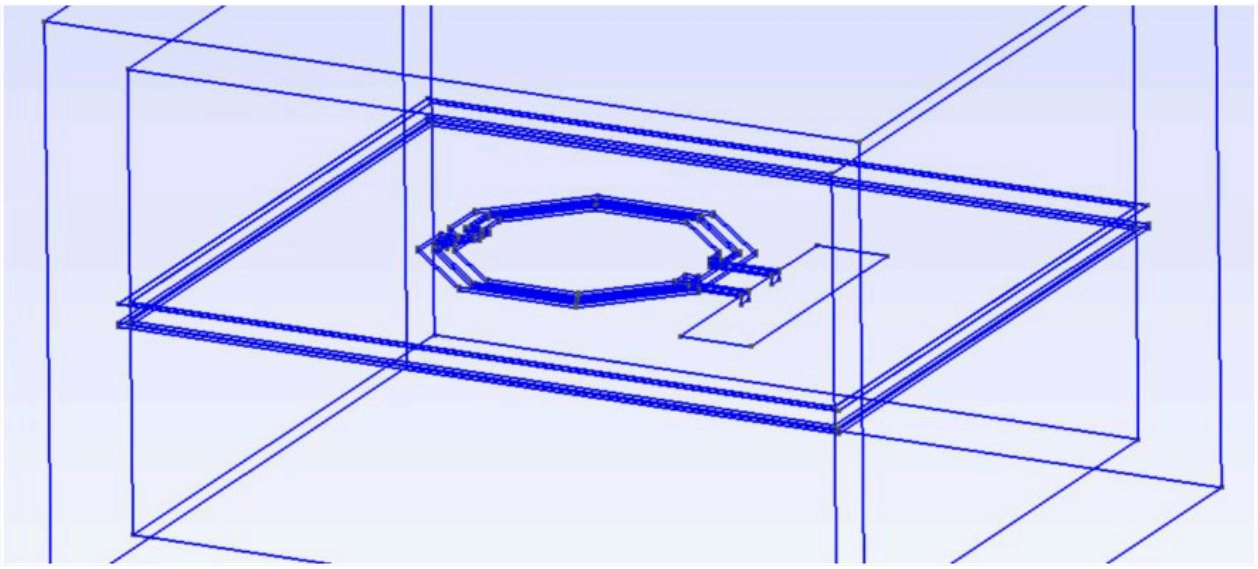


Figure 38: Example palace_L2n0.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_L2n0.py

Field dump at a single frequency is enabled in this model. The Paraview screenshot below shows current density using this file:

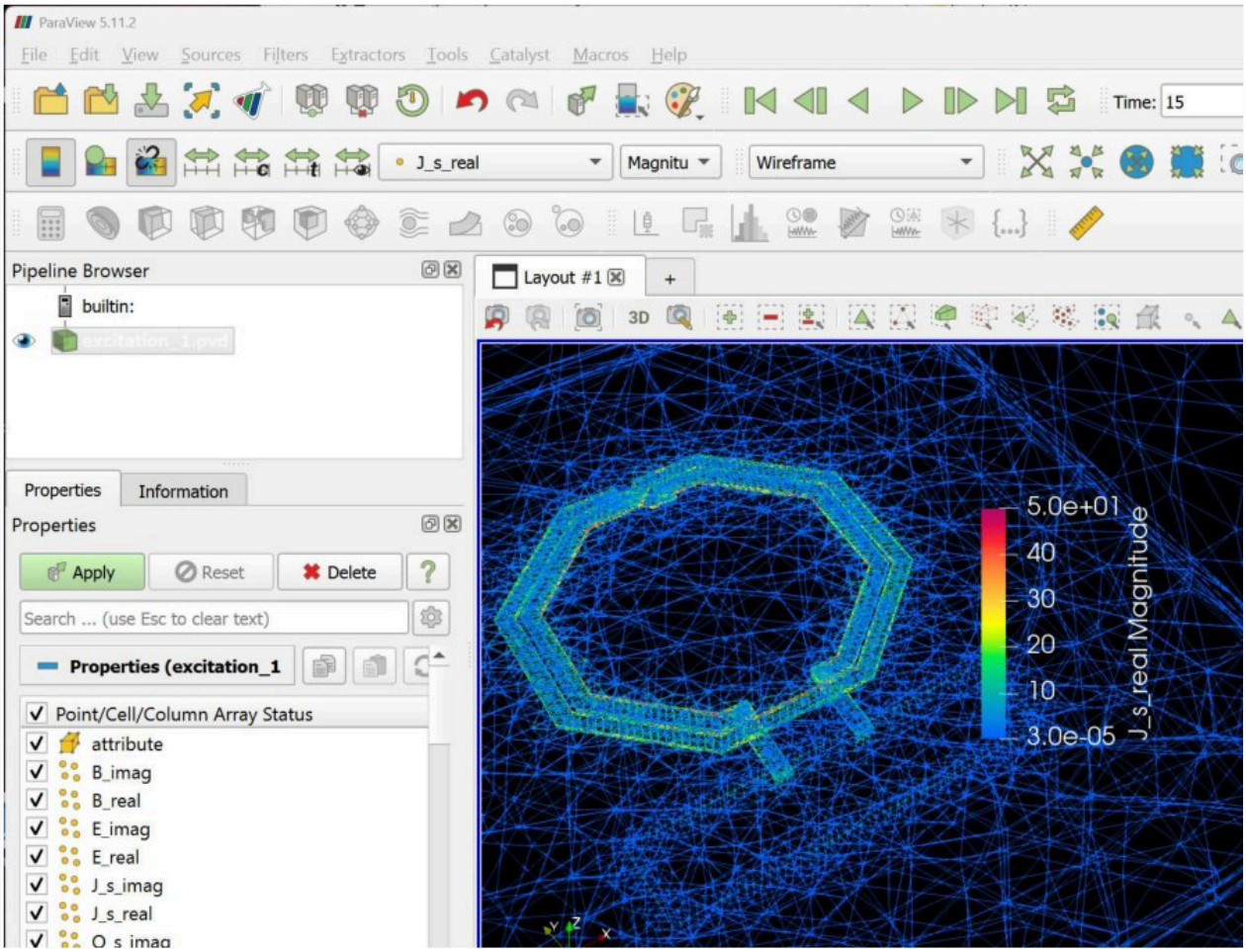


Figure 39: Example palace_L2n0.py: current density in ParaView

16.5 palace_butlermatrix.py

This is quite a big model, the Butler Matrix by Ardavan Rahimian in IHP Open PDK Tapeout - July 2025, available at https://github.com/IHP-GmbH/TO_July2025/tree/main/W_Band_Butler_Matrix_IC

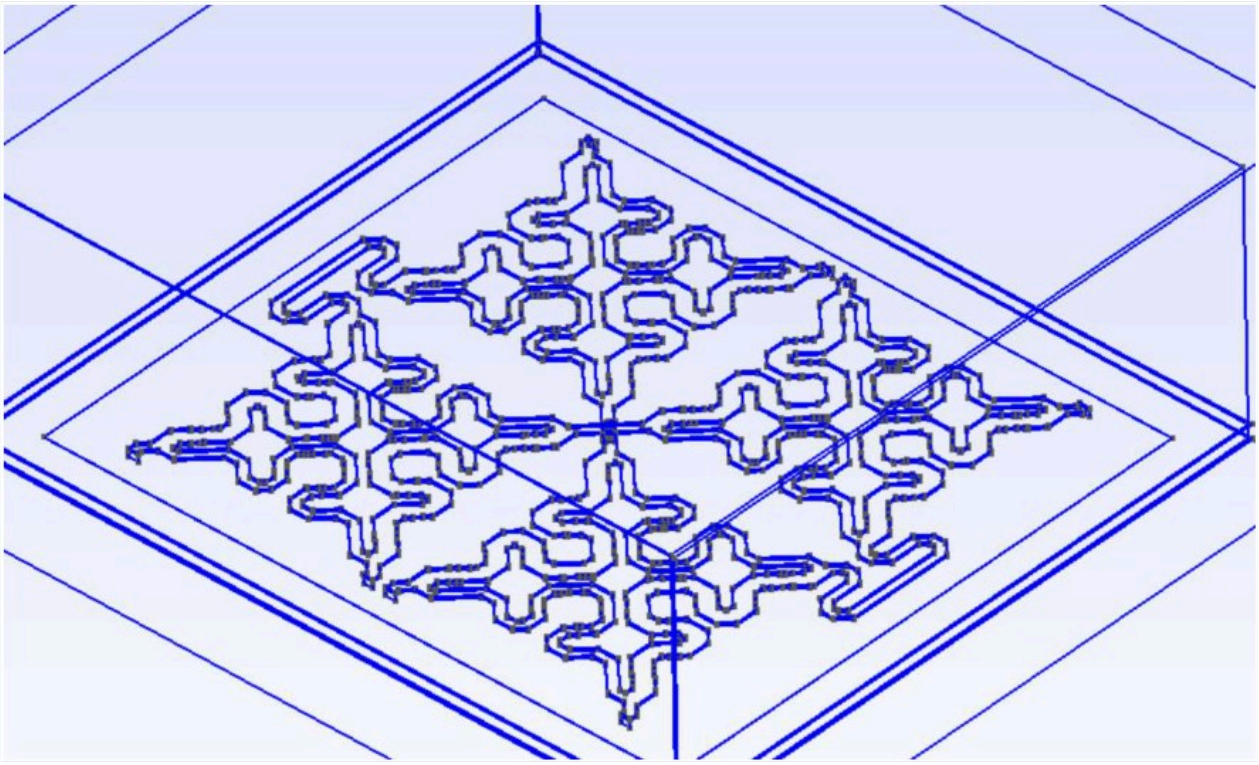


Figure 40: Example palace_butlermatrix.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_butlermatrix.py

Geometry is from GDSII, the model uses a total of 8 via ports between Metal3 and TopMetal2. However, to speed up simulation, only one port excitation is active in this model, and we only get S11,S21,S31,S41,S51,S61,S71 and S81 from this simulation run.

```
simulation_ports = simulation_setup.all_simulation_ports()
# instead of in-plane port specified with target_layername,
# we here use via port specified with from_layername and to_layername
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=1, voltage=1, port_Z0=50, source_layernum=201,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=2, voltage=0, port_Z0=50, source_layernum=202,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=3, voltage=0, port_Z0=50, source_layernum=203,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=4, voltage=0, port_Z0=50, source_layernum=204,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=5, voltage=0, port_Z0=50, source_layernum=205,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=6, voltage=0, port_Z0=50, source_layernum=206,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=7, voltage=0, port_Z0=50, source_layernum=207,
```

```
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
simulation_ports.add_port(simulation_setup.simulation_port(
    portnumber=8, voltage=0, port_Z0=50, source_layernum=208,
    from_layername='Metal3', to_layername='TopMetal2', direction='z'))
```

All other port excitations are set to voltage=0 in the model. To get full 8-port S-parameters, we would need to change all port voltages to 1 and re-run the model, so that all excitations are simulated, one after another.

16.6 palace_butlermatrix_dump93.py

Same as before, but a field dump setting is added in this model at 93 GHz.

```
settings['fstart']  = 90e9
settings['fstop']   = 95e9
settings['fstep']   = 0.1e9

# settings["fpoint"] = [93e9]  # list of discrete frequencies to be simulated

settings["fdump"] = [93e9]  # save field dump at these frequency points
```

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_butlermatrix_dump93.py

16.7 palace_core.py

This is the 4-port “core” model of the 60 GHz medium power amplifier from IHP Analog Academy online course. There are two via ports at the input and output, and two in-plane ports on Metal2 between the common emitter polygon and base/collector coming down from the via stacks.

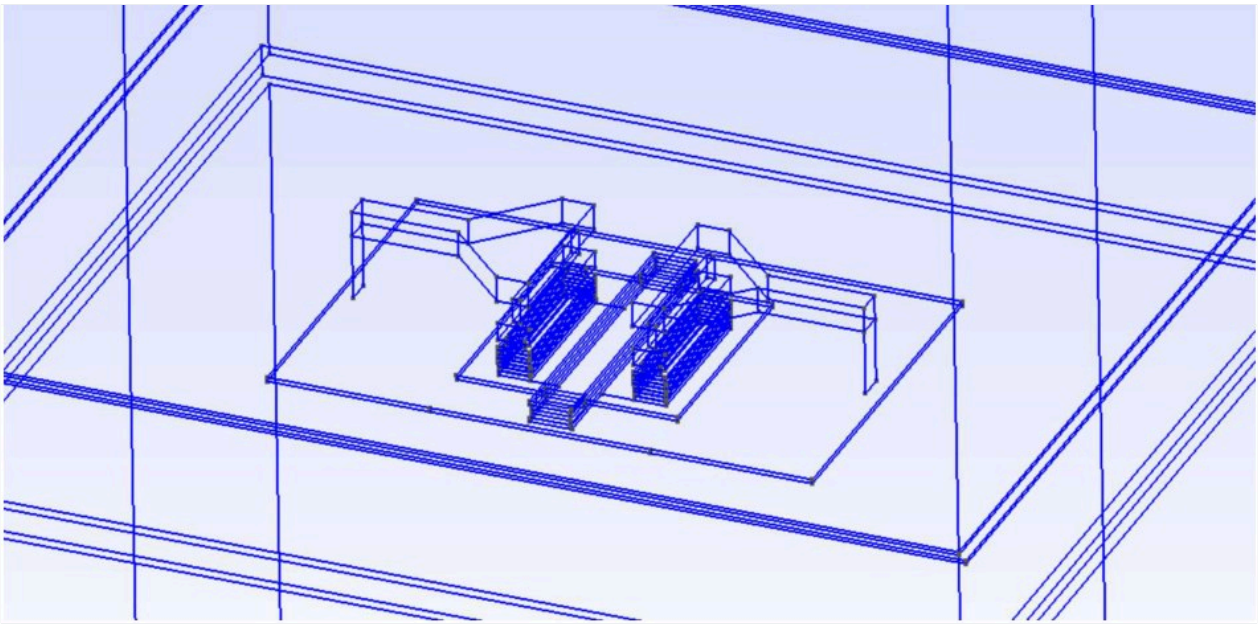


Figure 41: Example palace_core.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_core.py

16.8 palace_rfcmim.py

This is 2-port model of an rf_cmim component with some feedline length. Via ports are used at the input and output down to the Metal1 ground plane. Via arrays are merged by this setting: merge_polygon_size = 2

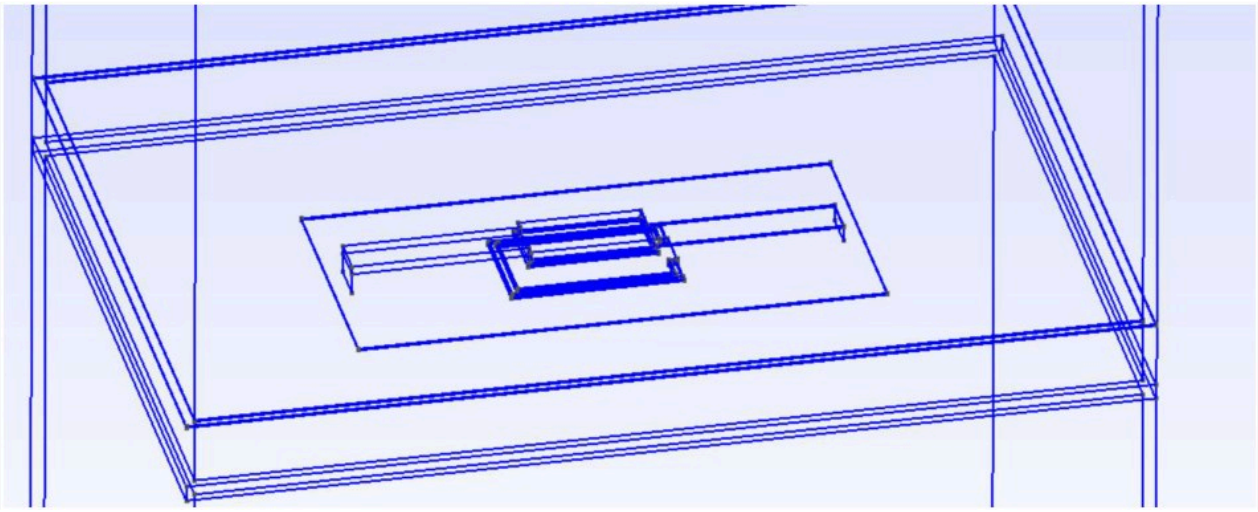


Figure 42: Example palace_rfcmim.py

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_rfcmim.py

16.9 palace_pcb_lowpass.py

This shows a use case of gds2palace outside the RFIC domain: the layout of a PCB lowpass on RO4003 substrate was imported to klayout and saved in GDSII format.

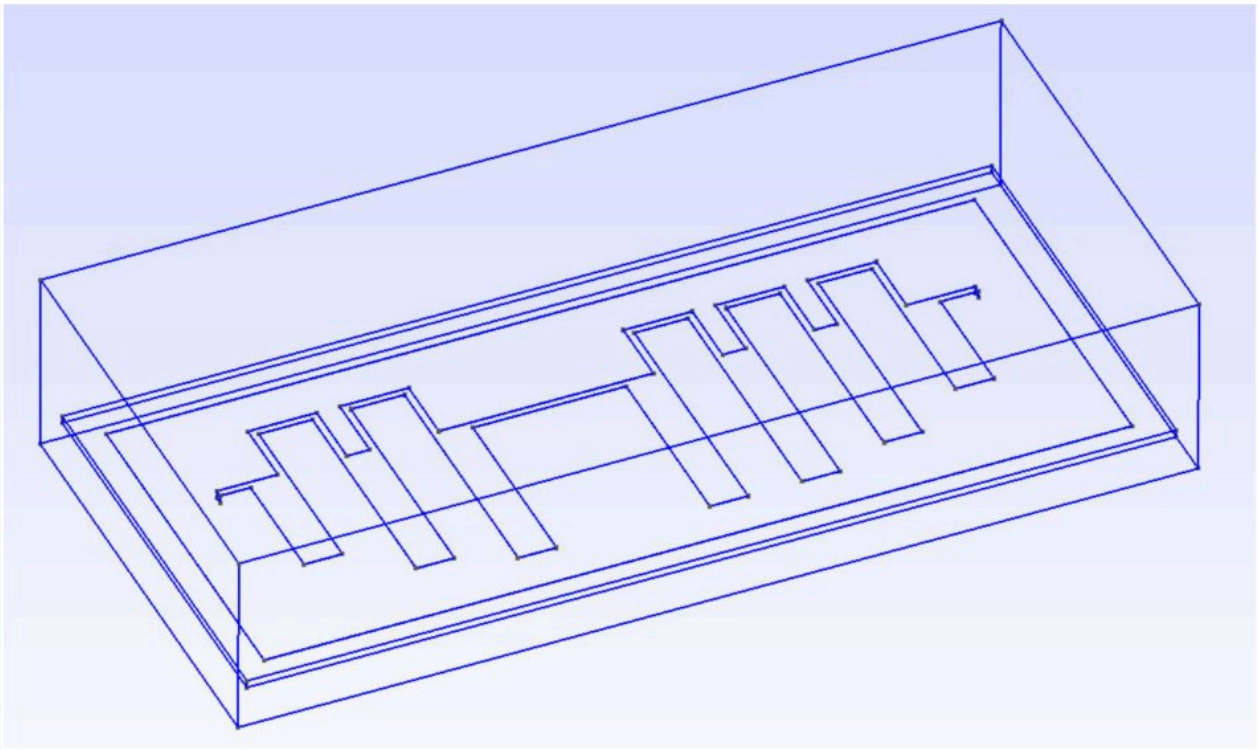


Figure 43: Example palace_pcb_lowpass.py

An XML stackup file was created for this substrate, with layout layers matching the layer numbers used in klayout.

```
<Stackup schemaVersion="2.0">
  <Materials>
    <Material Name="Copper" Type="Conductor" Permittivity="1" DielectricLossTangent="0" Conductivity="3.3e7"
Color="00ff00"/>
    <Material Name="RO4003" Type="Dielectric" Permittivity="3.38" DielectricLossTangent="0.0022"
Conductivity="0" Color="01e0ff"/>
    <Material Name="AIR" Type="Dielectric" Permittivity="1.0" DielectricLossTangent="0.0" Conductivity="0"
```

```
Color="d0d0d0"/>
  </Materials>
  <ELayers LengthUnit="um">
    <Dielectrics>
      <Dielectric Name="RO4003" Material="RO4003" Thickness="510"/>
    </Dielectrics>
    <Layers>
      <Substrate Offset="0"/>
      <Layer Name="Bottom" Type="Conductor" Zmin="-17" Zmax="0" Material="Copper" Layer="1"/>
      <Layer Name="Top" Type="Conductor" Zmin="510" Zmax="527" Material="Copper" Layer="10"/>
    </Layers>
  </ELayers>
</Stackup>
```

Here, setting “margins” only controls the (small) oversize of the dielectrics from the drawing layers. The optional “air_around” setting was used in the simulation model to define the air margins independently, without adding these air layers in the XML stackup.

```
settings['unit']   = 1e-6  # geometry is in MILLIMETER for PCB example
settings['margin'] = 2000    # distance in microns from GDSII geometry boundary to simulation boundary
settings['air_around'] = [1000,1000,1000,1000,2000, 10000] # airbox size to simulation boundary
```

https://github.com/VolkerMuehlhaus/gds2palace_ihp_sg13g2/blob/main/workflow/palace_pcb_lowpass.py